

Chapter VII

Zeta Functions and L -series

§ 1. The Riemann Zeta Function

One of the most astounding phenomena in number theory consists in the fact that a great number of deep arithmetic properties of a number field are hidden within a single analytic function, its **zeta function**. This function has a simple shape, but it is unwilling to yield its mysteries. Each time, however, that we succeed in stealing one of these well-guarded truths, we may expect to be rewarded by the revelation of some surprising and significant relationship. This is why zeta functions, as well as their generalizations, the L -series, have increasingly moved to the foreground of the arithmetic scene, and today are more than ever the focus of number-theoretic research. The fundamental prototype of such a function is **Riemann's zeta function**

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

where s is a complex variable. It is to this important function that we turn first.

(1.1) Proposition. *The series $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ is absolutely and uniformly convergent in the domain $\operatorname{Re}(s) \geq 1 + \delta$, for every $\delta > 0$. It therefore represents an analytic function in the half-plane $\operatorname{Re}(s) > 1$. One has **Euler's identity***

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}},$$

where p runs through the prime numbers.

Proof: For $\operatorname{Re}(s) = \sigma \geq 1 + \delta$, the series $\sum_{n=1}^{\infty} |1/n^s| = \sum_{n=1}^{\infty} 1/n^\sigma$ admits the convergent majorant $\sum_{n=1}^{\infty} 1/n^{1+\delta}$, i.e., $\zeta(s)$ is absolutely and uniformly convergent in this domain. In order to prove Euler's identity, we remind ourselves that an infinite product $\prod_{n=1}^{\infty} a_n$ of complex numbers a_n is said to converge if the sequence of partial products $P_n = a_1 \cdots a_n$ has a nonzero limit. This is the case if and only if the series $\sum_{n=1}^{\infty} \log a_n$ converges, where $\log$ denotes the principal branch of the logarithm (see [2], chap. V, 2.2). The

product is called absolutely convergent if the series converges absolutely. In this case the product converges to the same limit even after a reordering of its terms a_n .

Let us now formally take the logarithm of the product

$$E(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

We obtain the series

$$\log E(s) = \sum_p \sum_{n=1}^{\infty} \frac{1}{np^{ns}}.$$

It converges absolutely for $\operatorname{Re}(s) = \sigma \geq 1 + \delta$. In fact, since $|p^{ns}| = p^{n\sigma} \geq p^{(1+\delta)n}$, one has the convergent majorant

$$\sum_p \sum_{n=1}^{\infty} \left(\frac{1}{p^{1+\delta}} \right)^n = \sum_p \frac{1}{p^{1+\delta} - 1} \leq 2 \sum_p \frac{1}{p^{1+\delta}}.$$

This implies the absolute convergence of the product

$$E(s) = \prod_p \frac{1}{1 - p^{-s}} = \exp \left(\sum_p \left(\sum_{n=1}^{\infty} \frac{1}{np^{ns}} \right) \right).$$

In this product we now expand the product of the factors

$$\frac{1}{1 - p^{-s}} = 1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \dots$$

for all prime numbers $p_1, \dots, p_r \leq N$, and obtain the equality

$$(*) \quad \prod_{p \leq N} \frac{1}{1 - p^{-s}} = \sum_{v_1, \dots, v_r=0}^{\infty} \frac{1}{(p_1^{v_1} \dots p_r^{v_r})^s} = \sum_n' \frac{1}{n^s},$$

where $\sum'$ denotes the sum over all natural numbers which are divisible only by prime numbers $p \leq N$. Since the sum $\sum'$ contains in particular the terms corresponding to all $n \leq N$, we may also write

$$\prod_{p \leq N} \frac{1}{1 - p^{-s}} = \sum_{n \leq N} \frac{1}{n^s} + \sum_{n > N} \frac{1}{n^s}.$$

Comparing now in (*) the sum $\sum'$ with the series $\zeta(s)$, we get

$$\left| \prod_{p \leq N} \frac{1}{1 - p^{-s}} - \zeta(s) \right| \leq \left| \sum_{\substack{n > N \\ p_i \nmid n}} \frac{1}{n^s} \right| \leq \sum_{n > N} \frac{1}{n^{1+\delta}},$$

where the right hand side goes to zero as $N \rightarrow \infty$ because it is the remainder of a convergent series. This proves Euler's identity. $\square$

Euler's identity expresses the law of unique prime factorization of natural numbers in a single equation. This already demonstrates the number-theoretic significance of the zeta function. It challenges us to study its properties more closely. By its definition, the function is only given on the half-plane $\operatorname{Re}(s) > 1$. It does, however, admit an analytic continuation to the whole complex plane, with the point $s = 1$ removed, and it satisfies a functional equation which relates the argument s to the argument $1 - s$. These crucial facts will be proved next. The proof hinges on an **integral formula** for the zeta function $\zeta(s)$ which arises from the well-known **gamma function**. This latter is defined for $\operatorname{Re}(s) > 0$ by the absolutely convergent integral

$$\Gamma(s) = \int_0^{\infty} e^{-y} y^s \frac{dy}{y}$$

and obeys the following rules (see [34], vol. I, chap. I).

(1.2) Proposition. (i) *The gamma function is analytic and admits a meromorphic continuation to all of $\mathbb{C}$.*

(ii) *It is nowhere zero and has simple poles at $s = -n$, $n = 0, 1, 2, \dots$, with residues $(-1)^n/n!$. There are no poles anywhere else.*

(iii) *It satisfies the functional equations*

$$1) \Gamma(s+1) = s\Gamma(s),$$

$$2) \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s},$$

$$3) \Gamma(s)\Gamma(s + \tfrac{1}{2}) = \frac{2\sqrt{\pi}}{2^s} \Gamma(2s) \text{ (Legendre's duplication formula).}$$

(iv) *It has the special values $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(1) = 1$, $\Gamma(k+1) = k!$, $k = 0, 1, 2, \dots$.*

To relate the gamma function to the zeta function, start with the substitution $y \mapsto \pi n^2 y$, which gives the equation

$$\pi^{-s} \Gamma(s) \frac{1}{n^{2s}} = \int_0^{\infty} e^{-\pi n^2 y} y^s \frac{dy}{y}.$$

Now sum over all $n \in \mathbb{N}$ and get

$$\pi^{-s} \Gamma(s) \zeta(2s) = \int_0^{\infty} \sum_{n=1}^{\infty} e^{-\pi n^2 y} y^s \frac{dy}{y}.$$

Observe that it is legal to interchange the sum and the integral because

$$\begin{aligned} \sum_{n=1}^{\infty} \int_0^{\infty} |e^{-\pi n^2 y} y^s| \frac{dy}{y} &= \sum_{n=1}^{\infty} \int_0^{\infty} e^{-\pi n^2 y} y^{\operatorname{Re}(s)} \frac{dy}{y} \\ &= \pi^{-\operatorname{Re}(s)} \Gamma(\operatorname{Re}(s)) \zeta(2 \operatorname{Re}(s)) < \infty. \end{aligned}$$

Now the series under the integral,

$$g(y) = \sum_{n=1}^{\infty} e^{-\pi n^2 y},$$

arises from **Jacobi's classical theta series**

$$\theta(z) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 z} = 1 + 2 \sum_{n=1}^{\infty} e^{\pi i n^2 z},$$

i.e., we have $g(y) = \frac{1}{2}(\theta(iy) - 1)$. The function

$$Z(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

is called the **completed zeta function**. We obtain the

(1.3) Proposition. *The completed zeta function $Z(s)$ admits the integral representation*

$$Z(s) = \frac{1}{2} \int_0^{\infty} (\theta(iy) - 1) y^{s/2} \frac{dy}{y}.$$

The proof of the functional equation for the function $Z(s)$ is based on the following general principle. For a continuous function $f : \mathbb{R}_+^* \rightarrow \mathbb{C}$ on the group $\mathbb{R}_+^*$ of positive real numbers, we define the **Mellin transform** to be the improper integral

$$L(f, s) = \int_0^{\infty} (f(y) - f(\infty)) y^s \frac{dy}{y},$$

provided the limit $f(\infty) = \lim_{y \rightarrow \infty} f(y)$ and the integral exist. The following theorem is of pivotal importance, also for later applications. We will often refer to it as the **Mellin principle**.

(1.4) Theorem. Let $f, g : \mathbb{R}_+^* \rightarrow \mathbb{C}$ be continuous functions such that

$$f(y) = a_0 + O(e^{-cy^\alpha}), \quad g(y) = b_0 + O(e^{-cy^\alpha}),$$

for $y \rightarrow \infty$, with positive constants c, α . If these functions satisfy the equation

$$f\left(\frac{1}{y}\right) = Cy^k g(y),$$

for some real number $k > 0$ and some complex number $C \neq 0$, then one has:

(i) The integrals $L(f, s)$ and $L(g, s)$ converge absolutely and uniformly if s varies in an arbitrary compact domain contained in $\{s \in \mathbb{C} \mid \operatorname{Re}(s) > k\}$. They are therefore holomorphic functions on $\{s \in \mathbb{C} \mid \operatorname{Re}(s) > k\}$. They admit holomorphic continuations to $\mathbb{C} \setminus \{0, k\}$.

(ii) They have simple poles at $s = 0$ and $s = k$ with residues

$$\begin{aligned} \operatorname{Res}_{s=0} L(f, s) &= -a_0, & \operatorname{Res}_{s=k} L(f, s) &= Cb_0, & \text{resp.} \\ \operatorname{Res}_{s=0} L(g, s) &= -b_0, & \operatorname{Res}_{s=k} L(g, s) &= C^{-1}a_0. \end{aligned}$$

(iii) They satisfy the functional equation

$$L(f, s) = CL(g, k - s).$$

Remark 1: The symbol $\varphi(y) = O(\psi(y))$ means, as usual, that one has $\varphi(y) = c(y)\psi(y)$, for some function $c(y)$ which stays bounded under the limit in question, so in our case, as $y \rightarrow \infty$.

Remark 2: Condition (ii) is to be understood to say that there is no pole if $a_0 = 0$, resp. $b_0 = 0$. But there is a pole, which is simple, if $a_0 \neq 0$, resp. $b_0 \neq 0$.

Proof: If s varies over a compact subset of $\mathbb{C}$, then the function $e^{-cy^\alpha} y^\sigma$, $\sigma = \operatorname{Re}(s)$, is bounded for $y \geq 1$ by a constant which is independent of σ . Therefore the condition $f(y) = a_0 + O(e^{-cy^\alpha})$ gives the following upper bound for the integrand of the Mellin integral $L(f, s)$.

$$|(f(y) - a_0)y^{s-1}| \leq B e^{-cy^\alpha} y^{\sigma+1} y^{-2} \leq B' \frac{1}{y^2},$$

for all $y \geq 1$, with constants B, B' . The integral $\int_1^\infty (f(y) - a_0)y^{s-1} dy$ therefore admits the convergent majorant $\int_1^\infty \frac{B'}{y^2} dy$ which is independent of s . It therefore converges absolutely and uniformly, for all s in the compact subset. The same holds for $\int_1^\infty (g(y) - b_0)y^{s-1} dy$.

Now let $\operatorname{Re}(s) > k$. We cut the interval of integration $(0, \infty)$ into $(0, 1]$ and $(1, \infty)$ and write

$$L(f, s) = \int_1^{\infty} (f(y) - a_0) y^s \frac{dy}{y} + \int_0^1 (f(y) - a_0) y^s \frac{dy}{y}.$$

For the second integral, the substitution $y \mapsto 1/y$ and the equation $f(1/y) = C y^k g(y)$ give:

$$\begin{aligned} \int_0^1 (f(y) - a_0) y^s \frac{dy}{y} &= -a_0 \frac{y^s}{s} \Big|_0^1 + \int_1^{\infty} f\left(\frac{1}{y}\right) y^{-s} \frac{dy}{y} \\ &= -\frac{a_0}{s} + C \int_1^{\infty} (g(y) - b_0) y^{k-s-1} dy - \frac{C b_0}{k-s}. \end{aligned}$$

By the above, it also converges absolutely and uniformly for $\operatorname{Re}(s) > k$. We therefore obtain

$$L(f, s) = -\frac{a_0}{s} + \frac{C b_0}{s-k} + F(s),$$

where

$$F(s) = \int_1^{\infty} [(f(y) - a_0) y^s + C(g(y) - b_0) y^{k-s}] \frac{dy}{y}.$$

Swapping f and g , we see from $g(1/y) = C^{-1} y^k f(y)$ that:

$$L(g, s) = -\frac{b_0}{s} + \frac{C^{-1} a_0}{s-k} + G(s)$$

where

$$G(s) = \int_1^{\infty} [(g(y) - b_0) y^s + C^{-1}(f(y) - a_0) y^{k-s}] \frac{dy}{y}.$$

The integrals $F(s)$ and $G(s)$ converge absolutely and locally uniformly on the whole complex plane, as we saw above. So they represent holomorphic functions, and one obviously has $F(s) = C G(k-s)$. Thus $L(f, s)$ and $L(g, s)$ have been continued to all of $\mathbb{C} \setminus \{0, k\}$ and we have $L(f, s) = C L(g, k-s)$. This finishes the proof of the theorem. $\square$

The result can now be applied to the integral (1.3) representing the function $Z(s)$. In fact, Jacobi's theta function $\theta(z)$ is characterized by the following property.

(1.5) Proposition. *The series*

$$\theta(z) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 z}$$

converges absolutely and uniformly in the domain $\{z \in \mathbb{C} \mid \operatorname{Im}(z) \geq \delta\}$, for every $\delta > 0$. It therefore represents an analytic function on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} \mid \operatorname{Im}(z) > 0\}$, and satisfies the transformation formula

$$\theta(-1/z) = \sqrt{z/i} \theta(z).$$

We will prove this proposition in much greater generality in § 3 (see (3.6)), so we take it for granted here. Observe that if z lies in $\mathbb{H}$ then so does $-1/z$. The square root $\sqrt{z/i}$ is understood to be the holomorphic function

$$h(z) = e^{\frac{1}{2} \log z / i},$$

where $\log$ indicates the principal branch of the logarithm. It is determined uniquely by the conditions

$$h(z)^2 = z/i \quad \text{and} \quad h(iy) = \sqrt{y} > 0 \quad \text{for } y \in \mathbb{R}_+^*.$$

(1.6) Theorem. *The completed zeta function*

$$Z(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

admits an analytic continuation to $\mathbb{C} \setminus \{0, 1\}$, has simple poles at $s = 0$ and $s = 1$ with residues -1 and 1 , respectively, and satisfies the functional equation

$$Z(s) = Z(1-s).$$

Proof: By (1.3), we have

$$Z(2s) = \frac{1}{2} \int_0^\infty (\theta(iy) - 1) y^s \frac{dy}{y},$$

i.e., $Z(2s)$ is the Mellin transform

$$Z(2s) = L(f, s)$$

of the function $f(y) = \frac{1}{2} \theta(iy)$. Since

$$\theta(iy) = 1 + 2e^{-\pi y} \left(1 + \sum_{n=2}^\infty e^{-\pi(n^2-1)y} \right),$$

one has $f(y) = \frac{1}{2} + O(e^{-\pi y})$. From (1.5), we get the transformation formula

$$f(1/y) = \frac{1}{2} \theta(-1/iy) = \frac{1}{2} y^{1/2} \theta(iy) = y^{1/2} f(y).$$

By (1.4), $L(f, s)$ has a holomorphic continuation to $\mathbb{C} \setminus \{0, 1/2\}$ and simple poles at $s = 0, 1/2$ with residues $-1/2$ and $1/2$, respectively, and it satisfies the functional equation

$$L(f, s) = L\left(f, \frac{1}{2} - s\right).$$

Accordingly, $Z(s) = L(f, s/2)$ has a holomorphic continuation to $\mathbb{C} \setminus \{0, 1\}$ and simple poles at $s = 0, 1$ with residues -1 and 1 , respectively, and satisfies the functional equation

$$Z(s) = L\left(f, \frac{s}{2}\right) = L\left(f, \frac{1}{2} - \frac{s}{2}\right) = Z(1-s). \quad \square$$

For the Riemann zeta function itself, the theorem gives the

(1.7) Corollary. *The Riemann zeta function $\zeta(s)$ admits an analytic continuation to $\mathbb{C} \setminus \{1\}$, has a simple pole at $s = 1$ with residue 1 and satisfies the functional equation*

$$\zeta(1-s) = 2(2\pi)^{-s} \Gamma(s) \cos\left(\frac{\pi s}{2}\right) \zeta(s).$$

Proof: $Z(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ has a simple pole at $s = 0$, but so does $\Gamma(s/2)$. Hence $\zeta(s)$ has no pole. At $s = 1$, however, $Z(s)$ has a simple pole, and so does $\zeta(s)$, as $\Gamma(1/2) = \sqrt{\pi}$. The residue comes out to be

$$\text{Res}_{s=1} \zeta(s) = \pi^{1/2} \Gamma(1/2)^{-1} \text{Res}_{s=1} Z(s) = 1.$$

The equation $Z(1-s) = Z(s)$ translates into

$$(*) \quad \zeta(1-s) = \pi^{\frac{1}{2}-s} \frac{\Gamma(\frac{s}{2})}{\Gamma(\frac{1-s}{2})} \zeta(s).$$

Substituting $(1-s)/2$, resp. $s/2$, into the formulae (1.2), (iii), 2) and 3) gives

$$\begin{aligned} \Gamma\left(\frac{s}{2}\right) \Gamma\left(\frac{1+s}{2}\right) &= \frac{2\sqrt{\pi}}{2^s} \Gamma(s), \\ \Gamma\left(\frac{1-s}{2}\right) \Gamma\left(\frac{1+s}{2}\right) &= \frac{\pi}{\cos(\pi s/2)}, \end{aligned}$$

and after taking the quotient,

$$\Gamma\left(\frac{s}{2}\right) / \Gamma\left(\frac{1-s}{2}\right) = \frac{2}{2^s \sqrt{\pi}} \cos \frac{\pi s}{2} \Gamma(s).$$

Inserting this into (*) now yields the functional equation claimed. $\square$

At some point during the first months of studies every mathematics student has the surprise to discover the remarkable formula

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{6} \pi^2.$$

It is only the beginning of a sequence:

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{90} \pi^4, \quad \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{1}{945} \pi^6, \text{ etc.}$$

These are explicit evaluations of the special values of the Riemann zeta function at the points $s = 2k$, $k \in \mathbb{N}$. The phenomenon is explained via the functional equation by the fact that the values of the Riemann zeta function at the *negative* odd integers are given by **Bernoulli numbers**. These arise from the function

$$F(t) = \frac{t e^t}{e^t - 1}$$

and are defined by the series expansion

$$F(t) = \sum_{k=0}^{\infty} B_k \frac{t^k}{k!}.$$

Their relation to the zeta function gives them a special arithmetic significance. The first Bernoulli numbers are

$$B_0 = 1, \quad B_1 = \frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = -\frac{1}{30}, \quad B_5 = 0, \quad B_6 = \frac{1}{42}.$$

In general one has $B_{2\nu+1} = 0$ for $\nu \geq 1$, because $F(-t) = F(t) - t$. In the classical literature, it is usually the function $\frac{t}{e^t - 1}$, which serves for defining the Bernoulli numbers. As $F(t) = \frac{t}{e^t - 1} + t$, this does not change anything except for B_1 , where one finds $-\frac{1}{2}$ instead of $\frac{1}{2}$. But the above definition is more natural and better suited for the further development of the theory. We now prove the remarkable

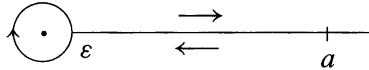
(1.8) Theorem. *For every integer $k > 0$ one has*

$$\zeta(1-k) = -\frac{B_k}{k}.$$

We prepare the proof proper by a function-theoretic lemma. For $\varepsilon > 0$ and $a \in [\varepsilon, \infty]$, we consider the path

$$C_{\varepsilon, a} = (a, \varepsilon] + K_{\varepsilon} + [\varepsilon, a),$$

which first follows the half-line from a to ε , then the circumference $K_{\varepsilon} = \{z \mid |z| = \varepsilon\}$ in the negative direction, and finally the half-line from ε to a :



(1.9) Lemma. *Let U be an open subset of $\mathbb{C}$ that contains the path $C_{\varepsilon, a}$ and also the interior of K_{ε} . Let $G(z)$ be a holomorphic function on $U \setminus \{0\}$ with a pole of order m at 0, and let $G(t)t^{ns-1}$ ($n \in \mathbb{N}$), for $\operatorname{Re}(s) > \frac{m}{n}$, be integrable on $(0, a)$. Then one has*

$$\int_{C_{\varepsilon, a}} G(z)z^{ns-1} dz = (e^{2\pi i ns} - 1) \int_0^a G(t)t^{ns-1} dt.$$

Proof: The integration does not actually take place in the complex plane but on the universal covering of $\mathbb{C}^*$,

$$X = \{(x, \alpha) \in \mathbb{C}^* \times \mathbb{R} \mid \arg x \equiv \alpha \bmod 2\pi\}.$$

z and z^{s-1} are holomorphic functions on X , namely

$$z(x, \alpha) = x, \quad z^{s-1}(x, \alpha) = e^{(s-1)(\log |x| + i\alpha)},$$

and $C_{\varepsilon, a}$ is the path

$$C_{\varepsilon, a} = I_{\varepsilon, a}^- + K_{\varepsilon} + I_{\varepsilon, a}^+$$

where $I_{\varepsilon, a}^- = (a, \varepsilon] \times \{0\}$, $K_{\varepsilon} = \{\varepsilon e^{-it} \mid t \in [0, 2\pi]\}$, $I_{\varepsilon, a}^+ = [\varepsilon, a) \times \{2\pi\}$ in X . We now have

$$\begin{aligned} \int_{I_{\varepsilon, a}^-} G(z)z^{ns-1} dz &= - \int_{\varepsilon}^a G(t)t^{ns-1} dt, \\ \int_{I_{\varepsilon, a}^+} G(z)z^{ns-1} dz &= e^{2\pi i ns} \int_{\varepsilon}^a G(t)t^{ns-1} dt, \end{aligned}$$

$$\begin{aligned}
\int_{K_\varepsilon} G(z) z^{ns-1} dz &= -i \int_0^{2\pi} G(\varepsilon e^{-it}) \varepsilon^{ns-1} e^{-it(ns-1)} \varepsilon e^{-it} dt \\
&= -i \int_0^{2\pi} \varepsilon^{ns} G(\varepsilon e^{-it}) e^{-itns} dt.
\end{aligned}$$

Since $\operatorname{Re}(s) > \frac{m}{n}$, i.e., $\operatorname{Re}(ns - m) > 0$, the last integral $I(\varepsilon)$ tends to zero as $\varepsilon \rightarrow 0$. In fact, one has $\lim_{\varepsilon \rightarrow 0} \varepsilon^{ns} G(\varepsilon e^{-it}) = 0$. This gives

$$\int_{C_{\varepsilon, a}} G(z) z^{ns-1} dz = (e^{2\pi i ns} - 1) \int_\varepsilon^a G(t) t^{ns-1} dt + I(\varepsilon),$$

and since the integral on the left is independent of ε , the lemma follows by passing to the limit as $\varepsilon \rightarrow 0$. $\square$

Proof of (1.8): The function

$$F(z) = \frac{z e^z}{e^z - 1} = \sum_{k=0}^{\infty} B_k \frac{z^k}{k!}$$

is a meromorphic function of the complex variable z , with poles only at $z = 2\pi i v$, $v \in \mathbb{Z}$, $v \neq 0$. B_k/k is the residue of $(k-1)! F(z) z^{-k-1}$ at 0, and the claim reduces to the identity

$$\operatorname{Res}_{z=0} F(z) z^{-k-1} = \frac{1}{2\pi i} \int_{|z|=\varepsilon} F(z) z^{-k-1} dz = -\frac{\zeta(1-k)}{(k-1)!},$$

for $0 < \varepsilon < 2\pi$, where the circle $|z| = \varepsilon$ is taken in the positive orientation. We may replace it with the path $-C_\varepsilon = (-\infty, -\varepsilon] + K_\varepsilon + [-\varepsilon, -\infty)$, which traces the half-line from $-\infty$ to $-\varepsilon$, followed by the circumference $K_\varepsilon = \{z \mid |z| = \varepsilon\}$ in the *positive* direction, from $-\varepsilon$ to $-\varepsilon$, and finally the half-line from $-\varepsilon$ to $-\infty$. In fact, the integrals over $(-\infty, -\varepsilon]$ and $[-\varepsilon, -\infty)$ cancel each other. We now consider on $\mathbb{C}$ the function

$$H(s) = \int_{-C_\varepsilon} F(z) z^{s-1} \frac{dz}{z}.$$

Here the integrals over $(-\infty, -\varepsilon]$ and $[-\varepsilon, -\infty)$ do not cancel each other any longer because the function z^{s-1} is multivalued. The integration takes place on the universal covering $X = \{(x, \alpha) \in \mathbb{C}^* \times \mathbb{R} \mid \arg x \equiv \alpha \bmod 2\pi\}$

of $\mathbb{C}^*$, as in (1.9), and z, z^{s-1} are the holomorphic functions $z(x, \alpha) = x$, $z^{s-1}(x, \alpha) = e^{(s-1)(\log|x|+i\alpha)}$. The integral converges absolutely and locally uniformly for all $s \in \mathbb{C}$. It thus defines a holomorphic function on $\mathbb{C}$, and we find that

$$\operatorname{Res}_{z=0} F(z)z^{-k-1} = \frac{1}{2\pi i} H(1-k).$$

Now substitute $z \mapsto -z$, or more precisely, apply the biholomorphic transformation

$$\varphi : X \longrightarrow X, \quad (x, \alpha) \longmapsto (-x, \alpha - \pi).$$

Since $z \circ \varphi = -z$ and

$$\begin{aligned} (z^{s-1} \circ \varphi)(x, \alpha) &= z^{s-1}(-x, \alpha - \pi) = e^{(s-1)(\log|x|+i\alpha-i\pi)} \\ &= -e^{-i\pi s} z^{s-1}(x, \alpha), \end{aligned}$$

we obtain

$$H(s) = -e^{-i\pi s} \int_{C_\varepsilon} F(-z)z^{s-1} \frac{dz}{z},$$

where the path $C_\varepsilon = \varphi^{-1} \circ (-C_\varepsilon)$ follows the half-line from ∞ to ε , then the circumference K_ε in *negative* direction from ε to ε , and finally the half-line from ε to ∞ . The function

$$G(z) = F(-z)z^{-1} = \frac{e^{-z}}{1 - e^{-z}} = \frac{1}{1 - e^{-z}} - 1 = \sum_{n=1}^{\infty} e^{-nz}$$

has a simple pole at $z = 0$ so that, for $\operatorname{Re}(s) > 1$, (1.9) yields

$$\begin{aligned} H(s) &= -e^{-\pi i s} \int_{C_\varepsilon} G(z)z^{s-1} dz \\ &= -(e^{\pi i s} - e^{-\pi i s}) \int_0^\infty G(t)t^s \frac{dt}{t} = -2i \sin \pi s \int_0^\infty G(t)t^s \frac{dt}{t}. \end{aligned}$$

The integral on the right will now be related to the zeta function. In the gamma integral

$$\Gamma(s) = \int_0^\infty e^{-t} t^s \frac{dt}{t},$$

we substitute $t \mapsto nt$ and get

$$\Gamma(s) \frac{1}{n^s} = \int_0^\infty e^{-nt} t^s \frac{dt}{t}.$$

Summing this over all $n \in \mathbb{N}$ yields

$$\Gamma(s)\zeta(s) = \int_0^\infty G(t)t^s \frac{dt}{t}.$$

The interchange of summation and integration is again justified because

$$\sum_{n=1}^\infty \int_0^\infty |e^{-nt}t^s| \frac{dt}{t} < \infty.$$

From this and (1.2), 2), we get

$$H(s) = -2i \sin \pi s \Gamma(s)\zeta(s) = -\frac{2\pi i}{\Gamma(1-s)} \zeta(s).$$

Since both sides are holomorphic on all of $\mathbb{C}$, this holds for all $s \in \mathbb{C}$. Putting $s = 1 - k$ we obtain, since $\Gamma(k) = (k-1)!$,

$$\operatorname{Res}_{z=0} F(z)z^{-k-1} = \frac{1}{2\pi i} H(1-k) = -\frac{\zeta(1-k)}{(k-1)!}, \quad \text{q.e.d.} \quad \square$$

Applying the functional equation (1.7) for $\zeta(s)$ and observing that $\Gamma(2k) = (2k-1)!$, the preceding theorem gives the following corollary, which goes back to *EULER*.

(1.10) Corollary. *The values of $\zeta(s)$ at the positive even integers $s = 2k$, $k = 1, 2, 3, \dots$, are given by*

$$\zeta(2k) = (-1)^{k-1} \frac{(2\pi)^{2k}}{2(2k)!} B_{2k}.$$

The values $\zeta(2k-1)$, $k > 1$, at the positive odd integers have been elucidated only recently. Surprisingly enough, it is the higher K -groups $K_i(\mathbb{Z})$ from algebraic K -theory, which take the lead. In fact, one has a mysterious canonical isomorphism

$$r : K_{4k-1}(\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{R} \xrightarrow{\cong} \mathbb{R}.$$

The image R_{2k} of a nonzero element in $K_{4k-1}(\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q}$ is called the $2k$ -th **regulator**. It is well-determined up to a rational factor, i.e., it is an element of $\mathbb{R}^*/\mathbb{Q}^*$, and one has

$$\zeta(2k-1) \equiv R_{2k} \bmod \mathbb{Q}^*.$$

This discovery of the Swiss mathematician *ARMAND BOREL* has had a tremendous influence on further arithmetical research, and has opened up deep insights into the arithmetic nature of zeta functions and L -series of the most general kind. These insights are summarized within the comprehensive **Beilinson conjecture** (see [117]). In the meantime, the mathematicians *SPENCER BLOCH* and *KAZUYA KATO* have found a complete description of the special zeta values $\zeta(2k - 1)$ (i.e., not just a description mod $\mathbb{Q}^*$) via a new theory of the *Tamagawa measure*.

The **zeroes** of the Riemann zeta function command special attention. Euler's identity (1.1) shows that one has $\zeta(s) \neq 0$ for $\text{Re}(s) > 1$. The gamma function $\Gamma(s)$ is nowhere 0 and has simple poles at $s = 0, -1, -2, \dots$. The functional equation $Z(s) = Z(1 - s)$, i.e.,

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \pi^{(s-1)/2} \Gamma((1-s)/2) \zeta(1-s),$$

therefore shows that the only zeroes of $\zeta(s)$ in the domain $\text{Re}(s) < 0$ are the poles of $\Gamma(s/2)$, i.e., the arguments $s = -2, -4, -6, \dots$. These are called the *trivial zeroes* of $\zeta(s)$. Other zeroes have to lie in the **critical strip** $0 \leq \text{Re}(s) \leq 1$, since $\zeta(s) \neq 0$ for $\text{Re}(s) > 1$. They are the subject of the famous, still unproven

Riemann Hypothesis: The non-trivial zeroes of $\zeta(s)$ lie on the line $\text{Re}(s) = \frac{1}{2}$.

This conjecture has been verified for 150 million zeroes. It has immediate consequences for the problem of the distribution of prime numbers within all the natural numbers. The distribution function

$$\pi(x) = \#\{p \text{ prime number} \leq x\}$$

may be written, according to *RIEMANN*, as the series

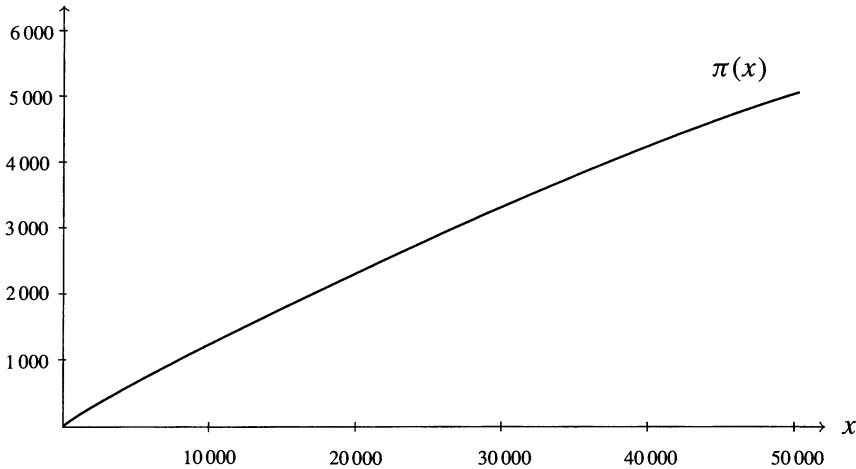
$$\pi(x) = R(x) - \sum_{\rho} R(x^{\rho}),$$

where ρ varies over all the zeroes of $\zeta(s)$, and $R(x)$ is the function

$$R(x) = 1 + \sum_{n=1}^{\infty} \frac{1}{n \zeta(n+1)} \frac{(\log x)^n}{n!}.$$

On a microscopic scale, the function $\pi(x)$ is a step-function with a highly irregular behaviour. But on a large scale it is its astounding smoothness

which poses one of the biggest mysteries in mathematics:



On this matter, we urge the reader to consult the essay [142] by *DON ZAGIER*.

Exercise 1. Let a, b be positive real numbers. Then the Mellin transforms of the functions $f(y)$ and $g(y) = f(ay^b)$ satisfy:

$$L(f, s/b) = ba^{s/b} L(g, s).$$

Exercise 2. The **Bernoulli polynomials** $B_k(x)$ are defined by

$$\frac{t e^{(1+x)t}}{e^t - 1} = F(t) e^{xt} = \sum_{k=0}^{\infty} B_k(x) \frac{t^k}{k!},$$

so that $B_k = B_k(0)$. Show that

$$B_m(x) = \sum_{k=0}^m \binom{m}{k} B_k x^{m-k}.$$

Exercise 3. $B_k(x) - B_k(x-1) = kx^{k-1}$.

Exercise 4. For the power sum

$$s_k(n) = 1^k + 2^k + 3^k + \dots + n^k$$

one has

$$s_k(n) = \frac{1}{k+1} (B_{k+1}(n) - B_{k+1}(0)).$$

Exercise 5. Let $\vartheta(z) = \theta(2z) = \sum_{n \in \mathbb{Z}} e^{2\pi i n^2 z}$. Then for all matrices $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in the group

$$\Gamma_0(4) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}) \mid c \equiv 0 \pmod{4} \right\}$$

one has the formula

$$\vartheta\left(\frac{az+b}{cz+d}\right) = j(\gamma, z)\vartheta(z), \quad z \in \mathbb{H},$$

where

$$j(\gamma, z) = \left(\frac{c}{d}\right) \varepsilon_d^{-1} (cz+d)^{1/2}.$$

The Legendre symbol $\left(\frac{c}{d}\right)$ and the constant ε_d are defined by

$$\left(\frac{c}{d}\right) = \begin{cases} -\left(\frac{c}{|d|}\right), & \text{if } c < 0, d < 0, \\ \left(\frac{c}{|d|}\right), & \text{otherwise,} \end{cases}$$

$$\varepsilon_d = \begin{cases} 1, & \text{if } d \equiv 1 \pmod{4}, \\ i, & \text{if } d \equiv 3 \pmod{4}. \end{cases}$$

Jacobi's theta function $\vartheta(z)$ is thus an example of a **modular form of weight $\frac{1}{2}$** for the group $\Gamma_0(4)$. The representation of L -series as Mellin transforms of modular forms, which we have introduced in the case of the Riemann zeta function, is one of the basic and seminal principles of current number-theoretic research (see [106]).

§ 2. Dirichlet L -series

The most immediate relatives of the Riemann zeta function are the Dirichlet L -series. They are defined as follows. Let m be a natural number. A **Dirichlet character mod m** is by definition a character

$$\chi : (\mathbb{Z}/m\mathbb{Z})^* \longrightarrow S^1 = \{z \in \mathbb{C} \mid |z| = 1\}.$$

It is called **primitive** if it does not arise as the composite

$$(\mathbb{Z}/m\mathbb{Z})^* \longrightarrow (\mathbb{Z}/m'\mathbb{Z})^* \xrightarrow{\chi'} S^1$$

of a Dirichlet character $\chi' \bmod m'$ for any proper divisor $m' \mid m$. In the general case, the gcd of all such divisors is called the **conductor** f of χ . So χ is always induced from a primitive character $\chi' \bmod f$. Given χ , we define the multiplicative function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ by

$$\chi(n) = \begin{cases} \chi(n \bmod m) & \text{for } (n, m) = 1, \\ 0 & \text{for } (n, m) \neq 1. \end{cases}$$

The *trivial character* $\chi^0 \bmod m$, $\chi^0(n) = 1$ for $(n, m) = 1$, $\chi^0(n) = 0$ for $(n, m) \neq 1$, plays a special role. When read $\bmod 1$, we denote it by $\chi = 1$.

It is also called the **principal character**. Considering it in the theory to be developed now has the effect of subsuming here everything we have done in the last section. For a Dirichlet character χ , we form the **Dirichlet L -series**

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

where s is a complex variable with $\operatorname{Re}(s) > 1$. In particular, for the principal character $\chi = \mathbf{1}$, we get back the Riemann zeta function $\zeta(s)$. All the results obtained for this special function in the last section can be transferred to the general L -series $L(\chi, s)$ using the same methods. This is the task of the present section.

(2.1) Proposition. *The series $L(\chi, s)$ converges absolutely and uniformly in the domain $\operatorname{Re}(s) \geq 1 + \delta$, for any $\delta > 0$. It therefore represents an analytic function on the half-plane $\operatorname{Re}(s) > 1$. We have **Euler's identity***

$$L(\chi, s) = \prod_p \frac{1}{1 - \chi(p)p^{-s}}.$$

In view of the multiplicativity of χ and since $|\chi(n)| \leq 1$, the proof is literally the same as for the Riemann zeta function. Since, moreover, we will have to give it again in a more general situation in § 8 below (see (8.1)), we may omit it here.

Like the Riemann zeta function, Dirichlet L -series also admit an analytic continuation to the whole complex plane (with a pole at $s = 1$ in the case $\chi = \chi^0$), and they satisfy a functional equation which relates the argument s to the argument $1 - s$. This particularly important property does in fact hold in a larger class of L -series, the *Hecke L -series*, the treatment of which is an essential goal of this chapter. In order to provide some preliminary orientation, the proof of the functional equation will be given here in the special case of the above L -series $L(\chi, s)$. We recommend it for careful study, also comparing it with the preceding section.

The proof again hinges on an integral representation of the function $L(\chi, s)$ which has the effect of realizing it as the Mellin transform of a theta series. We do, however, have to distinguish now between *even* and *odd* Dirichlet characters $\chi \bmod m$. This phenomenon will become increasingly important when we generalize further. We define the **exponent** $p \in \{0, 1\}$ of χ by

$$\chi(-1) = (-1)^p \chi(1).$$

Then the rule

$$\chi\left(\left(\frac{n}{|n|}\right)\right) = \chi(n) \left(\frac{n}{|n|}\right)^p$$

defines a multiplicative function on the semigroup of all ideals (n) which are relatively prime to m . This function is called a *Größencharakter* mod m . These *Größencharaktere* are capable of substantial generalization and will play the leading part when we consider higher algebraic number fields (see § 7).

We now consider the gamma integral

$$\Gamma(\chi, s) = \Gamma\left(\frac{s+p}{2}\right) = \int_0^\infty e^{-y} y^{(s+p)/2} \frac{dy}{y}.$$

Substituting $y \mapsto \pi n^2 y/m$, we obtain

$$\left(\frac{m}{\pi}\right)^{\frac{s+p}{2}} \Gamma(\chi, s) \frac{1}{n^s} = \int_0^\infty n^p e^{-\pi n^2 y/m} y^{(s+p)/2} \frac{dy}{y}.$$

We multiply this by $\chi(n)$, sum over all $n \in \mathbb{N}$, and get

$$(*) \quad \left(\frac{m}{\pi}\right)^{\frac{s+p}{2}} \Gamma(\chi, s) L(\chi, s) = \int_0^\infty \sum_{n=1}^\infty \chi(n) n^p e^{-\pi n^2 y/m} y^{(s+p)/2} \frac{dy}{y}.$$

Here, swapping the order of summation and integration is again justified, because

$$\begin{aligned} & \sum_{n=1}^\infty \int_0^\infty |\chi(n) n^p e^{-\pi n^2 y/m} y^{(s+p)/2}| \frac{dy}{y} \\ & \leq \left(\frac{m}{\pi}\right)^{(\operatorname{Re}(s)+p)/2} \Gamma\left(\frac{\operatorname{Re}(s)+p}{2}\right) \zeta(\operatorname{Re}(s)) < \infty. \end{aligned}$$

The series under the integral $(*)$,

$$g(y) = \sum_{n=1}^\infty \chi(n) n^p e^{-\pi n^2 y/m},$$

arises from the theta series

$$\theta(\chi, z) = \sum_{n \in \mathbb{Z}} \chi(n) n^p e^{\pi i n^2 z/m},$$

where we adopt the convention that $0^0 = 1$ in case $n = 0$, $p = 0$. Indeed, $\chi(n) n^p = \chi(-n)(-n)^p$ implies that

$$\theta(\chi, z) = \chi(0) + 2 \sum_{n=1}^\infty \chi(n) n^p e^{\pi i n^2 z/m},$$

so that $g(y) = \frac{1}{2}(\theta(\chi, iy) - \chi(0))$ with $\chi(0) = 1$, if χ is the trivial character **1**, and $\chi(0) = 0$ otherwise. When $m = 1$, this is Jacobi's theta function

$$\theta(z) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 z},$$

which is associated with Riemann's zeta function as we saw in §1. We view the factor

$$L_\infty(\chi, s) = \left(\frac{m}{\pi}\right)^{s/2} \Gamma(\chi, s)$$

in (*) as the “Euler factor” at the infinite prime. It joins with the Euler factors $L_p(s) = 1/(1 - \chi(p)p^{-s})$ of the product representation (2.1) of $L(\chi, s)$ to define the **completed L -series** of the character χ :

$$\Lambda(\chi, s) = L_\infty(\chi, s)L(\chi, s), \quad \operatorname{Re}(s) > 1.$$

For this function (*) gives us the

(2.2) Proposition. *The function $\Lambda(\chi, s)$ admits the integral representation*

$$\Lambda(\chi, s) = \frac{c(\chi)}{2} \int_0^\infty (\theta(\chi, iy) - \chi(0)) y^{(s+p)/2} \frac{dy}{y},$$

where $c(\chi) = (\frac{\pi}{m})^{p/2}$.

Let us emphasize the fact that the summation in the L -series is only over the natural numbers, whereas in the theta series we sum over *all* integers. This is why the factor n^p had to be included in order to link the L -series to the theta series.

We want to apply the Mellin principle to the above integral representation. So we have to show that the theta series $\theta(\chi, iy)$ satisfies a transformation formula as assumed in theorem (1.4). To do this, we use the following

(2.3) Proposition. *Let a, b, μ be real numbers, $\mu > 0$. Then the series*

$$\theta_\mu(a, b, z) = \sum_{g \in \mu\mathbb{Z}} e^{\pi i(a+g)^2 z + 2\pi i b g}$$

converges absolutely and uniformly in the domain $\operatorname{Im}(z) \geq \delta$, for every $\delta > 0$, and for $z \in \mathbb{H}$, one has the transformation formula

$$\theta_\mu(a, b, -1/z) = e^{-2\pi i a b} \frac{\sqrt{z/i}}{\mu} \theta_{1/\mu}(-b, a, z).$$

This proposition will be proved in § 3 in much greater generality (see (3.6)), so we take it for granted here. The series $\theta_\mu(a, b, z)$ is locally uniformly convergent in the variables a, b . This will also be shown in § 3. Differentiating p times ($p = 0, 1$) in the variable a , we obtain the function

$$\theta_\mu^p(a, b, z) = \sum_{g \in \mu\mathbb{Z}} (a + g)^p e^{\pi i(a+g)^2 z + 2\pi i b g}.$$

More precisely, we have

$$\frac{d^p}{da^p} \theta_\mu(a, b, z) = (2\pi i)^p z^p \theta_\mu^p(a, b, z)$$

and

$$\frac{d^p}{da^p} e^{-2\pi i ab} \theta_{1/\mu}(-b, a, z) = (2\pi i)^p e^{-2\pi i ab} \theta_{1/\mu}^p(-b, a, z).$$

Applying the differentiation d^p/da^p to the transformation formula (2.3), we get the

(2.4) Corollary. *For $a, b, \mu \in \mathbb{R}$, $\mu > 0$, one has the transformation formula*

$$\theta_\mu^p(a, b, -1/z) = [i^p e^{2\pi i ab} \mu]^{-1} (z/i)^{p+\frac{1}{2}} \theta_{1/\mu}^p(-b, a, z).$$

This corollary gives us the required transformation formula for the theta series $\theta(\chi, a)$, if we introduce the *Gauss sums* which are defined as follows.

(2.5) Definition. *For $n \in \mathbb{Z}$, the Gauss sum $\tau(\chi, n)$ associated to the Dirichlet character $\chi \bmod m$ is defined to be the complex number*

$$\tau(\chi, n) = \sum_{v=0}^{m-1} \chi(v) e^{2\pi i v n / m}.$$

For $n = 1$, we write $\tau(\chi) = \tau(\chi, 1)$.

(2.6) Proposition. *For a primitive Dirichlet character $\chi \bmod m$, one has*

$$\tau(\chi, n) = \bar{\chi}(n) \tau(\chi) \quad \text{and} \quad |\tau(\chi)| = \sqrt{m}.$$

Proof: The first identity in the case $(n, m) = 1$ follows from $\chi(vn) = \chi(n)\chi(v)$. When $d = (n, m) \neq 1$, both sides are zero. Indeed, since χ is primitive, we may in this case choose an $a \equiv 1 \pmod{m/d}$ such that $a \not\equiv 1 \pmod{m}$ and $\chi(a) \neq 1$. Multiplying $\tau(\chi, n)$ by $\chi(a)$ and observing that $e^{2\pi i a n/m} = e^{2\pi i v n/m}$ gives $\chi(a)\tau(\chi, n) = \tau(\chi, n)$, so that $\tau(\chi, n) = 0$. Further, we have

$$\begin{aligned} |\tau(\chi)|^2 &= \tau(\chi)\overline{\tau(\chi)} = \tau(\chi) \sum_{v=0}^{m-1} \bar{\chi}(v) e^{-2\pi i v/m} = \sum_{v=0}^{m-1} \tau(\chi, v) e^{-2\pi i v/m} \\ &= \sum_{v=0}^{m-1} \sum_{\mu=0}^{m-1} \chi(\mu) e^{2\pi i v \mu/m} e^{-2\pi i v/m} = \sum_{\mu=0}^{m-1} \chi(\mu) \sum_{v=0}^{m-1} e^{2\pi i v(\mu-1)/m}. \end{aligned}$$

The last sum equals m for $\mu = 1$. For $\mu \neq 1$, it vanishes because then $\xi = e^{2\pi i(\mu-1)/m}$ is an m -th root of unity $\neq 1$, hence a root of the polynomial

$$\frac{X^m - 1}{X - 1} = X^{m-1} + \cdots + X + 1.$$

Therefore $|\tau(\chi)|^2 = m\chi(1) = m$. □

We now obtain the following result for the theta series $\theta(\chi, z)$.

(2.7) Proposition. *If χ is a primitive Dirichlet character mod m , then we have the transformation formula*

$$\theta(\chi, -1/z) = \frac{\tau(\chi)}{i^p \sqrt{m}} (z/i)^{p+\frac{1}{2}} \theta(\bar{\chi}, z),$$

where $\bar{\chi}$ is the complex conjugate character to χ , i.e., its inverse.

Proof: We split up the series $\theta(\chi, z)$ according to the classes $a \pmod{m}$, $a = 0, 1, \dots, m-1$, and obtain

$$\theta(\chi, z) = \sum_{n \in \mathbb{Z}} \chi(n) n^p e^{\pi i n^2 z/m} = \sum_{a=0}^{m-1} \chi(a) \sum_{g \in m\mathbb{Z}} (a+g)^p e^{\pi i (a+g)^2 z/m},$$

hence

$$\theta(\chi, z) = \sum_{a=0}^{m-1} \chi(a) \theta_m^p(a, 0, z/m).$$

By (2.4), one has

$$\theta_m^p(a, 0, -1/mz) = \frac{1}{i^p m} (mz/i)^{p+\frac{1}{2}} \theta_{1/m}^p(0, a, mz),$$

and this gives

$$\theta_{1/m}^p(0, a, mz) = \sum_{g \in \frac{1}{m}\mathbb{Z}} g^p e^{\pi i g^2 mz + 2\pi i a g} = \frac{1}{m^p} \sum_{n \in \mathbb{Z}} e^{2\pi i a n/m} n^p e^{\pi i n^2 z/m}.$$

Multiplying this by $\chi(a)$, then summing over a , and observing that $\tau(\chi, n) = \bar{\chi}(n)\tau(\chi)$, we find:

$$\begin{aligned} \theta(\chi, -1/z) &= \frac{1}{i^p m} (mz/i)^{p+\frac{1}{2}} \sum_{a=0}^{m-1} \chi(a) \theta_{1/m}^p(0, a, mz) \\ &= \frac{1}{i^p m^{p+1}} (mz/i)^{p+\frac{1}{2}} \sum_{n \in \mathbb{Z}} \left(\sum_{a=0}^{m-1} \chi(a) e^{2\pi i a n/m} \right) n^p e^{\pi i n^2 z/m} \\ &= \frac{1}{i^p \sqrt{m}} (z/i)^{p+\frac{1}{2}} \tau(\chi) \sum_{n \in \mathbb{Z}} \bar{\chi}(n) n^p e^{\pi i n^2 z/m} \\ &= \frac{\tau(\chi)}{i^p \sqrt{m}} (z/i)^{p+\frac{1}{2}} \theta(\bar{\chi}, z). \quad \square \end{aligned}$$

The analytic continuation and functional equation for the function $\Lambda(\chi, s)$ now falls out immediately. We may restrict ourselves to the case of a *primitive* character mod m . For χ is always induced by a primitive character χ' mod f , where f is the conductor of χ (see p. 434), and we clearly have

$$L(\chi, s) = \prod_{\substack{p|m \\ p \nmid f}} (1 - \chi(p)p^{-s}) L(\chi', s),$$

so that the analytic continuation and functional equation of $\Lambda(\chi, s)$ follows from the one for $\Lambda(\chi', s)$. We may further exclude the case $m = 1$ (this is not really necessary, just to make life easy), this being the case of the Riemann zeta function which was settled in § 1. The poles in this case are different.

(2.8) Theorem. *If χ is a nontrivial primitive Dirichlet character, then the completed L -series $\Lambda(\chi, s)$ admits an analytic continuation to the whole complex plane $\mathbb{C}$ and satisfies the functional equation*

$$\Lambda(\chi, s) = W(\chi) \Lambda(\bar{\chi}, 1-s)$$

with the factor $W(\chi) = \frac{\tau(\chi)}{i^p \sqrt{m}}$. This factor has absolute value 1.

Proof: Let $f(y) = \frac{c(\chi)}{2}\theta(\chi, iy)$ and $g(y) = \frac{c(\chi)}{2}\theta(\bar{\chi}, iy)$, $c(\chi) = \left(\frac{\pi}{m}\right)^{p/2}$. We have $\chi(0) = \bar{\chi}(0) = 0$, so that

$$\theta(\chi, iy) = 2 \sum_{n=1}^{\infty} \chi(n) n^p e^{-\pi n^2 y/m},$$

and therefore $f(y) = O(e^{-\pi y/m})$, and likewise $g(y) = O(e^{-\pi y/m})$. By (2.2), one has

$$\Lambda(\chi, s) = \frac{c(\chi)}{2} \int_0^{\infty} \theta(\chi, iy) y^{\frac{s+p}{2}} \frac{dy}{y}.$$

We therefore obtain $\Lambda(\chi, s)$ and similarly also $\Lambda(\bar{\chi}, s)$ as Mellin transforms

$$\Lambda(\chi, s) = L(f, s') \quad \text{and} \quad \Lambda(\bar{\chi}, s) = L(g, s')$$

of the functions $f(y)$ and $g(y)$ at the point $s' = \frac{s+p}{2}$. The transformation formula (2.7) gives

$$f\left(\frac{1}{y}\right) = \frac{c(\chi)}{2} \theta(\chi, -1/iy) = \frac{c(\chi)\tau(\chi)}{2i^p\sqrt{m}} y^{p+\frac{1}{2}} \theta(\bar{\chi}, iy) = \frac{\tau(\chi)}{i^p\sqrt{m}} y^{p+\frac{1}{2}} g(y).$$

Theorem (1.4) therefore tells us that $\Lambda(\chi, s)$ admits an analytic continuation to all of $\mathbb{C}$ and that the equation

$$\begin{aligned} \Lambda(\chi, s) &= L\left(f, \frac{s+p}{2}\right) = W(\chi) L\left(g, p + \frac{1}{2} - \frac{s+p}{2}\right) = W(\chi) L\left(g, \frac{1-s+p}{2}\right) \\ &= W(\chi) \Lambda(\bar{\chi}, 1-s) \end{aligned}$$

holds with $W(\chi) = \frac{\tau(\chi)}{i^p\sqrt{m}}$. By (2.6), we have $|W(\chi)| = 1$. □

The behaviour of the special values at integer arguments of the Riemann zeta function generalizes to the Dirichlet L -series $L(\chi, s)$ if we introduce, for nontrivial primitive Dirichlet characters $\chi \bmod m$, the **generalized Bernoulli numbers** $B_{k, \chi}$ defined by the formula

$$F_{\chi}(t) = \sum_{a=1}^m \chi(a) \frac{t e^{at}}{e^{mt} - 1} = \sum_{k=0}^{\infty} B_{k, \chi} \frac{t^k}{k!}.$$

These are algebraic numbers which lie in the field $\mathbb{Q}(\chi)$ generated by the values of χ . Since

$$F_{\chi}(-t) = \sum_{a=1}^m \chi(-1)\chi(m-a) \frac{t e^{(m-a)t}}{e^{mt} - 1} = \chi(-1) F_{\chi}(t),$$

we find $(-1)^k B_{k, \chi} = \chi(-1) B_{k, \chi}$, so that

$$B_{k, \chi} = 0 \quad \text{for} \quad k \not\equiv p \pmod{2},$$

if $p \in \{0, 1\}$ is defined by $\chi(-1) = (-1)^p \chi(1)$.

(2.9) Theorem. *For any integer $k \geq 1$, one has*

$$L(\chi, 1 - k) = -\frac{B_{k, \chi}}{k}.$$

Proof: The proof is the same as for the Riemann zeta function (see (1.8)): the meromorphic function

$$F_\chi(z) = \sum_{a=1}^m \chi(a) \frac{z e^{az}}{e^{mz} - 1} = \sum_{k=0}^{\infty} B_{k, \chi} \frac{z^k}{k!}$$

has poles at most at $z = \frac{2\pi i \nu}{m}$, $\nu \in \mathbb{Z}$. The claim therefore reduces to showing that

$$(1) \quad -\frac{L(\chi, 1 - k)}{\Gamma(k)} = \text{residue of } F_\chi(z) z^{-k-1} \text{ at } z = 0.$$

Multiplying the equation

$$\Gamma(s) \frac{1}{n^s} = \int_0^\infty e^{-nt} t^s \frac{dt}{t}$$

by $\chi(n)$, and summing over all n , yields

$$(2) \quad \Gamma(s) L(\chi, s) = \int_0^\infty G_\chi(t) t^s \frac{dt}{t}$$

with the function

$$(3) \quad G_\chi(z) = \sum_{n=1}^{\infty} \chi(n) e^{-nz} = \sum_{a=1}^m \chi(a) \frac{e^{-az}}{1 - e^{-mz}} = F_\chi(-z) z^{-1}.$$

From the equations (2) and (3) one deduces equation (1) in exactly the same manner as in (1.8). $\square$

The theorem immediately gives that

$$L(\chi, 1 - k) = 0 \quad \text{for } k \not\equiv p \pmod{2},$$

$p \in \{0, 1\}$, $\chi(-1) = (-1)^p \chi(1)$, provided that χ is not the principal character **1**. From the functional equation (2.8) and the fact that $L(\chi, k) \neq 0$, we deduce for $k \geq 1$ that

$$L(\chi, 1 - k) = -\frac{B_{k, \chi}}{k} \neq 0 \quad \text{for } k \equiv p \pmod{2}.$$

The functional equation also gives the

(2.10) Corollary. For $k \equiv p \pmod{2}$, $k \geq 1$, one has

$$L(\chi, k) = (-1)^{1+(k-p)/2} \frac{\tau(\chi)}{2i^p} \left(\frac{2\pi}{m} \right)^k \frac{B_{k, \bar{\chi}}}{k!}.$$

For the values $L(\chi, k)$ at positive integer arguments $k \not\equiv p \pmod{2}$, similar remarks apply as the ones we made in § 1 about the Riemann zeta function at the points $2k$. Up to unknown algebraic factors, these values are certain “regulators” defined via canonical maps from higher K -groups into Minkowski space. A detailed treatment of this deep result of the Russian mathematician A.A. BEILINSON can be found in [110].

Exercise 1. Let $F_\chi(t, x) = \sum_{a=1}^m \chi(a) \frac{t e^{(a+x)t}}{e^{mt} - 1}$. The **Bernoulli polynomials** $B_{k, \chi}(x)$ associated to the Dirichlet character χ are defined by

$$F_\chi(t, x) = \sum_{k=0}^{\infty} B_{k, \chi}(x) \frac{t^k}{k!}.$$

Thus $B_{k, \chi}(0) = B_{k, \chi}$. Show that

$$B_{k, \chi}(x) = \sum_{i=0}^k \binom{k}{i} B_{i, \chi} x^{k-i}.$$

Exercise 2. $B_{k, \chi}(x) - B_{k, \chi}(x - m) = k \sum_{a=1}^m \chi(a)(a + x - m)^{k-1}$, $k \geq 0$.

Exercise 3. For the numbers $S_{k, \chi}(v) = \sum_{a=1}^v \chi(a)a^k$, $k \geq 0$, one has

$$S_{k, \chi}(vm) = \frac{1}{k+1} (B_{k+1, \chi}(vm) - B_{k+1, \chi}(0)).$$

Exercise 4. For a primitive odd character χ , one has

$$\sum_{a=1}^m \chi(a)a \neq 0.$$

§ 3. Theta Series

Riemann’s zeta function and Dirichlet’s L -series are attached to the field $\mathbb{Q}$. They have analogues for any algebraic number field K , and the results obtained in § 1 and 2 extend to these generalizations in the same way, with the same methods. In particular, the Mellin principle applies again, which allows us to view the L -series in question as integrals over theta series. But now higher dimensional theta series are required which live on a higher dimensional analogue of the upper half-plane $\mathbb{H}$. *A priori* they do not have any relation with number fields and deserve to be introduced in complete generality.

The familiar objects $\mathbb{C}$, $\mathbb{R}$, $\mathbb{R}_+^*$, $\mathbb{H}$, $|\cdot|$, $\log$, find their higher dimensional analogues as follows. Let X be a finite $G(\mathbb{C}|\mathbb{R})$ -set, i.e., a finite set with an involution $\tau \mapsto \bar{\tau}$ ($\tau \in X$), and let $n = \#X$. We consider the n -dimensional $\mathbb{C}$ -algebra

$$\mathbf{C} = \prod_{\tau \in X} \mathbb{C}$$

of all tuples $z = (z_\tau)_{\tau \in X}$, $z_\tau \in \mathbb{C}$, with componentwise addition and multiplication. If $z = (z_\tau) \in \mathbf{C}$, then the element $\bar{z} \in \mathbf{C}$ is defined to have the following components:

$$(\bar{z})_\tau = \bar{z}_{\bar{\tau}}.$$

We call the involution $z \mapsto \bar{z}$ the **conjugation** on $\mathbf{C}$. In addition, we have the involutions $z \mapsto z^*$ and $z \mapsto {}^*z$ given by

$$z_\tau^* = z_{\bar{\tau}}, \quad \text{resp.} \quad {}^*z_\tau = \bar{z}_{\bar{\tau}}.$$

One clearly has $\bar{z} = {}^*z^*$. The set

$$\mathbf{R} = \left[\prod_{\tau} \mathbb{C} \right]^+ = \{ z \in \mathbf{C} \mid z = \bar{z} \}$$

forms an n -dimensional commutative $\mathbb{R}$ -algebra, and $\mathbf{C} = \mathbf{R} \otimes_{\mathbb{R}} \mathbb{C}$.

If K is a number field of degree n and $X = \text{Hom}(K, \mathbb{C})$, then $\mathbf{R}$ is the **Minkowski space** $K_{\mathbb{R}} (\cong K \otimes_{\mathbb{Q}} \mathbb{R})$ which was introduced in chapter I, §5. The number-theoretic applications will occur there. But for the moment we leave all number-theoretic aspects aside.

For the additive, resp. multiplicative, group $\mathbf{C}$, resp. $\mathbf{C}^*$, we have the homomorphism

$$\text{Tr} : \mathbf{C} \rightarrow \mathbb{C}, \quad \text{Tr}(z) = \sum_{\tau} z_{\tau}, \quad \text{resp.}$$

$$N : \mathbf{C}^* \rightarrow \mathbb{C}^*, \quad N(z) = \prod_{\tau} z_{\tau}.$$

Here $\text{Tr}(z)$, resp. $N(z)$, denotes the trace, resp. the determinant, of the endomorphism $\mathbf{C} \rightarrow \mathbf{C}$, $x \mapsto zx$. Furthermore we have on $\mathbf{C}$ the hermitian scalar product

$$\langle x, y \rangle = \sum_{\tau} x_{\tau} \bar{y}_{\tau} = \text{Tr}(x {}^*y).$$

It is invariant under conjugation, $\overline{\langle x, y \rangle} = \langle \bar{x}, \bar{y} \rangle$, and restricting it yields a scalar product $\langle \cdot, \cdot \rangle$, i.e., a euclidean metric, on the $\mathbb{R}$ -vector space $\mathbf{R}$. If $z \in \mathbf{C}$, then *z is the adjoint element with respect to $\langle \cdot, \cdot \rangle$, i.e.,

$$\langle xz, y \rangle = \langle x, {}^*zy \rangle.$$

In $\mathbf{R}$, we consider the subspace

$$\mathbf{R}_{\pm} = \{x \in \mathbf{R} \mid x = x^*\} = \left[\prod_{\tau} \mathbb{R} \right]^+.$$

Thus we find for the components of $x = (x_{\tau}) \in \mathbf{R}_{\pm}$ that $x_{\bar{\tau}} = x_{\tau} \in \mathbb{R}$. If $\delta \in \mathbb{R}$, we simply write $x > \delta$ to signify that $x_{\tau} > \delta$ for all τ . The multiplicative group

$$\mathbf{R}_{+}^* = \{x \in \mathbf{R}_{\pm} \mid x > 0\} = \left[\prod_{\tau} \mathbb{R}_{+}^* \right]^+$$

will play a particularly important part. It consists of the tuples $x = (x_{\tau})$ of positive real numbers x_{τ} such that $x_{\bar{\tau}} = x_{\tau}$, and it occurs in the two homomorphisms

$$\begin{aligned} |\cdot| : \mathbf{R}^* &\longrightarrow \mathbf{R}_{+}^*, & x = (x_{\tau}) &\longmapsto |x| = (|x_{\tau}|), \\ \log : \mathbf{R}_{+}^* &\xrightarrow{\sim} \mathbf{R}_{\pm}, & x = (x_{\tau}) &\longmapsto \log x = (\log x_{\tau}). \end{aligned}$$

We finally define the **upper half-space** associated to the $G(\mathbb{C}|\mathbb{R})$ -set X by

$$\mathbf{H} = \mathbf{R}_{\pm} + i\mathbf{R}_{+}^*.$$

Putting $\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z})$, $\operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z})$, we may also write

$$\mathbf{H} = \{z \in \mathbb{C} \mid z = z^*, \operatorname{Im}(z) > 0\}.$$

If z lies in $\mathbf{H}$, then so does $-1/z$, because $z\bar{z} \in \mathbf{R}_{+}^*$, and $\operatorname{Im}(z) > 0$ implies $\operatorname{Im}(-1/z) > 0$, since $z\bar{z} \operatorname{Im}(-1/z) = -\operatorname{Im}(z^{-1}z\bar{z}) = \operatorname{Im}(z) > 0$.

For two tuples $z = (z_{\tau})$, $p = (p_{\tau}) \in \mathbb{C}$, the power

$$z^p = (z_{\tau}^{p_{\tau}}) \in \mathbb{C}$$

is well-defined by

$$z_{\tau}^{p_{\tau}} = e^{p_{\tau} \log z_{\tau}},$$

if we agree to take the principal branch of the logarithm and assume that the z_{τ} move only in the plane cut along the negative real axis. The table

$$\mathbb{H} \subseteq \mathbb{C} \supseteq \mathbb{R} = \mathbb{R} \supseteq \mathbb{R}_{+}^*, \quad |\cdot| : \mathbb{R}^* \rightarrow \mathbb{R}_{+}^*, \quad \log : \mathbb{R}_{+}^* \xrightarrow{\sim} \mathbb{R},$$

$$\mathbf{H} \subseteq \mathbb{C} \supseteq \mathbf{R} \supseteq \mathbf{R}_{\pm} \supseteq \mathbf{R}_{+}^*, \quad |\cdot| : \mathbf{R}^* \rightarrow \mathbf{R}_{+}^*, \quad \log : \mathbf{R}_{+}^* \xrightarrow{\sim} \mathbf{R}_{\pm},$$

shows the analogy of the notions introduced with the familiar ones in the case $n = 1$. We recommend that the reader memorize them well, for they will be used constantly in what follows without special cross-reference. This also includes the notation

$$\bar{z}, z^*, {}^*z, \operatorname{Tr}, N, \langle \cdot, \cdot \rangle, x > \delta, z^p.$$

The functional equations we are envisaging originate in a general formula from functional analysis, the *Poisson summation formula*. It will be proved first. A **Schwartz function** (or *rapidly decreasing function*) on a euclidean vector space $\mathbf{R}$ is by definition a C^∞ -function $f : \mathbf{R} \rightarrow \mathbb{C}$ which tends to zero as $x \rightarrow \infty$, even if multiplied by an arbitrary power $\|x\|^m$, $m \geq 0$, and which shares this behaviour with all its derivatives. For every Schwartz function f , one forms the **Fourier transform**

$$\widehat{f}(y) = \int_{\mathbf{R}} f(x) e^{-2\pi i \langle x, y \rangle} dx,$$

where dx is the Haar measure on $\mathbf{R}$ associated to $\langle \cdot, \cdot \rangle$ which ascribes the volume 1 to the cube spanned by an orthonormal basis, i.e., it is the Haar measure which is selfdual with respect to $\langle \cdot, \cdot \rangle$. The improper integral converges absolutely and uniformly and gives again a Schwartz function $\widehat{f}$. This is easily proved by elementary analytical techniques; we refer also to [98], chap. XIV. The prototype of a Schwartz function is the function

$$h(x) = e^{-\pi \langle x, x \rangle}.$$

All functional equations we are going to prove depend, in the final analysis, on the special property of this function of being its own Fourier transform:

(3.1) Proposition. (i) *The function $h(x) = e^{-\pi \langle x, x \rangle}$ is its own Fourier transform.*

(ii) *If f is an arbitrary Schwartz function and A is a linear transformation of $\mathbf{R}$, then the function $f_A(x) = f(Ax)$ has Fourier transform*

$$\widehat{f}_A(y) = \frac{1}{|\det A|} \widehat{f}({}^t A^{-1} y),$$

where ${}^t A$ is the adjoint transformation of A .

Proof: (i) We identify the euclidean vector space $\mathbf{R}$ with $\mathbb{R}^n$ via some isometry. Then the Haar measure dx turns into the Lebesgue measure $dx_1 \cdots dx_n$. Since $h(x) = \prod_{i=1}^n e^{-\pi x_i^2}$, we have $\widehat{h} = \prod_{i=1}^n (e^{-\pi x_i^2})^\wedge$, so we may assume $n = 1$. Differentiating

$$\widehat{h}(y) = \int_{-\infty}^{\infty} h(x) e^{-2\pi i xy} dx$$

in y under the integral, we find by partial integration that

$$\frac{d}{dy} \widehat{h}(y) = -2\pi i \int_{-\infty}^{\infty} x h(x) e^{-2\pi i x y} dx = -2\pi y \widehat{h}(y).$$

This implies that $\widehat{h}(y) = C e^{-\pi y^2}$ for some constant C . Putting $y = 0$ yields $C = 1$, since it is well-known that $\int e^{-\pi x^2} dx = 1$.

(ii) Substituting $x \mapsto Ax$ gives the Fourier transform of $f_A(x)$ as:

$$\begin{aligned} \widehat{f}_A(y) &= \int f(Ax) e^{-2\pi i \langle x, y \rangle} dx = \int f(x) e^{-2\pi i \langle A^{-1}x, y \rangle} |\det A|^{-1} dx \\ &= \frac{1}{|\det A|} \int f(x) e^{-2\pi i \langle x, {}^t A^{-1}y \rangle} dx = \frac{1}{|\det A|} \widehat{f}({}^t A^{-1}y). \quad \square \end{aligned}$$

From the proposition ensues the following result, which will be crucial for the sequel.

(3.2) Poisson Summation Formula. *Let Γ be a complete lattice in $\mathbf{R}$ and let*

$$\Gamma' = \{g' \in \mathbf{R} \mid \langle g, g' \rangle \in \mathbb{Z} \text{ for all } g \in \Gamma\}$$

be the lattice dual to Γ . Then for any Schwartz function f , one has:

$$\sum_{g \in \Gamma} f(g) = \frac{1}{\text{vol}(\Gamma)} \sum_{g' \in \Gamma'} \widehat{f}(g'),$$

where $\text{vol}(\Gamma)$ is the volume of a fundamental mesh of Γ .

Proof: We identify as before $\mathbf{R}$ with the euclidean vector space $\mathbb{R}^n$ via some isometry. This turns the measure dx into the Lebesgue measure $dx_1 \cdots dx_n$. Let A be an invertible $n \times n$ -matrix which maps the lattice $\mathbb{Z}^n$ onto Γ . Hence $\Gamma = A\mathbb{Z}^n$ and $\text{vol}(\Gamma) = |\det A|$. The lattice $\mathbb{Z}^n$ is dual to itself, and we get $\Gamma' = A^*\mathbb{Z}^n$ where $A^* = {}^t A^{-1}$, as

$$\begin{aligned} g' \in \Gamma' &\iff {}^t(A\mathbf{n})g' = {}^t\mathbf{n} {}^tAg' \in \mathbb{Z} \quad \text{for all } \mathbf{n} \in \mathbb{Z}^n \\ &\iff {}^tAg' \in \mathbb{Z}^n \iff g' \in {}^tA^{-1}\mathbb{Z}^n. \end{aligned}$$

Substituting the equations

$$\Gamma = A\mathbb{Z}^n, \quad \Gamma' = A^*\mathbb{Z}^n, \quad f_A(x) = f(Ax), \quad \widehat{f}_A(y) = \frac{1}{\text{vol}(\Gamma)} \widehat{f}(A^*y)$$

into the identity we want to prove, gives

$$\sum_{\mathbf{n} \in \mathbb{Z}^n} f_A(\mathbf{n}) = \sum_{\mathbf{n} \in \mathbb{Z}^n} \widehat{f}_A(\mathbf{n}).$$

In order to prove this, let us write f instead of f_A and take the series

$$g(x) = \sum_{\mathbf{k} \in \mathbb{Z}^n} f(x + \mathbf{k}).$$

It converges absolutely and locally uniformly. For since f is a Schwartz function, we have, if x varies in a compact domain,

$$|f(x + \mathbf{k})| \cdot \|\mathbf{k}\|^{n+1} \leq C$$

for almost all $\mathbf{k} \in \mathbb{Z}^n$. Hence $g(x)$ is majorized by a constant multiple of the convergent series $\sum_{\mathbf{k} \neq 0} \frac{1}{\|\mathbf{k}\|^{n+1}}$. This argument works just as well for all partial derivatives of f . So $g(x)$ is a C^∞ -function. It is clearly periodic,

$$g(x + \mathbf{n}) = g(x) \quad \text{for all } \mathbf{n} \in \mathbb{Z}^n,$$

and therefore admits a Fourier expansion

$$g(x) = \sum_{\mathbf{n} \in \mathbb{Z}^n} a_{\mathbf{n}} e^{2\pi i' \mathbf{n}x},$$

whose Fourier coefficients are given by the well-known formula

$$a_{\mathbf{n}} = \int_0^1 \cdots \int_0^1 g(x) e^{-2\pi i' \mathbf{n}x} dx_1 \cdots dx_n.$$

Swapping summation and integration gives

$$\begin{aligned} a_{\mathbf{n}} &= \int_0^1 \cdots \int_0^1 g(x) e^{-2\pi i' \mathbf{n}x} dx = \sum_{\mathbf{k} \in \mathbb{Z}^n} \int_0^1 \cdots \int_0^1 f(x + \mathbf{k}) e^{-2\pi i' \mathbf{n}x} dx \\ &= \widehat{f}(\mathbf{n}). \end{aligned}$$

It follows that

$$\sum_{\mathbf{n} \in \mathbb{Z}^n} f(\mathbf{n}) = g(0) = \sum_{\mathbf{n} \in \mathbb{Z}^n} a_{\mathbf{n}} = \sum_{\mathbf{n} \in \mathbb{Z}^n} \widehat{f}(\mathbf{n}), \quad \text{q.e.d.} \quad \square$$

We apply the Poisson summation formula to the functions

$$f_p(a, b, x) = N((x + a)^p) e^{-\pi(a+x, a+x) + 2\pi i(b, x)}$$

with the parameters $a, b \in \mathbf{R}$ and a tuple $p = (p_\tau)$ of nonnegative integers, such that $p_\tau \in \{0, 1\}$ if $\tau = \bar{\tau}$, and $p_\tau p_{\bar{\tau}} = 0$ if $\tau \neq \bar{\tau}$. Such an element $p \in \prod_\tau \mathbb{Z}$ will henceforth be called **admissible**.

(3.3) Proposition. *The function $f(x) = f_p(a, b, x)$ is a Schwartz function on $\mathbf{R}$. Its Fourier transform is*

$$\widehat{f}(y) = [i^{\text{Tr}(p)} e^{2\pi i \langle a, b \rangle}]^{-1} f_p(-b, a, y).$$

Proof: It is clear that $f_p(a, b, x)$ is a Schwartz function, because

$$|f_p(a, b, x)| = |P(x)| e^{-\pi \langle a+x, a+x \rangle},$$

for some polynomial $P(x)$.

Let $p = 0$. By (3.1), the function $h(x) = e^{-\pi \langle x, x \rangle}$ equals its own Fourier transform and one has

$$f(x) = f_0(a, b, x) = h(a+x) e^{2\pi i \langle b, x \rangle}.$$

We therefore obtain

$$\begin{aligned} \widehat{f}(y) &= \int_{\mathbf{R}} h(a+x) e^{2\pi i \langle b, x \rangle} e^{-2\pi i \langle x, y \rangle} dx \\ &= \int_{\mathbf{R}} h(x) e^{-2\pi i \langle y-b, x-a \rangle} dx \\ &= e^{2\pi i \langle y-b, a \rangle} \widehat{h}(y-b) \\ &= e^{-2\pi i \langle a, b \rangle} e^{-\pi \langle y-b, y-b \rangle + 2\pi i \langle y, a \rangle} \\ &= e^{-2\pi i \langle a, b \rangle} f_0(-b, a, y). \end{aligned}$$

For an arbitrary admissible p , we get the formula by differentiating p times the identity

$$(*) \quad \widehat{f}_0(a, b, y) = e^{-2\pi i \langle a, b \rangle} f_0(-b, a, y)$$

in the variable a . Now the functions are neither analytic in the individual components a_τ of a , nor are these independent of each other, when there exists a couple $\tau \neq \bar{\tau}$. We therefore proceed as follows. Let ρ vary over the elements of X such that $\rho = \bar{\rho}$, and let σ run through a system of representatives of the conjugation classes $\{\tau, \bar{\tau}\}$ such that $\tau \neq \bar{\tau}$. Since $p_\tau p_{\bar{\tau}} = 0$, we may choose σ in such a way that $p_{\bar{\sigma}} = 0$. Then one has

$$\langle a+x, a+x \rangle = \sum_{\rho} (a_{\rho} + x_{\rho})^2 + 2 \sum_{\sigma} (a_{\sigma} + x_{\sigma})(a_{\bar{\sigma}} + x_{\bar{\sigma}}).$$

We now differentiate p_{ρ} times both sides of $(*)$ in the real variable a_{ρ} , for all ρ , and apply p_{σ} times the differential operator

$$\frac{\partial}{\partial a_{\bar{\sigma}}} = \frac{1}{2} \left(\frac{\partial}{\partial \xi_{\bar{\sigma}}} - i \frac{\partial}{\partial \eta_{\bar{\sigma}}} \right),$$

for all σ . Here we consider $a_{\bar{\sigma}} = \xi_{\bar{\sigma}} + i\eta_{\bar{\sigma}}$ as a function in the real variables $\xi_{\bar{\sigma}}, \eta_{\bar{\sigma}}$ ("Wirtinger calculus"). On the left-hand side

$$\widehat{f}_0(a, b, y) = \int e^{-\pi \langle a+x, a+x \rangle + 2\pi i \langle b, x \rangle} e^{-2\pi i \langle x, y \rangle} dx,$$

we may differentiate under the integral. Then, observing that $p_{\bar{\sigma}} = 0$ and $\frac{\partial}{\partial a_{\bar{\sigma}}}((a_{\sigma} + x_{\sigma})(a_{\bar{\sigma}} + x_{\bar{\sigma}})) = (a_{\sigma} + x_{\sigma})$, we obtain

$$\begin{aligned} & \int \prod_{\rho} (-2\pi(a_{\rho} + x_{\rho}))^{p_{\rho}} \\ & \quad \cdot \prod_{\sigma} (-2\pi(a_{\sigma} + x_{\sigma}))^{p_{\sigma}} e^{-\pi \langle a+x, a+x \rangle + 2\pi i \langle b, x \rangle - 2\pi i \langle x, y \rangle} dx \\ &= N((-2\pi)^p) \int N((a+x)^p) e^{-\pi \langle a+x, a+x \rangle + 2\pi i \langle b, x \rangle} e^{-2\pi i \langle x, y \rangle} dx \\ &= N((-2\pi)^p) \widehat{f}_p(a, b, y). \end{aligned}$$

The right-hand side of (*),

$$e^{-2\pi i \langle a, b \rangle - \pi \langle -b+y, -b+y \rangle + 2\pi i \langle a, y \rangle} = e^{2\pi i \langle a, -b+y \rangle - \pi \langle -b+y, -b+y \rangle},$$

in view of

$$\langle a, -b+y \rangle = \sum_{\rho} a_{\rho}(-b_{\rho} + y_{\rho}) + \sum_{\sigma} (a_{\sigma}(-b_{\bar{\sigma}} + y_{\bar{\sigma}}) + a_{\bar{\sigma}}(-b_{\sigma} + y_{\sigma})),$$

and as $p_{\bar{\sigma}} = 0$, becomes accordingly

$$\begin{aligned} & N((2\pi i)^p) N((-b+y)^p) e^{-2\pi i \langle a, b \rangle} f_0(-b, a, y) \\ &= N((2\pi i)^p) e^{-2\pi i \langle a, b \rangle} f_p(-b, a, y). \end{aligned}$$

Hence

$$\widehat{f}_p(a, b, y) = N(i^{-p}) e^{-2\pi i \langle a, b \rangle} f_p(-b, a, y). \quad \square$$

We now create our general theta series on the upper half-space

$$\mathbf{H} = \{z \in \mathbf{C} \mid z = z^*, \operatorname{Im}(z) > 0\} = \mathbf{R}_{\pm} + i\mathbf{R}_{+}^*.$$

(3.4) Definition. For every complete lattice Γ of $\mathbf{R}$, we define the **theta series**

$$\theta_{\Gamma}(z) = \sum_{g \in \Gamma} e^{\pi i \langle g z, g \rangle}, \quad z \in \mathbf{H}.$$

More generally, for $a, b \in \mathbf{R}$ and any admissible $p \in \prod_{\tau} \mathbb{Z}$, we put

$$\theta_{\Gamma}^p(a, b, z) = \sum_{g \in \Gamma} N((a+g)^p) e^{\pi i \langle (a+g)z, a+g \rangle + 2\pi i \langle b, g \rangle}.$$

(3.5) Proposition. *The series $\theta_F^p(a, b, z)$ converges absolutely and uniformly on every compact subset of $\mathbf{R} \times \mathbf{R} \times \mathbf{H}$.*

Proof: Let $\delta \in \mathbb{R}$, $\delta > 0$. For all $z \in \mathbf{H}$ such that $\text{Im}(z) \geq \delta$, we find

$$|N((a+g)^p) e^{\pi i \langle (a+g)z, a+g \rangle + 2\pi i \langle b, g \rangle}| \leq |N((a+g)^p)| e^{-\pi \delta \langle a+g, a+g \rangle}.$$

Let

$$f_g(a) = N((a+g)^p) e^{-\pi \delta \langle a+g, a+g \rangle} \quad (a \in \mathbf{R}, g \in \Gamma).$$

For $\mathbf{K} \subseteq \mathbf{R}$ compact, put $|f_g|_{\mathbf{K}} = \sup_{x \in \mathbf{K}} |f_g(x)|$. We have to show that

$$\sum_{g \in \Gamma} |f_g|_{\mathbf{K}} < \infty.$$

Let $g_1, \dots, g_n$ be a $\mathbb{Z}$ -basis of Γ , and for $g = \sum_{i=1}^n m_i g_i \in \Gamma$, let $\mu_g = \max_i |m_i|$. Furthermore, define $\|x\| = \sqrt{\langle x, x \rangle}$. If $\|g\| \geq 4 \sup_{x \in \mathbf{K}} \|x\|$, then for all $a \in \mathbf{K}$:

$$\begin{aligned} \langle a+g, a+g \rangle &\geq (\|a\| - \|g\|)^2 \geq \|g\|^2 - 2\|a\| \cdot \|g\| \\ &\geq \frac{1}{2} \|g\|^2 \geq \frac{1}{2} \varepsilon \sum_{i=1}^n m_i^2 \geq \frac{1}{2} \varepsilon \mu_g^2, \end{aligned}$$

where $\varepsilon = \inf_{\sum y_i^2 = 1} \sum_{i,j=1}^n \langle g_i, g_j \rangle y_i y_j$ is the smallest eigenvalue of the matrix $(\langle g_i, g_j \rangle)$.

$N((a + \sum m_i g_i)^p)$ is a polynomial of degree q in the m_i , ($q = \text{Tr}(p)$), the coefficients of which are continuous functions of a . It follows that

$$|N((a+g)^p)| \leq \mu_g^{q+1} \quad \text{for all } a \in \mathbf{K},$$

provided μ_g is sufficiently big. One therefore finds a subset $\Gamma' \subseteq \Gamma$ with finite complement such that

$$\sum_{g \in \Gamma'} |f_g|_{\mathbf{K}} \leq \sum_{\mu=0}^{\infty} P(\mu) \mu^{q+1} e^{-\frac{\pi}{2} \delta \varepsilon \mu^2},$$

where $P(\mu) = \#\{\mathbf{m} \in \mathbb{Z}^n \mid \max_i |m_i| = \mu\} = (2\mu + 1)^n - (2\mu - 1)^n$. The series on the right is clearly convergent. $\square$

From the Poisson summation formula we now get the general

(3.6) Theta Transformation Formula. *One has*

$$\theta_F^p(a, b, -1/z) = [i^{\text{Tr}(p)} e^{2\pi i \langle a, b \rangle} \text{vol}(\Gamma)]^{-1} N((z/i)^{p+\frac{1}{2}}) \theta_{F'}^p(-b, a, z).$$

In particular, one has for the function $\theta_\Gamma(z) = \theta_\Gamma^0(0, 0, z)$:

$$\theta_\Gamma(-1/z) = \frac{\sqrt{N(z/i)}}{\text{vol}(\Gamma)} \theta_{\Gamma'}(z).$$

Proof: Both sides of the transformation formula are holomorphic in z by (3.5). Therefore it suffices to check the identity for $z = iy$, with $y \in \mathbf{R}_+^*$. Put $t = y^{-1/2}$, so that

$$z = i \frac{1}{t^2} \quad \text{and} \quad -1/z = it^2.$$

Observing that $t = t^* = {}^*t$, so that $\langle \xi t, \eta \rangle = \langle \xi, {}^*t\eta \rangle = \langle \xi, t\eta \rangle$, we obtain

$$\theta_F^p(a, b, -1/z) = N(t^{-p}) \sum_{g \in \Gamma} N((ta + tg)^p) e^{-\pi \langle ta + tg, ta + tg \rangle + 2\pi i \langle t^{-1}b, tg \rangle}.$$

Let $\alpha = ta$, $\beta = t^{-1}b$. We consider the function

$$f_p(\alpha, \beta, x) = N((\alpha + x)^p) e^{-\pi \langle \alpha + x, \alpha + x \rangle + 2\pi i \langle \beta, x \rangle},$$

and put

$$\varphi_t(\alpha, \beta, x) = f_p(\alpha, \beta, tx).$$

This gives

$$(1) \quad \theta_F^p(a, b, -1/z) = N(t^{-p}) \sum_{g \in \Gamma} \varphi_t(\alpha, \beta, g)$$

and similarly $z = i \frac{1}{t^2}$ gives that

$$(2) \quad \theta_{F'}^p(-b, a, z) = N(t^p) \sum_{g' \in \Gamma'} \varphi_{t^{-1}}(-\beta, \alpha, g').$$

Now apply the Poisson summation formula

$$(3) \quad \sum_{g \in \Gamma} f(g) = \frac{1}{\text{vol}(\Gamma)} \sum_{g' \in \Gamma'} \widehat{f}(g')$$

to the function

$$f(x) = \varphi_t(\alpha, \beta, x) = f_p(\alpha, \beta, tx).$$

Its Fourier transform is computed as follows. Let $h(x) = f_p(\alpha, \beta, x)$, so that $f(x) = h(tx) = h_t(x)$. The transformation $A : x \mapsto tx$ of $\mathbf{R}$ is self-adjoint and has determinant $N(t)$. Thus (3.1), (ii), gives

$$\widehat{f}(y) = \frac{1}{N(t)} \widehat{h}(t^{-1}y).$$

The Fourier transform $\widehat{h}$ has been computed in (3.3). This yields

$$\begin{aligned}\widehat{f}(y) &= [N(i^p)N(t)e^{2\pi i\langle a, b \rangle}]^{-1} f_p(-\beta, \alpha, t^{-1}y) \\ &= [N(i^p)N(t)e^{2\pi i\langle a, b \rangle}]^{-1} \varphi_{t^{-1}}(-\beta, \alpha, y).\end{aligned}$$

Substituting this into (3) and multiplying by $N(t^{-p})$ gives, by (1) and (2):

$$\theta_F^p(a, b, -1/z) = [N(i^p t^{2p+1})e^{2\pi i\langle a, b \rangle} \text{vol}(\Gamma)]^{-1} \theta_{F'}^p(-b, a, z).$$

Since $t = (z/i)^{-1/2}$, i.e., $(t^{2p+1})^{-1} = (z/i)^{p+\frac{1}{2}}$, this is indeed the transformation formula sought. $\square$

For $n = 1$, we obtain proposition (2.3), which at the time was used without proof for proving the functional equation of the Dirichlet L -series (and Riemann's zeta function).

§ 4. The Higher-dimensional Gamma Function

The passage from theta series to L -series in § 1 and § 2 was afforded by the gamma function

$$\Gamma(s) = \int_0^\infty e^{-y} y^s \frac{dy}{y}.$$

In order to generalize this process, we now introduce a higher-dimensional gamma function for every finite $G(\mathbb{C}|\mathbb{R})$ -set X , building upon the notation of the last section. First we fix a Haar measure on the multiplicative group $\mathbf{R}_+^*$:

Let $\mathfrak{p} = \{\tau, \bar{\tau}\}$ be the conjugation classes in X . We call $\mathfrak{p}$ real or complex, depending whether $\#\mathfrak{p} = 1$ or $\#\mathfrak{p} = 2$. We then have

$$\mathbf{R}_+^* = \prod_{\mathfrak{p}} \mathbf{R}_{+\mathfrak{p}}^*,$$

where

$$\mathbf{R}_{+\mathfrak{p}}^* = \mathbb{R}_+^*, \quad \text{resp.} \quad \mathbf{R}_{+\mathfrak{p}}^* = [\mathbb{R}_+^* \times \mathbb{R}_+^*]^+ = \{(y, y) \mid y \in \mathbb{R}_+^*\}.$$

We define isomorphisms

$$\mathbf{R}_{+\mathfrak{p}}^* \xrightarrow{\sim} \mathbb{R}_+^*$$

by $y \mapsto y$, resp. $(y, y) \mapsto y^2$, and obtain an isomorphism

$$\varphi : \mathbf{R}_+^* \xrightarrow{\sim} \prod_{\mathfrak{p}} \mathbb{R}_+^*.$$

We now denote by $\frac{dy}{y}$ the Haar measure on $\mathbf{R}_+^*$ which corresponds to the product measure

$$\prod_{\mathfrak{p}} \frac{dt}{t},$$

where $\frac{dt}{t}$ is the usual Haar measure on $\mathbb{R}_+^*$. The Haar measure thus defined is called the **canonical measure** on $\mathbf{R}_+^*$. Under the logarithm

$$\log : \mathbf{R}_+^* \xrightarrow{\sim} \mathbf{R}_+,$$

it is mapped to the Haar measure dx on $\mathbf{R}_\pm$ which under the isomorphism

$$\mathbf{R}_\pm = \prod_{\mathfrak{p}} \mathbf{R}_{\pm\mathfrak{p}} \xrightarrow{\varphi} \prod_{\mathfrak{p}} \mathbb{R},$$

$x_{\mathfrak{p}} \mapsto x_{\mathfrak{p}}$, resp. $(x_{\mathfrak{p}}, x_{\mathfrak{p}}) \mapsto 2x_{\mathfrak{p}}$, corresponds to the Lebesgue measure on $\prod_{\mathfrak{p}} \mathbb{R}$.

(4.1) Definition. For $s = (s_{\tau}) \in \mathbb{C}$ such that $\operatorname{Re}(s_{\tau}) > 0$, we define the **gamma function** associated to the $G(\mathbb{C}|\mathbb{R})$ -set X by

$$\Gamma_X(s) = \int_{\mathbf{R}_+^*} N(e^{-y}y^s) \frac{dy}{y}.$$

The integrand is well-defined, according to our conventions from p. 445, and the convergence of the integral can be reduced to the case of the ordinary gamma function as follows.

(4.2) Proposition. Decomposing the $G(\mathbb{C}|\mathbb{R})$ -set X into its conjugation classes $\mathfrak{p}$, one has

$$\Gamma_X(s) = \prod_{\mathfrak{p}} \Gamma_{\mathfrak{p}}(s_{\mathfrak{p}}),$$

where $s_{\mathfrak{p}} = s_{\tau}$ for $\mathfrak{p} = \{\tau\}$, resp. $s_{\mathfrak{p}} = (s_{\tau}, s_{\bar{\tau}})$ for $\mathfrak{p} = \{\tau, \bar{\tau}\}$, $\tau \neq \bar{\tau}$. The factors are given explicitly by

$$\Gamma_{\mathfrak{p}}(s_{\mathfrak{p}}) = \begin{cases} \Gamma(s_{\mathfrak{p}}), & \text{if } \mathfrak{p} \text{ real,} \\ 2^{1-\operatorname{Tr}(s_{\mathfrak{p}})} \Gamma(\operatorname{Tr}(s_{\mathfrak{p}})), & \text{if } \mathfrak{p} \text{ complex,} \end{cases}$$

where $\operatorname{Tr}(s_{\mathfrak{p}}) = s_{\tau} + s_{\bar{\tau}}$.

Proof: The first statement is clear in view of the product decomposition

$$\left(\mathbf{R}_+^*, \frac{dy}{y}\right) = \left(\prod_{\mathfrak{p}} \mathbf{R}_{+\mathfrak{p}}^*, \prod_{\mathfrak{p}} \frac{dy_{\mathfrak{p}}}{y_{\mathfrak{p}}}\right).$$

The second is relative to a $G(\mathbb{C}|\mathbb{R})$ -set X which has only one conjugation class. If $\#X = 1$, then trivially $\Gamma_X(s) = \Gamma(s)$. So let $X = \{\tau, \bar{\tau}\}$, $\tau \neq \bar{\tau}$. Mapping

$$\psi : \mathbb{R}_+^* \longrightarrow \mathbf{R}_+^*, \quad t \longmapsto (\sqrt{t}, \sqrt{t}),$$

one then gets

$$\begin{aligned} \int_{\mathbf{R}_+^*} N(e^{-y} y^s) \frac{dy}{y} &= \int_{\mathbf{R}_+^*} N(e^{-(\sqrt{t}, \sqrt{t})} (\sqrt{t}, \sqrt{t})^{(s_{\tau}, s_{\bar{\tau}})}) \frac{dt}{t} \\ &= \int_0^{\infty} e^{-2\sqrt{t}} \sqrt{t}^{Tr(s)} \frac{dt}{t}, \end{aligned}$$

and, since $d(t/2)^2/(t/2)^2 = 2dt/t$, the substitution $t \mapsto (t/2)^2$ yields

$$\int_{\mathbf{R}_+^*} N(e^{-y} y^s) \frac{dy}{y} = 2^{1-Tr(s)} \Gamma(Tr(s)). \quad \square$$

The proposition shows that the gamma integral $\Gamma(s)$ converges for $s = (s_{\tau})$ with $\text{Re}(s_{\tau}) > 0$, and admits an analytic continuation to all of $\mathbb{C}$, except for poles at points dictated in the obvious way by the ordinary gamma function $\Gamma(s)$.

We call the function

$$L_X(s) = N(\pi^{-s/2}) \Gamma_X(s/2)$$

the **L -function** of the $G(\mathbb{C}|\mathbb{R})$ -set X . Decomposing X into the conjugation classes $\mathfrak{p}$, yields

$$L_X(s) = \prod_{\mathfrak{p}} L_{\mathfrak{p}}(s_{\mathfrak{p}}),$$

where as before we write $s_{\mathfrak{p}} = s_{\tau}$ for $\mathfrak{p} = \{\tau\}$ and $s_{\mathfrak{p}} = (s_{\tau}, s_{\bar{\tau}})$ for $\mathfrak{p} = \{\tau, \bar{\tau}\}$, $\tau \neq \bar{\tau}$. The factors $L_{\mathfrak{p}}(s_{\mathfrak{p}})$ are given explicitly, by (4.2), as

$$L_{\mathfrak{p}}(s_{\mathfrak{p}}) = \begin{cases} \pi^{-s_{\mathfrak{p}}/2} \Gamma(s_{\mathfrak{p}}/2), & \text{if } \mathfrak{p} \text{ real,} \\ 2(2\pi)^{-Tr(s_{\mathfrak{p}})/2} \Gamma(Tr(s_{\mathfrak{p}})/2), & \text{if } \mathfrak{p} \text{ complex.} \end{cases}$$

For a single complex variable $s \in \mathbb{C}$, we put

$$\Gamma_X(s) = \Gamma_X(s\mathbf{1}),$$

where $\mathbf{1} = (1, \dots, 1)$ is the unit element of $\mathbf{C}$. Denoting r_1 , resp. r_2 , the number of real, resp. complex, conjugation classes of X , we find

$$\Gamma_X(s) = 2^{(1-2s)r_2} \Gamma(s)^{r_1} \Gamma(2s)^{r_2}.$$

In the same way we put

$$L_X(s) = L_X(s\mathbf{1}) = \pi^{-ns/2} \Gamma_X(s/2), \quad n = \#X,$$

and in particular

$$L_{\mathbb{R}}(s) = L_X(s) = \pi^{-s/2} \Gamma(s/2), \quad \text{if } X = \{\tau\},$$

$$L_{\mathbb{C}}(s) = L_X(s) = 2(2\pi)^{-s} \Gamma(s), \quad \text{if } X = \{\tau, \bar{\tau}\}, \tau \neq \bar{\tau}.$$

Then we have, for an arbitrary $G(\mathbb{C}|\mathbb{R})$ -set X :

$$L_X(s) = L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2}.$$

With this notation, (1.2) implies the

(4.3) Proposition. (i) $L_{\mathbb{R}}(1) = 1$, $L_{\mathbb{C}}(1) = \frac{1}{\pi}$.

(ii) $L_{\mathbb{R}}(s+2) = \frac{s}{2\pi} L_{\mathbb{R}}(s)$, $L_{\mathbb{C}}(s+1) = \frac{s}{2\pi} L_{\mathbb{C}}(s)$.

(iii) $L_{\mathbb{R}}(1-s)L_{\mathbb{R}}(1+s) = \frac{1}{\cos \pi s/2}$, $L_{\mathbb{C}}(s)L_{\mathbb{C}}(1-s) = \frac{2}{\sin \pi s}$.

(iv) $L_{\mathbb{R}}(s)L_{\mathbb{R}}(s+1) = L_{\mathbb{C}}(s)$ (**Legendre's duplication formula**).

As a consequence we obtain the following functional equation for the L -function $L_X(s)$:

(4.4) Proposition. $L_X(s) = A(s)L_X(1-s)$ with the factor

$$A(s) = (\cos \pi s/2)^{r_1+r_2} (\sin \pi s/2)^{r_2} L_{\mathbb{C}}(s)^n.$$

Proof: On the one hand we have

$$\frac{L_{\mathbb{R}}(s)}{L_{\mathbb{R}}(1-s)} = \frac{L_{\mathbb{R}}(s)L_{\mathbb{R}}(1+s)}{L_{\mathbb{R}}(1-s)L_{\mathbb{R}}(1+s)} = \cos \pi s/2 L_{\mathbb{C}}(s),$$

and on the other

$$\begin{aligned} \frac{L_{\mathbb{C}}(s)}{L_{\mathbb{C}}(1-s)} &= \frac{L_{\mathbb{C}}(s)^2}{L_{\mathbb{C}}(1-s)L_{\mathbb{C}}(s)} = \frac{1}{2} \sin \pi s L_{\mathbb{C}}(s)^2 \\ &= \cos \pi s/2 \sin \pi s/2 L_{\mathbb{C}}(s)^2. \end{aligned}$$

The proposition therefore results from the identity $L_X(s) = L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2}$. □

This concludes the purely function-theoretic preparations. They will now be applied to number theory.

§ 5. The Dedekind Zeta Function

The Riemann zeta function $\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}$ is associated with the field $\mathbb{Q}$ of rational numbers. It generalizes in the following way to an arbitrary number field K of degree $n = [K : \mathbb{Q}]$.

(5.1) Definition. *The Dedekind zeta function of the number field K is defined by the series*

$$\zeta_K(s) = \sum_{\mathfrak{a}} \frac{1}{\mathfrak{N}(\mathfrak{a})^s},$$

where $\mathfrak{a}$ varies over the integral ideals of K , and $\mathfrak{N}(\mathfrak{a})$ denotes their absolute norm.

(5.2) Proposition. *The series $\zeta_K(s)$ converge absolutely and uniformly in the domain $\operatorname{Re}(s) \geq 1 + \delta$ for every $\delta > 0$, and one has*

$$\zeta_K(s) = \prod_{\mathfrak{p}} \frac{1}{1 - \mathfrak{N}(\mathfrak{p})^{-s}},$$

where $\mathfrak{p}$ runs through the prime ideals of K .

The proof proceeds in the same way as for the Riemann zeta function (see (1.1)), because the absolute norm $\mathfrak{N}(\mathfrak{a})$ is multiplicative. We do not go into it here, because it is the same argument that also applies to *Hecke L -series*, which will be introduced in §8 as a common generalization of Dirichlet L -series and of the Dedekind zeta function.

Just like the Riemann zeta function, the Dedekind zeta function also admits an analytic continuation to the complex plane with 1 removed, and it satisfies a functional equation relating the argument s to $1 - s$. This is what we are now going to prove. The argument will turn out to be a higher dimensional generalization of the one used in §1 for the Riemann zeta function.

First we split up the series $\zeta_K(s)$, according to the classes $\mathfrak{K}$ of the usual ideal class group $Cl_K = J/P$ of K , into the **partial zeta functions**

$$\zeta(\mathfrak{K}, s) = \sum_{\substack{\mathfrak{a} \in \mathfrak{K} \\ \text{integral}}} \frac{1}{\mathfrak{N}(\mathfrak{a})^s}$$

so that

$$\zeta_K(s) = \sum_{\mathfrak{K}} \zeta(\mathfrak{K}, s).$$

The functional equation is then proved for the individual functions $\zeta(\mathfrak{K}, s)$. The integral ideals in $\mathfrak{K}$ are described as follows. If $\mathfrak{a}$ is a fractional ideal, then the unit group $\mathcal{O}^*$ of $\mathcal{O}$ operates on the set $\mathfrak{a}^* = \mathfrak{a} \setminus \{0\}$, and we denote by $\mathfrak{a}^*/\mathcal{O}^*$ the set of orbits, i.e., the set of classes of non-zero associated elements in $\mathfrak{a}$.

(5.3) Lemma. *Let $\mathfrak{a}$ be an integral ideal of K and $\mathfrak{K}$ the class of the ideal $\mathfrak{a}^{-1}$. Then there is a bijection*

$$\mathfrak{a}^*/\mathcal{O}^* \xrightarrow{\sim} \{ \mathfrak{b} \in \mathfrak{K} \mid \mathfrak{b} \text{ integral} \}, \quad \bar{a} \mapsto \mathfrak{b} = a\mathfrak{a}^{-1}.$$

Proof: If $a \in \mathfrak{a}^*$, then $a\mathfrak{a}^{-1} = (a)\mathfrak{a}^{-1}$ is an integral ideal in $\mathfrak{K}$, and if $a\mathfrak{a}^{-1} = b\mathfrak{a}^{-1}$, then $(a) = (b)$, so that $ab^{-1} \in \mathcal{O}^*$. This shows the injectivity of the mapping. But it is surjective as well, since for every integral $\mathfrak{b} \in \mathfrak{K}$, one has $\mathfrak{b} = a\mathfrak{a}^{-1}$ with $a \in \mathfrak{a}\mathfrak{b} \subseteq \mathfrak{a}$. $\square$

To the $G(\mathbb{C}|\mathbb{R})$ -set $X = \text{Hom}(K, \mathbb{C})$ corresponds the Minkowski space

$$K_{\mathbb{R}} = \mathbf{R} = \left[\prod_{\tau} \mathbb{C} \right]^+.$$

The field K may be embedded into $K_{\mathbb{R}}$. Then one finds for $a \in K^*$ that

$$\mathfrak{N}((a)) = |N_{K|\mathbb{Q}}(a)| = |N(a)|,$$

where N denotes the norm on $\mathbf{R}^*$ (see chap. I, §5). The lemma therefore yields the

(5.4) Proposition. $\zeta(\mathfrak{K}, s) = \mathfrak{N}(\mathfrak{a})^s \sum_{\bar{a} \in \mathfrak{a}^*/\mathcal{O}^*} \frac{1}{|N(\bar{a})|^s}.$

By chap. I, (5.2), the ideal $\mathfrak{a}$ forms a complete lattice in $\mathbf{R}$ whose fundamental mesh has volume

$$\text{vol}(\mathfrak{a}) = \sqrt{d_{\mathfrak{a}}},$$

where $d_a = \mathfrak{N}(\mathfrak{a})^2 |d_K|$ denotes the absolute value of the discriminant of $\mathfrak{a}$, and d_K is the discriminant of K . To the series $\zeta(\mathfrak{K}, s)$ we associate the theta series

$$\theta(\mathfrak{a}, z) = \theta_{\mathfrak{a}}(z/d_a^{1/n}) = \sum_{a \in \mathfrak{a}} e^{\pi i \langle az/d_a^{1/n}, a \rangle}.$$

It is related to $\zeta(\mathfrak{K}, s)$ via the gamma integral associated to the $G(\mathbb{C}|\mathbb{R})$ -set $X = \text{Hom}(K, \mathbb{C})$,

$$\Gamma_K(s) = \Gamma_X(s) = \int_{\mathbf{R}_+^*} N(e^{-y} y^s) \frac{dy}{y},$$

where $s \in \mathbb{C}$, $\text{Re}(s) > 0$ (see (4.1)). In the integral, we substitute

$$y \mapsto \pi |a|^2 y/d_a^{1/n}$$

with $|\cdot|$ denoting the map $\mathbf{R}^* \rightarrow \mathbf{R}_+^*$, $(x_\tau) \mapsto (|x_\tau|)$. We then obtain

$$|d_K|^s \pi^{-ns} \Gamma_K(s) \frac{\mathfrak{N}(\mathfrak{a})^{2s}}{|N(\mathfrak{a})|^{2s}} = \int_{\mathbf{R}_+^*} e^{-\pi \langle ay/d_a^{1/n}, a \rangle} N(y)^s \frac{dy}{y}.$$

Summing this over a full system $\mathfrak{R}$ of representatives of $\mathfrak{a}^*/\mathcal{O}^*$, yields

$$|d_K|^s \pi^{-ns} \Gamma_K(s) \zeta(\mathfrak{K}, 2s) = \int_{\mathbf{R}_+^*} g(y) N(y)^s \frac{dy}{y}$$

with the series

$$g(y) = \sum_{a \in \mathfrak{R}} e^{-\pi \langle ay/d_a^{1/n}, a \rangle}.$$

Swapping summation and integration is legal, for the same reason as in the case of the Riemann zeta function (see p. 422). We view the function

$$Z_\infty(s) = |d_K|^{s/2} \pi^{-ns/2} \Gamma_K(s/2) = |d_K|^{s/2} L_X(s)$$

as the “Euler factor at infinity” of the zeta function $\zeta(\mathfrak{K}, s)$ (see § 4, p. 455) and define

$$Z(\mathfrak{K}, s) = Z_\infty(s) \zeta(\mathfrak{K}, s).$$

The desire to realize this function as an integral over the theta series $\theta(\mathfrak{a}, s)$ is frustrated by the fact that in the theta series we sum over *all* $a \in \mathfrak{a}$, whereas summation in the series $g(y)$ is only over a system of representatives of $\mathfrak{a}^*/\mathcal{O}^*$. This difficulty — which was already hinted at in the case of the Riemann zeta function — will now be overcome in the general case as follows.

The image $|\mathcal{O}^*|$ of the unit group $\mathcal{O}^*$ under the mapping $|\cdot| : \mathbf{R}^* \rightarrow \mathbf{R}_+^*$ is contained in the **norm-one hypersurface**

$$\mathbf{S} = \{x \in \mathbf{R}_+^* \mid N(x) = 1\}.$$

Writing every $y \in \mathbf{R}_+^*$ in the form

$$y = xt^{1/n}, \quad x = \frac{y}{N(y)^{1/n}}, \quad t = N(y),$$

we obtain a direct decomposition

$$\mathbf{R}_+^* = \mathbf{S} \times \mathbb{R}_+^*.$$

Let d^*x be the unique Haar measure on the multiplicative group $\mathbf{S}$ such that the canonical Haar measure dy/y on $\mathbf{R}_+^*$ becomes the product measure

$$\frac{dy}{y} = d^*x \times \frac{dt}{t}.$$

We will not need any more explicit description of d^*x .

We now choose a fundamental domain F for the action of the group $|\mathcal{O}^*|^2 = \{|\varepsilon|^2 \mid \varepsilon \in \mathcal{O}^*\}$ on $\mathbf{S}$ as follows. The logarithm map

$$\log : \mathbf{R}_+^* \longrightarrow \mathbf{R}_\pm, \quad (x_\tau) \longmapsto (\log x_\tau),$$

takes the norm-one hypersurface $\mathbf{S}$ to the **trace-zero space** $H = \{x \in \mathbf{R}_\pm \mid \text{Tr}(x) = 0\}$, and the group $|\mathcal{O}^*|$ is taken to a complete lattice G in H (Dirichlet's unit theorem). Choose F to be the preimage of an arbitrary fundamental mesh of the lattice $2G$. Any such choice satisfies the

(5.5) Proposition. *The function $Z(\mathfrak{K}, 2s)$ is the Mellin transform*

$$Z(\mathfrak{K}, 2s) = L(f, s)$$

of the function

$$f(t) = f_F(\mathfrak{a}, t) = \frac{1}{w} \int_F \theta(\mathfrak{a}, ixt^{1/n}) d^*x,$$

where $w = \#\mu(K)$ denotes the number of roots of unity in K .

Proof: Decomposing $\mathbf{R}_+^* = \mathbf{S} \times \mathbb{R}_+^*$, we find

$$Z(\mathfrak{K}, 2s) = \int_0^\infty \int_{\mathbf{S}} \sum_{a \in \mathfrak{A}} e^{-\pi \langle axt', a \rangle} d^*x t^s \frac{dt}{t},$$

with $t' = (t/d_{\mathfrak{a}})^{1/n}$. The fundamental domain F cuts up the norm-one hypersurface $\mathbf{S}$ into the disjoint union

$$\mathbf{S} = \bigcup_{\eta \in |\mathcal{O}^*|} \eta^2 F.$$

The transformation $x \mapsto \eta^2 x$ of $\mathbf{S}$ leaves the Haar measure d^*x invariant and maps F to $\eta^2 F$, so that

$$\begin{aligned} \int_{\mathbf{S}} \sum_{a \in \mathfrak{R}} e^{-\pi \langle axt', a \rangle} d^*x &= \sum_{\eta \in |\mathcal{O}^*|} \int_{\eta^2 F} \sum_{a \in \mathfrak{R}} e^{-\pi \langle axt', a \rangle} d^*x \\ &= \frac{1}{w} \int_F \sum_{\varepsilon \in \mathcal{O}^*} \sum_{a \in \mathfrak{R}} e^{-\pi \langle a\varepsilon xt', a\varepsilon \rangle} d^*x \\ &= \frac{1}{w} \int_F (\theta(\mathfrak{a}, ixt^{1/n}) - 1) d^*x = f(t) - f(\infty). \end{aligned}$$

Observe here that we have to divide by $w = \#\mu(K)$, because $\mu(K)$ is just the kernel of $\mathcal{O}^* \rightarrow |\mathcal{O}^*|$ (see chap. I, (7.1)), hence $\sum_{|\varepsilon|} = \frac{1}{w} \sum_{\varepsilon}$. Observe furthermore that $a\varepsilon$ runs through the set $\mathfrak{a}^* = \mathfrak{a} \setminus \{0\}$ exactly once, and finally that $f(\infty) = \frac{1}{w} \int_F d^*x$, as $\theta(\mathfrak{a}, ixt) = 1$. This result does indeed show that

$$Z(\mathfrak{R}, 2s) = \int_0^\infty (f(t) - f(\infty)) t^s \frac{dt}{t} = L(f, s). \quad \square$$

Using this proposition, the functional equation for the function $Z(\mathfrak{R}, s)$ follows via the Mellin principle from a corresponding transformation formula for the function $f_F(\mathfrak{a}, t)$, which in turn derives from the general theta transformation formula (3.6). In order to find the precise equation, we have to compute the volume $\text{vol}(F)$ of the fundamental domain F with respect to d^*x , and the lattice which is dual to $\mathfrak{a}$ in $\mathbf{R}$. This is achieved by the following two lemmas.

(5.6) Lemma. *The fundamental domain F of $\mathbf{S}$ has the following volume with respect to d^*x :*

$$\text{vol}(F) = 2^{r-1} R,$$

where r is the number of infinite places and R is the **regulator** of K (see chap. I, (7.5)).

Proof: The canonical measure dy/y on $\mathbf{R}_+^*$ is transformed into the product measure $d^*x \times dt/t$ by the isomorphism

$$\alpha : \mathbf{S} \times \mathbb{R}_+^* \longrightarrow \mathbf{R}_+^*, \quad (x, t) \longmapsto xt^{1/n}.$$

Since $I = \{t \in \mathbb{R}_+^* \mid 1 \leq t \leq e\}$ has measure 1 with respect to dt/t , the quantity $\text{vol}(F)$ is also the volume of $F \times I$ with respect to $d^*x \times dt/t$, i.e., the volume of $\alpha(F \times I)$ with respect to dy/y . The composite ψ of the isomorphisms

$$\mathbf{R}_+^* \xrightarrow{\log} \mathbf{R}_\pm \xrightarrow{\varphi} \prod_{p|\infty} \mathbb{R} = \mathbb{R}^r$$

(see §4, p. 454) transforms dy/y into the Lebesgue measure of $\mathbb{R}^r$,

$$\text{vol}(F) = \text{vol}_{\mathbb{R}^r}(\psi\alpha(F \times I)).$$

Let us compute the image $\psi\alpha(F \times I)$. Let $\mathbf{1} = (1, \dots, 1) \in \mathbf{S}$. Then we find

$$\psi\alpha((\mathbf{1}, t)) = \epsilon \log t^{1/n} = \frac{1}{n} \epsilon \log t$$

with the vector $\epsilon = (e_{p_1}, \dots, e_{p_r}) \in \mathbb{R}^r$, $e_{p_i} = 1$, resp. $= 2$, depending whether p_i is real or complex. By definition of F , we also have

$$\psi\alpha(F \times \{1\}) = 2\Phi,$$

where Φ denotes a fundamental mesh of the unit lattice G in trace-zero space $H = \{(x_i) \in \mathbb{R}^r \mid \sum x_i = 0\}$. This gives

$$\psi\alpha(F \times I) = 2\Phi + \left[0, \frac{1}{n}\right] \epsilon,$$

the parallelepiped spanned by the vectors $2\epsilon_1, \dots, 2\epsilon_{r-1}, \frac{1}{n}\epsilon$, if $\epsilon_1, \dots, \epsilon_{r-1}$ span the fundamental mesh Φ . Its volume is $\frac{1}{n}2^{r-1}$ times the absolute value of the determinant

$$\det \begin{pmatrix} \epsilon_{11} & \cdots & \epsilon_{r-1,1} & e_{p_1} \\ \vdots & & \vdots & \vdots \\ \epsilon_{1r} & \cdots & \epsilon_{r-1,r} & e_{p_r} \end{pmatrix}.$$

Adding the first $r-1$ lines to the last one, all entries of the last line become zero, except the last one, which is $n = \sum e_{p_i}$. The matrix above these zeroes has the absolute value of its determinant by definition equal to the regulator R . Thus we get

$$\text{vol}(F) = 2^{r-1}R.$$

□

(5.7) Lemma. *The lattice Γ' in $\mathbf{R}$ which is dual to the lattice $\Gamma = \mathfrak{a}$ is given by*

$$*\Gamma' = (\mathfrak{a}\mathfrak{d})^{-1},$$

where the asterisk denotes the involution $(x_\tau) \mapsto (\bar{x}_\tau)$ on $K_{\mathbb{R}}$ and $\mathfrak{d}$ the different of $K|\mathbb{Q}$.

Proof: As $\langle x, y \rangle = \text{Tr}(*xy)$, we have

$$*\Gamma' = \{ *g \in \mathbf{R} \mid \langle g, a \rangle \in \mathbb{Z} \text{ for all } a \in \mathfrak{a} \} = \{ x \in \mathbf{R} \mid \text{Tr}(xa) \subseteq \mathbb{Z} \}.$$

$\text{Tr}(xa) \subseteq \mathbb{Z}$ implies immediately $x \in K$, for if $a_1, \dots, a_n$ is a $\mathbb{Z}$ -basis of $\mathfrak{a}$ and $x = x_1a_1 + \dots + x_na_n$, with $x_i \in \mathbb{R}$, then $\text{Tr}(xa_j) = \sum_i x_i \text{Tr}(a_ia_j) = n_j \in \mathbb{Z}$ is a system of linear equations with coefficients $\text{Tr}(a_ia_j) = \text{Tr}_{K|\mathbb{Q}}(a_ia_j) \in \mathbb{Q}$, so all $x_i \in \mathbb{Q}$, and thus $x \in K$. It follows that

$$*\Gamma' = \{ x \in K \mid \text{Tr}(xa) \subseteq \mathbb{Z} \}.$$

By definition we have $\mathfrak{d}^{-1} = \{ x \in K \mid \text{Tr}_{K|\mathbb{Q}}(x\mathfrak{o}) \subseteq \mathbb{Z} \}$, and we obtain the equivalences $x \in *\Gamma' \iff \text{Tr}_{K|\mathbb{Q}}(xa\mathfrak{o}) \subseteq \mathbb{Z} \text{ for all } a \in \mathfrak{a} \iff xa \subseteq \mathfrak{d}^{-1} \iff x \in (\mathfrak{a}\mathfrak{d})^{-1}$. $\square$

(5.8) Proposition. *The functions $f_F(\mathfrak{a}, t)$ satisfy the transformation formula*

$$f_F\left(\mathfrak{a}, \frac{1}{t}\right) = t^{1/2} f_{F^{-1}}((\mathfrak{a}\mathfrak{d})^{-1}, t),$$

and one has

$$f_F(\mathfrak{a}, t) = \frac{2^{r-1}}{w} R + O(e^{-ct^{1/n}}) \quad \text{for } t \rightarrow \infty, c > 0.$$

Proof: We make use of formula (3.6)

$$\theta_\Gamma(-1/z) = \frac{\sqrt{N(z/i)}}{\text{vol}(\Gamma)} \theta_{\Gamma'}(z)$$

for the lattice $\Gamma = \mathfrak{a}$ in $\mathbf{R}$, whose fundamental mesh has volume $\text{vol}(\Gamma) = \mathfrak{N}(\mathfrak{a})|d_K|^{1/2}$. The lattice Γ' dual to Γ is given by (5.7) as $*\Gamma' = (\mathfrak{a}\mathfrak{d})^{-1}$. The compatibility $\langle *gz, *g \rangle = \langle gz, g \rangle$ implies that $\theta_{\Gamma'}(z) = \theta_{*\Gamma'}(z)$. Furthermore we have

$$d_{(\mathfrak{a}\mathfrak{d})^{-1}} = \mathfrak{N}(\mathfrak{a})^{-2} \mathfrak{N}(\mathfrak{d})^{-2} |d_K| = 1/(\mathfrak{N}(\mathfrak{a})^2 |d_K|) = 1/d_{\mathfrak{a}}.$$

The transformation $x \mapsto x^{-1}$ of the multiplicative group $\mathbf{S}$ fixes the Haar measure d^*x (in the same way as $x \mapsto -x$ fixes a Haar measure on $\mathbb{R}^n$)

and maps the fundamental domain F onto the fundamental domain F^{-1} , whose image $\log(F^{-1})$ is again a fundamental mesh of the lattice $2 \log |\mathcal{O}^*|$. Observing that $N(x(td_a)^{1/n}) = td_a$ for $x \in S$, we obtain

$$\begin{aligned} f_F\left(a, \frac{1}{t}\right) &= \frac{1}{w} \int_F \theta_a(ix/\sqrt[n]{td_a}) d^*x \\ &= \frac{1}{w} \int_{F^{-1}} \theta_a(-1/ix\sqrt[n]{td_a}) d^*x \\ &= \frac{1}{w} \frac{(td_a)^{1/2}}{\text{vol}(a)} \int_{F^{-1}} \theta_{(a\mathfrak{d})^{-1}}(ix\sqrt[n]{td_a}) d^*x \\ &= \frac{t^{1/2}}{w} \int_{F^{-1}} \theta_{(a\mathfrak{d})^{-1}}(ix\sqrt[n]{t/d_{(a\mathfrak{d})^{-1}}}) d^*x \\ &= t^{1/2} f_{F^{-1}}((a\mathfrak{d})^{-1}, t). \end{aligned}$$

This shows the first formula. To prove the second, we write

$$f_F(a, t) = \frac{1}{w} \int_F d^*x + \frac{1}{w} \int_F (\theta(a, ixt^{1/n}) - 1) d^*x = \frac{\text{vol}(F)}{w} + r(t).$$

The function $r(t)$ satisfies $r(t) = O(e^{-ct^{1/n}})$, $c > 0$, $t \rightarrow \infty$, as the summands of $\theta(a, ixt^{1/n}) - 1$ are of the form

$$e^{-\pi \langle ax, a \rangle \sqrt[n]{t'}} , \quad a \in \mathfrak{a}, a \neq 0, t' = t/d_a.$$

The point $x = (x_\tau)$ varies in the compact closure $\bar{F} \subseteq [\prod_\tau \mathbb{R}_+^*]^+$ of F . Hence $x_\tau \geq \delta > 0$ for all τ , i.e.,

$$\langle ax, a \rangle = \sum_\tau |\tau a|^2 x_\tau \geq \delta \langle a, a \rangle$$

and so

$$r(t) \leq \frac{\text{vol}(F)}{w} (\theta_a(i\delta \sqrt[n]{t'}) - 1).$$

Writing $m = \min\{\langle a, a \rangle \mid a \in \mathfrak{a}, a \neq 0\}$ and $M = \#\{a \in \mathfrak{a} \mid \langle a, a \rangle = m\}$, it follows that

$$\theta_a(i\delta \sqrt[n]{t'}) - 1 = e^{-\pi \delta m \sqrt[n]{t'}} \left(M + \sum_{\langle a, a \rangle > m} e^{-\pi \delta (\langle a, a \rangle - m) \sqrt[n]{t'}} \right) = O(e^{-ct^{1/n}})$$

where $c = \pi \delta m / d_a^{1/n}$. We thus get as claimed

$$f_F(a, t) = \frac{\text{vol}(F)}{w} + O(e^{-ct^{1/n}}) = \frac{2^{r-1}}{w} R + O(e^{-ct^{1/n}}). \quad \square$$

This last proposition now enables us to apply the Mellin principle (1.4) to the functions $f_F(\mathfrak{a}, t)$. For the partial zeta functions

$$\zeta(\mathfrak{K}, s) = \sum_{\substack{\mathfrak{b} \in \mathfrak{K} \\ \text{integral}}} \frac{1}{\mathfrak{N}(\mathfrak{b})^s},$$

this yields the following result, where the notations d_K , R , w , and r signify as before the discriminant, the regulator, the number of roots of unity, and the number of infinite places, respectively.

(5.9) Theorem. *The function*

$$Z(\mathfrak{K}, s) = Z_\infty(s) \zeta(\mathfrak{K}, s), \quad \operatorname{Re}(s) > 1,$$

$Z_\infty(s) = |d_K|^{s/2} \pi^{-ns/2} \Gamma_K(s/2)$, admits an analytic continuation to $\mathbb{C} \setminus \{0, 1\}$ and satisfies the functional equation

$$Z(\mathfrak{K}, s) = Z(\mathfrak{K}', 1 - s),$$

where the ideal classes $\mathfrak{K}$ and $\mathfrak{K}'$ correspond to each other via $\mathfrak{K}\mathfrak{K}' = [\mathfrak{d}]$. It has simple poles at $s = 0$ and $s = 1$ with residues

$$-\frac{2^r}{w} R, \quad \text{resp.} \quad \frac{2^r}{w} R.$$

Proof: Let $f(t) = f_F(\mathfrak{a}, t)$ and $g(t) = f_{F^{-1}}((\mathfrak{a}\mathfrak{d})^{-1}, t)$. Then (5.8) implies

$$f\left(\frac{1}{t}\right) = t^{1/2} g(t)$$

and

$$f(t) = a_0 + O(e^{-ct^{1/n}}), \quad g(t) = a_0 + O(e^{-ct^{1/n}}),$$

with $a_0 = \frac{2^{r-1}}{w} R$. Proposition (1.4) thus ensures the analytic continuation of the Mellin transforms of f and g , and the functional equation

$$L(f, s) = L\left(g, \frac{1}{2} - s\right)$$

with simple poles of $L(f, s)$ at $s = 0$ and $s = \frac{1}{2}$ with residues $-a_0$, resp. a_0 . Therefore

$$Z(\mathfrak{K}, s) = L\left(f, \frac{s}{2}\right)$$

admits an analytic continuation to $\mathbb{C} \setminus \{0, 1\}$ with simple poles at $s = 0$ and $s = 1$ and residues

$$-2a_0 = -\frac{2^r}{w} R, \quad \text{resp.} \quad 2a_0 = \frac{2^r}{w} R$$

and satisfies the functional equation

$$Z(\mathfrak{K}, s) = L\left(f, \frac{s}{2}\right) = L\left(g, \frac{1-s}{2}\right) = Z(\mathfrak{K}', 1-s). \quad \square$$

This theorem about the partial zeta functions immediately implies an analogous result for the **completed zeta function** of the number field K ,

$$Z_K(s) = Z_\infty(s)\zeta_K(s) = \sum_{\mathfrak{K}} Z(\mathfrak{K}, s).$$

(5.10) Corollary. *The completed zeta function $Z_K(s)$ admits an analytic continuation to $\mathbb{C} \setminus \{0, 1\}$ and satisfies the functional equation*

$$Z_K(s) = Z_K(1-s).$$

It has simple poles at $s = 0$ and $s = 1$ with residues

$$-\frac{2^r hR}{w}, \quad \text{resp.} \quad \frac{2^r hR}{w},$$

where h is the class number of K .

The last result can be immediately generalized as follows. For every character

$$\chi : J/P \longrightarrow S^1$$

of the ideal class group, one may form the zeta function

$$Z(\chi, s) = Z_\infty(s)\zeta(\chi, s),$$

where

$$\zeta(\chi, s) = \sum_{\mathfrak{a} \text{ integral}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s}$$

and $\chi(\mathfrak{a})$ denotes the value $\chi(\mathfrak{K})$ of the class $\mathfrak{K} = [\mathfrak{a}]$ of an ideal $\mathfrak{a}$. Then clearly

$$Z(\chi, s) = \sum_{\mathfrak{K}} \chi(\mathfrak{K})Z(\mathfrak{K}, s),$$

and in view of $\mathfrak{K}' = \mathfrak{K}^{-1}[\mathfrak{d}]$, we obtain from (5.9) the functional equation

$$Z(\chi, s) = \chi(\mathfrak{d})Z(\bar{\chi}, 1-s).$$

If $\chi \neq 1$, then $Z(\chi, s)$ is holomorphic on all of $\mathbb{C}$, as $\sum_{\mathfrak{K}} \chi(\mathfrak{K}) = 0$.

We now conclude with the original Dedekind zeta function

$$\zeta_K(s) = \sum_{\mathfrak{a}} \frac{1}{\mathfrak{N}(\mathfrak{a})^s}, \quad \operatorname{Re}(s) > 1.$$

The Euler factor at infinity, $Z_\infty(s)$, is given explicitly by § 4 as

$$Z_\infty(s) = |d_K|^{s/2} L_X(s) = |d_K|^{s/2} L_{\mathbb{R}}(s)^{r_1} L_{\mathbb{C}}(s)^{r_2},$$

where r_1 , resp. r_2 , denotes the number of real, resp. complex, places. By (4.3), (i), one has $Z_\infty(1) = |d_K|^{1/2}/\pi^{r_2}$. As

$$\zeta_K(s) = Z_\infty(s)^{-1} Z_K(s) = |d_K|^{-s/2} L_X(s)^{-1} Z_K(s),$$

we obtain from (4.4) the

(5.11) Corollary. (i) *The Dedekind zeta function $\zeta_K(s)$ has an analytic continuation to $\mathbb{C} \setminus \{1\}$.*

(ii) *At $s = 1$ it has a simple pole with residue*

$$\kappa = \frac{2^{r_1} (2\pi)^{r_2}}{w |d_K|^{1/2}} hR = hR / e^g.$$

Here h denotes the class number and

$$g = \log \frac{w |d_K|^{1/2}}{2^{r_1} (2\pi)^{r_2}}$$

the genus of the number field K (see chap. III, (3.5)).

(iii) *It satisfies the functional equation*

$$\zeta_K(1-s) = A(s) \zeta_K(s)$$

with the factor

$$A(s) = |d_K|^{s-\frac{1}{2}} \left(\cos \frac{\pi s}{2} \right)^{r_1+r_2} \left(\sin \frac{\pi s}{2} \right)^{r_2} L_{\mathbb{C}}(s)^n.$$

The proof of the analytic continuation and functional equation of the Dedekind zeta function was first given by the mathematician *ERICH HECKE* (1887–1947), along the same general lines we have presented here, albeit in a somewhat different formulation. Further, the theory we are about to develop in the following sections §§ 6–8 also substantially goes back to *HECKE*.

The formula for the residue

$$\operatorname{Res}_{s=1} \zeta_K(s) = \frac{2^{r_1} (2\pi)^{r_2}}{w |d_K|^{1/2}} hR$$

is commonly known as the analytic **class number formula**. It does allow us to determine the class number h of the field K , provided we know the law for the decomposition of primes in this field sufficiently well to lay our hands on the Euler product and thus compute the zeta function.

The following application of corollary (5.11) to Dirichlet L -series $L(\chi, s)$ (see §2) is highly remarkable. It results from studying the Dedekind zeta function $\zeta_K(s)$ for the field $K = \mathbb{Q}(\mu_m)$ of m -th roots of unity, and is based on the

(5.12) Proposition. *If $K = \mathbb{Q}(\mu_m)$ is the field of m -th roots of unity, then*

$$\zeta_K(s) = G(s) \prod_{\chi} L(\chi, s),$$

where χ varies over all Dirichlet characters mod m , and

$$G(s) = \prod_{\mathfrak{p}|m} (1 - \mathfrak{N}(\mathfrak{p})^{-s})^{-1}.$$

Proof: The proof hinges on the law of decomposition of prime numbers p in the field K . Let $p = (\mathfrak{p}_1 \dots \mathfrak{p}_r)^e$ be the decomposition of the prime number p in K , and let f be the degree of the $\mathfrak{p}_i$, i.e., $\mathfrak{N}(\mathfrak{p}_i) = p^f$. Then $\zeta_K(s)$ contains the factor

$$\prod_{\mathfrak{p}|p} (1 - \mathfrak{N}(\mathfrak{p})^{-s})^{-1} = (1 - p^{-fs})^{-r}.$$

On the other hand, the L -series give the factor $\prod_{\chi} (1 - \chi(p)p^{-s})^{-1}$. For $p|m$ this is 1. So let $p \nmid m$. By chap. I, (10.3), f is the order of p mod m in $(\mathbb{Z}/m\mathbb{Z})^*$ and $e = 1$. Since $efr = \varphi(m)$, the quotient $r = \varphi(m)/f$ is the index of the subgroup G_p generated by p in $G = (\mathbb{Z}/m\mathbb{Z})^*$. Associating $\chi \mapsto \chi(p)$ defines an isomorphism $\widehat{G}_p \cong \mu_f$, and gives the exact sequence

$$1 \longrightarrow \widehat{G/G_p} \longrightarrow \widehat{G} \longrightarrow \mu_f \longrightarrow 1,$$

where $\widehat{}$ indicates character groups. We therefore find $r = \#(\widehat{G/G_p}) = (G : G_p)$ elements in the preimage of $\chi(p)$. It follows that

$$\begin{aligned} \prod_{\chi} (1 - \chi(p)p^{-s})^{-1} &= \prod_{\zeta \in \mu_f} (1 - \zeta p^{-s})^{-r} = (1 - p^{-fs})^{-r} \\ &= \prod_{\mathfrak{p}|p} (1 - \mathfrak{N}(\mathfrak{p})^{-s})^{-1}. \end{aligned}$$

Finally, taking the product over all p , we get $\zeta_K(s) = G(s) \prod_{\chi} L(\chi, s)$. $\square$

For the trivial character $\chi^0 \bmod m$, we have $L(\chi^0, s) = \prod_{p|m} (1 - p^{-s}) \zeta(s)$, so that

$$\zeta_K(s) = G(s) \prod_{p|m} (1 - p^{-s}) \zeta(s) \prod_{\chi \neq \chi^0} L(\chi, s).$$

Since $\zeta(s)$ and $\zeta_K(s)$ both have a simple pole at $s = 1$, we obtain the

(5.13) Proposition. *For every non-trivial Dirichlet character χ , one has*

$$L(\chi, 1) \neq 0.$$

This innocuous looking result is in fact rather profound, and yields as a concrete consequence

(5.14) Dirichlet's Prime Number Theorem. *Every arithmetic progression*

$$a, a \pm m, a \pm 2m, a \pm 3m, \dots, \text{ with } (a, m) = 1,$$

i.e., every class $a \bmod m$, contains infinitely many prime numbers.

Proof: Let χ be a Dirichlet character $\bmod m$. Then one has, for $\operatorname{Re}(s) > 1$,

$$\log L(\chi, s) = - \sum_p \log(1 - \chi(p)p^{-s}) = \sum_p \sum_{m=1}^{\infty} \frac{\chi(p^m)}{mp^{ms}} = \sum_p \frac{\chi(p)}{p^s} + g_\chi(s),$$

where $g_\chi(s)$ is holomorphic for $\operatorname{Re}(s) > \frac{1}{2}$ — this follows from a trivial estimate. Multiplying by $\chi(a^{-1})$ and summing over all characters $\bmod m$, yields

$$\begin{aligned} \sum_{\chi} \chi(a^{-1}) \log L(\chi, s) &= \sum_{\chi} \sum_p \frac{\chi(a^{-1}p)}{p^s} + g(s) \\ &= \sum_{b=1}^m \sum_{\chi} \chi(a^{-1}b) \sum_{p \equiv b(m)} \frac{1}{p^s} + g(s) \\ &= \sum_{p \equiv a(m)} \frac{\varphi(m)}{p^s} + g(s). \end{aligned}$$

Note here that

$$\sum_{\chi} \chi(a^{-1}b) = \begin{cases} 0, & \text{if } a \neq b, \\ \varphi(m) = \#(\mathbb{Z}/m\mathbb{Z})^*, & \text{if } a = b. \end{cases}$$

When we pass to the limit $s \rightarrow 1$ (s real > 1), $\log L(\chi, s)$ stays bounded for $\chi \neq \chi^0$ because $L(\chi, 1) \neq 0$, whereas $\log L(\chi^0, s) =$

$\sum_{p|m} \log(1 - p^{-s}) + \log \zeta(s)$ tends to ∞ because $\zeta(s)$ has a pole. The left-hand side of the above equation therefore tends to ∞ , and since $g(s)$ is holomorphic at $s = 1$, we find

$$\lim_{s \rightarrow 1} \sum_{p \equiv a(m)} \frac{\varphi(m)}{p^s} = \infty.$$

Thus the sum cannot consist of only finitely many terms, and the theorem is proved. $\square$

For $a = 1$, Dirichlet's prime number theorem may be proved by pure algebra (see chap. I, § 10, exercise 1). Searching for a proof in the general case Dirichlet was led to the study of the L -series $L(\chi, s)$. This analytic method gives sharper results on the distribution of prime numbers among the classes $a \bmod m$. We will come back to this in a more general context in § 13.

§ 6. Hecke Characters

Let $\mathfrak{m}$ be an integral ideal of the number field K , and let $J^{\mathfrak{m}}$ be the group of all ideals of K which are relatively prime to $\mathfrak{m}$. Given any character

$$\chi : J^{\mathfrak{m}} \rightarrow S^1 = \{z \in \mathbb{C} \mid |z| = 1\},$$

we may associate to it, as a common generalization of the Dirichlet L -series as well as the Dedekind zeta function, the L -series

$$L(\chi, s) = \sum_{\mathfrak{a}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s}.$$

Here $\mathfrak{a}$ varies over all integral ideals of K , and one defines $\chi(\mathfrak{a}) = 0$ whenever $(\mathfrak{a}, \mathfrak{m}) \neq 1$. Searching for the most comprehensive class of characters χ for which the corresponding L -series could be shown to have a functional equation, HECKE was led to the notion of *Größencharaktere*, which we define as follows.

(6.1) Definition. A **Größencharakter mod $\mathfrak{m}$** is a character $\chi : J^{\mathfrak{m}} \rightarrow S^1$ for which there exists a pair of characters

$$\chi_f : (\mathcal{O}/\mathfrak{m})^* \rightarrow S^1, \quad \chi_\infty : \mathbf{R}^* \rightarrow S^1,$$

such that

$$\chi((a)) = \chi_f(a) \chi_\infty(a)$$

for every algebraic integer $a \in \mathcal{O}$ relatively prime to $\mathfrak{m}$.

A character χ of $J^{\mathfrak{m}}$ is a *Größencharakter* mod $\mathfrak{m}$ as soon as there exists a character χ_{∞} of $\mathbf{R}^*$ such that

$$\chi((a)) = \chi_{\infty}(a)$$

for all $a \in \mathcal{O}$ such that $a \equiv 1 \pmod{\mathfrak{m}}$. For if this is the case, then the rule $\chi_{\mathfrak{f}}(a) = \chi((a))\chi_{\infty}(a)^{-1}$ defines a character $\chi_{\mathfrak{f}}$ of $(\mathcal{O}/\mathfrak{m})^*$ which satisfies

$$\chi((a)) = \chi_{\mathfrak{f}}(a)\chi_{\infty}(a)$$

for all algebraic integers $a \in \mathcal{O}$ relatively prime to $\mathfrak{m}$. This last identity underlines the fact that the restriction of a *Größencharakter* to principal ideals breaks up into a finite and an infinite part. From

$$\mathcal{O}^{(\mathfrak{m})} = \{a \in \mathcal{O} \mid (a, \mathfrak{m}) = 1\},$$

it extends uniquely to the group

$$K^{(\mathfrak{m})} = \{a \in K^* \mid (a, \mathfrak{m}) = 1\}$$

of all fractions relatively prime to $\mathfrak{m}$, because every $a \in K^{(\mathfrak{m})}$ determines a well-defined class in $(\mathcal{O}/\mathfrak{m})^*$. The character χ_{∞} , and thus also the character $\chi_{\mathfrak{f}}$, are determined uniquely by the *Größencharakter* χ , since the group

$$K^{\mathfrak{m}} = \{a \in K^{(\mathfrak{m})} \mid a \equiv 1 \pmod{\mathfrak{m}}\}$$

is dense in $\mathbf{R}^*$, by the approximation theorem, and one has $\chi_{\infty}(a) = \chi((a))$ for $a \in K^{\mathfrak{m}}$. Let us recall that the congruence $a \equiv 1 \pmod{\mathfrak{m}}$ signifies that $a = b/c$, for two integers b, c relatively prime to $\mathfrak{m}$, such that $b \equiv c \pmod{\mathfrak{m}}$ or, equivalently, $a \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})} \subseteq K_{\mathfrak{p}}$ for $\mathfrak{p} \mid \mathfrak{m}$, if $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$.

The character χ_{∞} factors automatically through $\mathbf{R}^*/\mathcal{O}^{\mathfrak{m}}$, where

$$\mathcal{O}^{\mathfrak{m}} = \{\varepsilon \in \mathcal{O}^* \mid \varepsilon \equiv 1 \pmod{\mathfrak{m}}\}.$$

In fact, for $\varepsilon \in \mathcal{O}^{\mathfrak{m}}$ we have $\chi_{\mathfrak{f}}(\varepsilon) = 1$, and thus $\chi_{\infty}(\varepsilon) = \chi_{\mathfrak{f}}(\varepsilon)\chi_{\infty}(\varepsilon) = \chi((\varepsilon)) = 1$. The two characters $\chi_{\mathfrak{f}}$ and χ_{∞} of $(\mathcal{O}/\mathfrak{m})^*$, resp. $\mathbf{R}^*/\mathcal{O}^{\mathfrak{m}}$, associated with a *Größencharakter* χ satisfy the relation

$$\chi_{\mathfrak{f}}(\varepsilon)\chi_{\infty}(\varepsilon) = 1 \quad \text{for all } \varepsilon \in \mathcal{O}^*,$$

and it can be shown that every such pair of characters $(\chi_{\mathfrak{f}}, \chi_{\infty})$ comes from a *Größencharakter* χ (exercise 5).

The attempt to understand *Größencharaktere* in a conceptual way leads one to introduce **idèles**. In fact, all *Größencharaktere* arise as characters of the **idèle class group** of the number field K . We will not use this more abstract interpretation in what follows, but it will be explained at the end of this section.

(6.2) Proposition. Let χ be a *Größencharakter* mod m , and let m' be a divisor of m . Then the following conditions are equivalent.

- (i) χ is the restriction of a *Größencharakter* $\chi' : J^{m'} \rightarrow S^1$ mod m' .
- (ii) χ_f factors through $(\mathcal{O}/m')^*$.

Proof: (i) $\Rightarrow$ (ii). Let χ be the restriction of the *Größencharakter* $\chi' : J^{m'} \rightarrow S^1$, and let χ'_f, χ'_∞ be the pair of characters associated with χ' . Let $\tilde{\chi}_f$, resp. $\tilde{\chi}_\infty$, be the composite of

$$(\mathcal{O}/m)^* \longrightarrow (\mathcal{O}/m')^* \xrightarrow{\chi'_f} S^1, \quad \text{resp.} \quad \mathbf{R}^*/\mathcal{O}^m \longrightarrow \mathbf{R}^*/\mathcal{O}^{m'} \xrightarrow{\chi'_\infty} S^1.$$

We then find for $a \in \mathcal{O}^{(m)} \subseteq \mathcal{O}^{(m')}$:

$$\chi(a) = \chi'(a) = \chi'_f(a)\chi'_\infty(a) = \tilde{\chi}_f(a)\tilde{\chi}_\infty(a),$$

so that $\chi_f = \tilde{\chi}_f$ and $\chi_\infty = \tilde{\chi}_\infty$ because χ_f and χ_∞ are uniquely determined by χ . Thus χ_f factors through $(\mathcal{O}/m')^*$ (and χ_∞ through $\mathbf{R}^*/\mathcal{O}^{m'}$).

(ii) $\Rightarrow$ (i). Let χ_f be the composite of $(\mathcal{O}/m)^* \rightarrow (\mathcal{O}/m')^* \xrightarrow{\chi'_f} S^1$. In every class $a' \bmod P^{m'} \in J^{m'}/P^{m'}$, there is an ideal $\mathfrak{a} \in J^m$ which is relatively prime to m , i.e., $a' = \mathfrak{a}a$ for some $(a) \in P^{m'}$. We put

$$\chi'(a') = \chi(\mathfrak{a})\chi'_f(a)\chi'_\infty(a).$$

This definition does not depend on the choice of the ideal $\mathfrak{a} \in J^m$, for if $a' = \mathfrak{a}_1 a_1$, $\mathfrak{a}_1 \in J^m$, $(a_1) \in P^{m'}$, then one has $(\mathfrak{a}\mathfrak{a}_1^{-1}) \in J^m$, and

$$\begin{aligned} \chi(\mathfrak{a})\chi'_f(a)\chi'_\infty(a) &= \chi(\mathfrak{a})\chi((\mathfrak{a}\mathfrak{a}_1^{-1}))\chi_f(a^{-1}a_1)\chi_\infty(a^{-1}a_1)\chi'_f(a)\chi'_\infty(a) \\ &= \chi(\mathfrak{a}_1)\chi'_f(a_1)\chi'_\infty(a_1). \end{aligned}$$

The restriction of the character χ' from $J^{m'}$ to J^m is the *Größencharakter* χ of J^m , and if (a') is a principal ideal prime to m' and $a' = ab$, $(a) \in J^m$, $(b) \in P^{m'}$, then we have

$$\begin{aligned} \chi'((a')) &= \chi((a))\chi'((b)) = \chi((a))\chi'_f(b)\chi'_\infty(b) \\ &= \chi_f(a)\chi_\infty(a)\chi'_f(b)\chi'_\infty(b) = \chi'_f(ab)\chi'_\infty(ab) = \chi'_f(a')\chi'_\infty(a'). \end{aligned}$$

Thus χ' is a *Größencharakter* mod m' with corresponding pair of characters χ'_f, χ'_∞ . □

The *Größencharakter* χ mod m is called **primitive** if it is not the restriction of a *Größencharakter* χ' mod m' for any proper divisor $m'|m$.

According to (6.2), this is the case if and only if the character χ_f of $(\mathcal{O}/\mathfrak{m})^*$ is primitive in the sense that it does not factorize through $(\mathcal{O}/\mathfrak{m}')^*$ for any proper divisor $\mathfrak{m}'|\mathfrak{m}$. The **conductor** of χ is the smallest divisor $\mathfrak{f}$ of $\mathfrak{m}$ such that χ is the restriction of a *Größencharakter* mod $\mathfrak{f}$. By (6.2), $\mathfrak{f}$ is the conductor of χ_f , i.e., the smallest divisor of $\mathfrak{m}$ such that χ_f factors through $(\mathcal{O}/\mathfrak{f})^*$.

Let us now have a closer look at the character χ_f , and then at the character χ_∞ .

(6.3) Definition. Let χ_f be a character of $(\mathcal{O}/\mathfrak{m})^*$ and $y \in \mathfrak{m}^{-1}\mathfrak{d}^{-1}$, where $\mathfrak{d}$ is the different of $K|\mathbb{Q}$. Then we define the **Gauss sum** of χ_f to be

$$\tau_{\mathfrak{m}}(\chi_f, y) = \sum_{\substack{x \bmod \mathfrak{m} \\ (x, \mathfrak{m})=1}} \chi_f(x) e^{2\pi i \operatorname{Tr}(xy)},$$

where x varies over a system of representatives of $(\mathcal{O}/\mathfrak{m})^*$.

The Gauss sum does not depend on the choice of representatives x , for if $x' \equiv x \bmod \mathfrak{m}$, then $x'y - xy \in \mathfrak{m}\mathfrak{m}^{-1}\mathfrak{d}^{-1} = \mathfrak{d}^{-1} = \{a \in K \mid \operatorname{Tr}(a) \in \mathbb{Z}\}$, so that

$$\operatorname{Tr}(x'y) \equiv \operatorname{Tr}(xy) \bmod \mathbb{Z}$$

and therefore $e^{2\pi i \operatorname{Tr}(x'y)} = e^{2\pi i \operatorname{Tr}(xy)}$. The same argument shows that $\tau_{\mathfrak{m}}(\chi_f, y)$ depends only on the coset $y + \mathfrak{d}^{-1}$, i.e., it defines a function on the $\mathcal{O}/\mathfrak{m}$ -module $\mathfrak{m}^{-1}\mathfrak{d}^{-1}/\mathfrak{d}^{-1}$. In the case $K = \mathbb{Q}$, $\mathfrak{m} = (m)$, we get back the Gauss sum introduced in (2.5) by $\tau(\chi_f, n) = \tau_{\mathfrak{m}}(\chi_f, \frac{n}{m})$. We will have to define theta series and L -series attached to Hecke's *Größencharaktere* with a view to proving functional equations. For this, the following properties of Gauss sums will play a crucial rôle.

(6.4) Theorem. Let χ_f be a primitive character of $(\mathcal{O}/\mathfrak{m})^*$, let $y \in \mathfrak{m}^{-1}\mathfrak{d}^{-1}$ and $a \in \mathcal{O}$. Then one has

$$\tau_{\mathfrak{m}}(\chi_f, ay) = \begin{cases} \overline{\chi_f}(a) \tau_{\mathfrak{m}}(\chi_f, y), & \text{if } (a, \mathfrak{m}) = 1, \\ 0, & \text{if } (a, \mathfrak{m}) \neq 1, \end{cases}$$

and furthermore

$$|\tau_{\mathfrak{m}}(\chi_f, y)| = \sqrt{\mathfrak{N}(\mathfrak{m})}, \quad \text{if } (y\mathfrak{m}\mathfrak{d}, \mathfrak{m}) = 1.$$

The most difficult part of the theorem is the last claim. To prove it, we make the following preparations. For integral ideals $\mathfrak{a} = \mathfrak{p}_1^{v_1} \cdots \mathfrak{p}_r^{v_r}$, $v_i \geq 1$, consider the **Möbius function**

$$\mu(\mathfrak{a}) = \begin{cases} 1, & \text{if } r = 0, \text{ i.e., } \mathfrak{a} = (1), \\ (-1)^r, & \text{if } v_1 = \cdots = v_r = 1, \\ 0, & \text{otherwise.} \end{cases}$$

For this function we have the

(6.5) Proposition. *If $\mathfrak{a} \neq 1$, then $\sum_{\mathfrak{b}|\mathfrak{a}} \mu(\mathfrak{b}) = 0$.*

Proof: If $\mathfrak{a} = \mathfrak{p}_1^{v_1} \cdots \mathfrak{p}_r^{v_r}$, $v_i \geq 1$, then

$$\begin{aligned} \sum_{\mathfrak{b}|\mathfrak{a}} \mu(\mathfrak{b}) &= \mu(1) + \sum_i \mu(\mathfrak{p}_i) + \sum_{i_1 < i_2} \mu(\mathfrak{p}_{i_1} \mathfrak{p}_{i_2}) + \cdots + \mu(\mathfrak{p}_1 \cdots \mathfrak{p}_r) \\ &= 1 + \binom{r}{1}(-1) + \binom{r}{2}(-1)^2 + \cdots + \binom{r}{r}(-1)^r \\ &= (1 + (-1))^r = 0. \end{aligned}$$

□

Now, for $y \in \mathfrak{m}^{-1}\mathfrak{d}^{-1}$ and for every integral divisor $\mathfrak{a}$ of $\mathfrak{m}$, we look at the sums

$$T_{\mathfrak{a}}(y) = \sum_{\substack{x \bmod \mathfrak{m} \\ (x, \mathfrak{m}) = \mathfrak{a}}} e^{2\pi i \operatorname{Tr}(xy)} \quad \text{and} \quad S_{\mathfrak{a}}(y) = \sum_{\substack{x \bmod \mathfrak{m} \\ \mathfrak{a}|x}} e^{2\pi i \operatorname{Tr}(xy)}.$$

These sums do not depend on the choice of representatives x , for if $x' \equiv x \bmod \mathfrak{m}$, then $(x' - x)y \in \mathfrak{d}^{-1}$, hence $\operatorname{Tr}(x'y) \equiv \operatorname{Tr}(xy) \bmod \mathbb{Z}$. We find the

(6.6) Lemma. *One has*

$$T_1(y) = \sum_{\mathfrak{a}|\mathfrak{m}} \mu(\mathfrak{a}) S_{\mathfrak{a}}(y),$$

and for every divisor $\mathfrak{a}|\mathfrak{m}$,

$$S_{\mathfrak{a}}(y) = \begin{cases} \mathfrak{N}(\frac{\mathfrak{m}}{\mathfrak{a}}), & \text{if } y \in \mathfrak{a}^{-1}\mathfrak{d}^{-1}, \\ 0, & \text{if } y \notin \mathfrak{a}^{-1}\mathfrak{d}^{-1}. \end{cases}$$

Proof: In view of (6.5), we have

$$\sum_{\mathfrak{a}|\mathfrak{m}} \mu(\mathfrak{a}) S_{\mathfrak{a}}(y) = \sum_{\mathfrak{a}|\mathfrak{m}} \mu(\mathfrak{a}) \sum_{\substack{\mathfrak{b} \\ \mathfrak{a}|\mathfrak{b}|\mathfrak{m}}} T_{\mathfrak{b}}(y) = \sum_{\mathfrak{b}|\mathfrak{m}} T_{\mathfrak{b}}(y) \sum_{\mathfrak{a}|\mathfrak{b}} \mu(\mathfrak{a}) = T_1(y).$$

If $y \in \mathfrak{a}^{-1}\mathfrak{d}^{-1}$ and $\mathfrak{a} \mid x$, then $xy \in \mathfrak{d}^{-1}$, so that $\text{Tr}(xy) \in \mathbb{Z}$, i.e., all summands of $S_{\mathfrak{a}}$ are 1 and there are $\#(\mathfrak{a}/\mathfrak{m}) = \mathfrak{N}(\frac{\mathfrak{m}}{\mathfrak{a}})$ of them. If on the other hand $y \notin \mathfrak{a}^{-1}\mathfrak{d}^{-1}$, then we can find in $\mathfrak{a}/\mathfrak{m}$ a class $z \bmod \mathfrak{m}$ such that $zy \notin \mathfrak{d}^{-1}$, i.e., $\text{Tr}(zy) \notin \mathbb{Z}$, so that $e^{2\pi i \text{Tr}(zy)} \neq 1$, and we obtain

$$e^{2\pi i \text{Tr}(zy)} S_{\mathfrak{a}}(y) = \sum_{\substack{x \bmod \mathfrak{m} \\ \mathfrak{a} \mid x}} e^{2\pi i \text{Tr}((x+z)y)} = S_{\mathfrak{a}}(y),$$

since $x + z$ varies over all the classes of $\mathfrak{a}/\mathfrak{m}$ as x does, so that we do find $S_{\mathfrak{a}}(y) = 0$. $\square$

Proof of Theorem (6.4): Let $a \in \mathcal{O}$, $(a, \mathfrak{m}) = 1$. As x runs through a system of representatives of $(\mathcal{O}/\mathfrak{m})^*$, so does xa . We get

$$\begin{aligned} \tau_{\mathfrak{m}}(\chi_f, ay) &= \sum_{\substack{x \bmod \mathfrak{m} \\ (x, \mathfrak{m})=1}} \chi_f(x) e^{2\pi i \text{Tr}(xay)} \\ &= \overline{\chi_f}(a) \sum_{\substack{x \bmod \mathfrak{m} \\ (x, \mathfrak{m})=1}} \chi_f(xa) e^{2\pi i \text{Tr}(xay)} \\ &= \overline{\chi_f}(a) \tau_{\mathfrak{m}}(\chi_f, y). \end{aligned}$$

Let $(a, \mathfrak{m}) = \mathfrak{m}_1 \neq 1$. Since χ_f is primitive, we can find a class $b \bmod \mathfrak{m} \in (\mathcal{O}/\mathfrak{m})^*$ such that

$$\chi_f(b) \neq 1 \quad \text{and} \quad b \equiv 1 \bmod \frac{\mathfrak{m}}{\mathfrak{m}_1}.$$

As a consequence, $ab \equiv a \bmod \mathfrak{m}$, so that $aby - ay \in \mathfrak{d}^{-1}$, and by what we have just shown,

$$\overline{\chi_f}(b) \tau_{\mathfrak{m}}(\chi_f, ay) = \tau_{\mathfrak{m}}(\chi_f, bay) = \tau_{\mathfrak{m}}(\chi_f, ay).$$

Finally, in view of $\overline{\chi_f}(b) \neq 1$, we find $\tau_{\mathfrak{m}}(\chi_f, ay) = 0$.

As for the absolute value of the Gauss sum, we see from (6.6) that

$$\begin{aligned}
|\tau_m(\chi_f, y)|^2 &= \tau_m(\chi_f, y) \overline{\tau_m(\chi_f, y)} \\
&= \sum_{\substack{x \bmod m \\ (x, m)=1}} \tau_m(\chi_f, y) \overline{\chi_f(x)} e^{-2\pi i \operatorname{Tr}(xy)} \\
&= \sum_{\substack{x \bmod m \\ (x, m)=1}} \tau_m(\chi_f, xy) e^{-2\pi i \operatorname{Tr}(xy)} \\
&= \sum_{\substack{z \bmod m \\ (z, m)=1}} \sum_{\substack{x \bmod m \\ (x, m)=1}} \chi_f(z) e^{2\pi i \operatorname{Tr}(xy(z-1))} \\
&= \sum_{\substack{z \bmod m \\ (z, m)=1}} \chi_f(z) T_1(y(z-1)) \\
&= \sum_{\substack{z \bmod m \\ (z, m)=1}} \chi_f(z) \sum_{a|m} \mu(a) S_a(y(z-1)).
\end{aligned}$$

We now make use of the condition $(ym\mathfrak{d}, m) = 1$. It implies that

$$y(z-1) \in \mathfrak{a}^{-1}\mathfrak{d}^{-1} \iff z \equiv 1 \pmod{\frac{m}{a}}.$$

Indeed, if $z-1 \in \mathfrak{a}^{-1}m$, then $y(z-1) \in m^{-1}\mathfrak{d}^{-1}\mathfrak{a}^{-1}m = \mathfrak{a}^{-1}\mathfrak{d}^{-1}$. If on the other hand $z \not\equiv 1 \pmod{\frac{m}{a}}$, i.e., $\frac{m}{a} \nmid (z-1)$, then $v_p(z-1) < v_p(\frac{m}{a})$ for a prime divisor p of $\frac{m}{a}$. Since $(ym\mathfrak{d}, m) = 1$, we have $v_p(ym\mathfrak{d}) = 0$, so that $v_p(y) = -v_p(m) - v_p(\mathfrak{d})$ and

$$v_p(y(z-1)) < v_p(m) - v_p(a) + v_p(y) = -v_p(a) - v_p(\mathfrak{d}) = v_p(\mathfrak{a}^{-1}\mathfrak{d}^{-1}),$$

and thus $y(z-1) \notin \mathfrak{a}^{-1}\mathfrak{d}^{-1}$. This, together with (6.6), gives

$$|\tau_m(\chi_f, y)|^2 = \sum_{a|m} \mu(a) \mathfrak{N}\left(\frac{m}{a}\right) \sum_{\substack{z \bmod m \\ z \equiv 1 \pmod{m/a}} \chi_f(z).$$

For $a \neq 1$, the last character sum vanishes since χ_f is primitive, and therefore nonzero on the subgroup of $z \bmod m \in (\mathcal{O}/m)^*$ such that $z \equiv 1 \pmod{m/a}$: the sum reproduces itself under multiplication with a value $\chi_f(x) \neq 1$ of the character. So we finally have that $|\tau_m(\chi_f, y)|^2 = \mathfrak{N}(m)$. This proves all the statements of the theorem. $\square$

Having studied the characters χ_f of $(\mathcal{O}/m)^*$, we now turn to the characters χ_∞ of $\mathbf{R}^*$. They are given explicitly as follows.

(6.7) Proposition. *The characters λ of $\mathbf{R}^*$, i.e., the continuous homomorphisms*

$$\lambda : \mathbf{R}^* \longrightarrow S^1,$$

are given explicitly by

$$\lambda(x) = N(x^p |x|^{-p+iq}),$$

for some admissible $p \in \prod_{\tau} \mathbb{Z}$ (see § 3, p. 448) and a $q \in \mathbf{R}_{\pm}$. p and q are uniquely determined by λ .

Proof: For every $x \in \mathbf{R}^*$ we may write $x = \frac{x}{|x|} |x|$, and obtain in this way a decomposition

$$\mathbf{R}^* = \mathbf{U} \times \mathbf{R}_+^*,$$

where $\mathbf{U} = \{x \in \mathbf{R}^* \mid |x| = 1\}$. It therefore suffices to determine separately the characters of $\mathbf{U}$ and those of $\mathbf{R}_+^*$. We write ρ instead of τ for elements of $\text{Hom}(K, \mathbb{C})$ to indicate that $\tau = \bar{\tau}$, and we choose an element σ from each pair $\{\tau, \bar{\tau}\}$ such that $\tau \neq \bar{\tau}$. Then we have

$$\mathbf{U} = \left[\prod_{\tau} S^1 \right]^+ = \prod_{\rho} \{\pm 1\} \times \prod_{\sigma} [S^1 \times S^1]^+,$$

and $S^1 \rightarrow [S^1 \times S^1]^+$, $x_{\sigma} \mapsto (x_{\sigma}, \bar{x}_{\sigma})$, is a topological isomorphism. The characters of $\{\pm 1\}$ correspond one-to-one to exponentiating by a $p_{\rho} \in \{0, 1\}$, and the characters of S^1 correspond one-to-one to the mappings $x_{\sigma} \mapsto x_{\sigma}^k$, for $k \in \mathbb{Z}$. From the correspondence $k \mapsto (k, 0)$, resp. $(0, -k)$, for $k \geq 0$, resp. $k \leq 0$, we obtain the characters of $[S^1 \times S^1]^+$ in a one-to-one way from the pairs $(p_{\tau}, p_{\bar{\tau}})$ with $p_{\tau}, p_{\bar{\tau}} \geq 0$ and $p_{\tau} p_{\bar{\tau}} = 0$. The characters of $\mathbf{U}$ are therefore given by

$$\lambda(x) = N(x^p),$$

with a uniquely determined admissible $p \in \prod_{\tau} \mathbb{Z}$.

The characters of $\mathbf{R}_+^*$ are obtained via the topological isomorphism

$$\log : \mathbf{R}_+^* \longrightarrow \mathbf{R}_{\pm}.$$

Writing as above

$$\mathbf{R}_{\pm} = \prod_{\rho} \mathbb{R} \times \prod_{\sigma} [\mathbb{R} \times \mathbb{R}]^+,$$

and observing the isomorphism $[\mathbb{R} \times \mathbb{R}]^+ \xrightarrow{\sim} \mathbb{R}$, $(x_{\sigma}, x_{\sigma}) \mapsto 2x_{\sigma}$, we see that a character of $\mathbf{R}_{\pm}$ corresponds one-to-one to a system (q_{ρ}, q_{σ}) via the rule

$$x \mapsto \prod_{\rho} e^{iq_{\rho} x_{\rho}} \prod_{\sigma} e^{2iq_{\sigma} x_{\sigma}}.$$

It is therefore given by an element $q \in \mathbf{R}_{\pm}$ via $x \mapsto N(e^{iqx})$. The isomorphism $\log$ then gives a character λ of $\mathbf{R}_+^*$ via $y \mapsto N(e^{iq \log y}) =$

$N(y^{iq})$, with a uniquely determined $q \in \mathbf{R}_{\pm}$. In view of the decomposition $x = \frac{x}{|x|}|x|$, we finally obtain the characters λ of $\mathbf{R}^*$ as

$$\lambda(x) = N\left(\left(\frac{x}{|x|}\right)^p |x|^{iq}\right) = N(x^p |x|^{-p+iq}). \quad \square$$

If the character χ_{∞} associated to the *Größencharakter* $\chi : J^{\mathfrak{m}} \rightarrow S^1$ is given by

$$\chi_{\infty}(x) = N(x^p |x|^{-p+iq}),$$

then we say that χ is of **type** (p, q) , and we call $p - iq$ the **exponent** of χ . Since χ_{∞} factors through $\mathbf{R}^*/\mathcal{O}^{\mathfrak{m}}$, not all exponents actually occur (see exercise 3).

The class of all *Größencharaktere* subsumes in particular the generalized *Dirichlet characters* defined as follows. To the **module**

$$\mathfrak{m} = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{n_{\mathfrak{p}}},$$

we associate the **ray class group** $J^{\mathfrak{m}}/P^{\mathfrak{m}} \bmod \mathfrak{m}$ (see chap. VI, § 1). Here $J^{\mathfrak{m}}$ is the group of all ideals relatively prime to $\mathfrak{m}$, and $P^{\mathfrak{m}}$ is the group of fractional principal ideals (a) such that

$$a \equiv 1 \bmod \mathfrak{m} \quad \text{and } a \text{ totally positive.}$$

This last condition means that $\tau a > 0$ for every real embedding $\tau : K \rightarrow \mathbb{R}$.

(6.8) Definition. A **Dirichlet character** $\bmod \mathfrak{m}$ is a character

$$\chi : J^{\mathfrak{m}}/P^{\mathfrak{m}} \longrightarrow S^1$$

of the ray class group $\bmod \mathfrak{m}$, i.e., a character $\chi : J^{\mathfrak{m}} \rightarrow S^1$ such that $\chi(P^{\mathfrak{m}}) = 1$.

The **conductor** of a Dirichlet character $\chi \bmod \mathfrak{m}$ is defined to be the smallest module $\mathfrak{f}$ dividing $\mathfrak{m}$ such that χ factors through $J^{\mathfrak{f}}/P^{\mathfrak{f}}$.

(6.9) Proposition. The Dirichlet characters $\chi \bmod \mathfrak{m}$ are precisely the *Größencharaktere* $\bmod \mathfrak{m}$ of type $(p, 0)$, $p = (p_{\tau})$, such that $p_{\tau} = 0$ for all complex τ . In other words, one has

$$\chi((a)) = \chi_{\mathfrak{f}}(a) N\left(\left(\frac{a}{|a|}\right)^p\right),$$

for some character $\chi_{\mathfrak{f}}$ of $(\mathcal{O}/\mathfrak{m})^*$. The conductor of the Dirichlet character is at the same time also the conductor of the corresponding *Größencharakter*.

Proof: Let χ be a *Größencharakter* mod $\mathfrak{m}$ with corresponding characters χ_f, χ_∞ of $(\mathcal{O}/\mathfrak{m})^*, \mathbf{R}^*/\mathcal{O}^\mathfrak{m}$, such that χ_∞ is of type $(p, 0)$ with $p_\tau = 0$ for τ complex. For totally positive $a \in \mathcal{O}$ such that $a \equiv 1 \pmod{\mathfrak{m}}$, we then obviously have $\chi_f(a) = 1$, and $\chi_\infty(a) = 1$, and then $\chi((a)) = \chi_f(a)\chi_\infty(a) = 1$. Therefore χ factorizes through $J^\mathfrak{m}/P^\mathfrak{m}$, and is thus a Dirichlet character mod $\mathfrak{m}$.

Conversely, let χ be a Dirichlet character mod $\mathfrak{m}$, i.e., a character of $J^\mathfrak{m}$ such that $\chi(P^\mathfrak{m}) = 1$. Let $K^\mathfrak{m} = \{a \in K^* \mid a \equiv 1 \pmod{\mathfrak{m}}\}$, $K_+^\mathfrak{m} = \{a \in K^\mathfrak{m} \mid a \text{ totally positive}\}$ and $\mathbf{R}_{(+)}^* = \{(x_\tau) \in \mathbf{R}^* \mid x_\tau > 0 \text{ for } \tau \text{ real}\}$. Then we have an isomorphism

$$K^\mathfrak{m}/K_+^\mathfrak{m} \longrightarrow \mathbf{R}^*/\mathbf{R}_{(+)}^* \cong \prod_{p \text{ real}} \{\pm 1\}.$$

Then the composite

$$K^\mathfrak{m}/K_+^\mathfrak{m} \xrightarrow{(\cdot)} J^\mathfrak{m}/P^\mathfrak{m} \xrightarrow{\chi} S^1$$

defines a character of $\mathbf{R}^*/\mathbf{R}_{(+)}^*$. It is induced by a character χ_∞ of $\mathbf{R}^*$ which — because $\chi_\infty(\mathbf{R}_{(+)}^*) = 1$ — is of the form $\chi_\infty(x) = N((\frac{x}{|x|})^p)$ with $p = (p_\tau)$, $p_\tau \in \{0, 1\}$ for τ real, and $p_\tau = 0$ for τ complex. We have $\chi((a)) = \chi_\infty(a)$ for $a \in K^\mathfrak{m}$, and

$$\chi_f(a) = \chi((a))\chi_\infty(a)^{-1}$$

gives us a character of $(\mathcal{O}/\mathfrak{m})^*$. Therefore χ is indeed a *Größencharakter* of the type claimed.

Let $\mathfrak{f}$ be the conductor of the Dirichlet character χ mod $\mathfrak{m}$, and let $\mathfrak{f}'$ be the conductor of the corresponding *Größencharakter* mod $\mathfrak{m}$. $\chi : J^\mathfrak{m}/P^\mathfrak{m} \rightarrow S^1$ is then induced by a character $\chi' : J^{\mathfrak{f}}/P^{\mathfrak{f}} \rightarrow S^1$, so the *Größencharakter* $\chi : J^\mathfrak{m} \rightarrow S^1$ mod $\mathfrak{m}$ is the restriction of the *Größencharakter* $\chi' : J^{\mathfrak{f}} \rightarrow S^1$. This implies that $\mathfrak{f}' \mid \mathfrak{f}$. On the other hand, the *Größencharakter* $\chi : J^\mathfrak{m} \rightarrow S^1$ is the restriction of a *Größencharakter* $\chi'' : J^{\mathfrak{f}'} \rightarrow S^1$, so χ_f is the composite of $(\mathcal{O}/\mathfrak{m})^* \rightarrow (\mathcal{O}/\mathfrak{f}')^* \xrightarrow{\chi_f''} S^1$ (see (6.2)). By the above, χ'' gives a character $J^{\mathfrak{f}'} / P^{\mathfrak{f}'} \rightarrow S^1$ such that the Dirichlet character $\chi : J^\mathfrak{m} / P^\mathfrak{m} \rightarrow S^1$ factors through $J^{\mathfrak{f}'} / P^{\mathfrak{f}'}$. Hence $\mathfrak{f} \mid \mathfrak{f}'$, so that $\mathfrak{f} = \mathfrak{f}'$. $\square$

(6.10) Corollary. *The characters of the ideal class group $Cl_K = J/P$, i.e., the characters $\chi : J \rightarrow S^1$ such that $\chi(P) = 1$, are precisely the Größencharaktere χ mod 1 satisfying $\chi_\infty = 1$.*

Proof: For $\mathfrak{m} = 1$ we have $(\mathcal{O}/\mathfrak{m})^* = \{1\}$. A character χ of J/P is a *Größencharakter* mod 1. The associated character χ_f is trivial, so $\chi_\infty(a) = \chi_f(a)^{-1}\chi((a)) = 1$, and thus $\chi_\infty = 1$, because K^* is dense in $\mathbf{R}^*$. If conversely χ is a *Größencharakter* mod 1 satisfying $\chi_\infty = 1$, then

$$\chi((a)) = \chi_f(a)\chi_\infty(a) = \chi_f(a) = 1,$$

for $a \in K^*$. Therefore $\chi(P) = 1$, and χ is a character of the ideal class group. $\square$

To conclude this section, let us study the relation of *Größencharaktere* to characters of the idèle class group.

(6.11) Definition. A **Hecke character** is a character of the idèle class group $C = I/K^*$ of the number field K , i.e., a continuous homomorphism

$$\chi : I \longrightarrow S^1$$

of the idèle group $I = \prod_{\mathfrak{p}} K_{\mathfrak{p}}^*$ such that $\chi(K^*) = 1$.

In order to deal with Hecke characters concretely, consider an integral ideal $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$ of K , i.e., $n_{\mathfrak{p}} \geq 0$ and $n_{\mathfrak{p}} = 0$ for $\mathfrak{p} \mid \infty$. We associate to this ideal the *subgroup* $\bar{I}^{\mathfrak{m}}$ of I ,

$$\bar{I}^{\mathfrak{m}} = I_f^{\mathfrak{m}} \times I_{\infty} \quad \text{where} \quad I_f^{\mathfrak{m}} = \prod_{\mathfrak{p} \nmid \infty} U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}, \quad I_{\infty} = \prod_{\mathfrak{p} \mid \infty} K_{\mathfrak{p}}^*.$$

If $\mathfrak{p} \nmid \infty$, then $U_{\mathfrak{p}}^{(n)}$ is the group of units $U_{\mathfrak{p}}$ if $n = 0$, and the n -th group of higher units for $n \geq 1$. We interpret I_{∞} as the multiplicative group $\mathbf{R}^*$ of the $\mathbb{R}$ -algebra $\mathbf{R} = K \otimes_{\mathbb{Q}} \mathbb{R} = \prod_{\mathfrak{p} \mid \infty} K_{\mathfrak{p}}$. Observe that $\bar{I}^{\mathfrak{m}}$ differs slightly from the congruence subgroup $I^{\mathfrak{m}} = \prod_{\mathfrak{p}} U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$ introduced in chap. VI, § 1, in that, for real $\mathfrak{p}$, we have the factor $U_{\mathfrak{p}}^{(0)} = \mathbb{R}_+^*$ instead of the component $K_{\mathfrak{p}}^*$. The effect is that $I/\bar{I}^{\mathfrak{m}}K^*$ is not the ray class group $J^{\mathfrak{m}}/P^{\mathfrak{m}} \bmod \mathfrak{m}$, but isomorphic to the quotient $J^{\mathfrak{m}}/\bar{P}^{\mathfrak{m}}$ by the group $\bar{P}^{\mathfrak{m}}$ of all principal ideals (a) such that $a \equiv 1 \bmod \mathfrak{m}$ — this is seen as in chap. VI, (1.9). We will refer to $J^{\mathfrak{m}}/\bar{P}^{\mathfrak{m}}$ as the *small ray class group*.

We call $\mathfrak{m}$ a **module of definition** for the Hecke character χ if

$$\chi(I_f^{\mathfrak{m}}) = 1.$$

Every Hecke character admits a module of definition, since the image of $\chi : \prod_{\mathfrak{p} \nmid \infty} U_{\mathfrak{p}} \rightarrow S^1$ is a compact and totally disconnected subgroup of

S^1 , hence finite, and so the kernel has to contain a subgroup of the form $\prod_{p \nmid \infty} U_p^{(n_p)}$ where $n_p = 0$ for almost all p . For it we can take the ideal $\mathfrak{m} = \prod_{p \nmid \infty} p^{n_p}$ as a module of definition.

Since $\chi(I_f^{\mathfrak{m}}) = 1$, the character $\chi : C = I/K^* \rightarrow S^1$ induces a character

$$\chi : C(\mathfrak{m}) \longrightarrow S^1$$

of the group

$$C(\mathfrak{m}) = I/I_f^{\mathfrak{m}}K^*.$$

But it will not in general factor through the small ray class group $I/\bar{I}^{\mathfrak{m}}K^* \cong J^{\mathfrak{m}}/\bar{P}^{\mathfrak{m}}$ (see chap. VI, (1.7), (1.9)), which bears the following relation to $C(\mathfrak{m})$.

(6.12) Proposition. *There is an exact sequence*

$$1 \longrightarrow \mathbf{R}^*/\mathcal{O}^{\mathfrak{m}} \longrightarrow C(\mathfrak{m}) \longrightarrow J^{\mathfrak{m}}/\bar{P}^{\mathfrak{m}} \longrightarrow 1.$$

Proof: The claim follows immediately from the two exact sequences

$$1 \longrightarrow \bar{I}^{\mathfrak{m}}K^*/I_f^{\mathfrak{m}}K^* \longrightarrow I/I_f^{\mathfrak{m}}K^* \longrightarrow I/\bar{I}^{\mathfrak{m}}K^* \longrightarrow 1,$$

$$1 \longrightarrow \bar{I}^{\mathfrak{m}} \cap K^*/I_f^{\mathfrak{m}} \cap K^* \longrightarrow \bar{I}^{\mathfrak{m}}/I_f^{\mathfrak{m}} \longrightarrow \bar{I}^{\mathfrak{m}}K^*/I_f^{\mathfrak{m}}K^* \longrightarrow 1.$$

In the second one, one has $\bar{I}^{\mathfrak{m}} \cap K^* = \mathcal{O}^{\mathfrak{m}}$, $I_f^{\mathfrak{m}} \cap K^* = 1$ and $\bar{I}^{\mathfrak{m}}/I_f^{\mathfrak{m}} = I_{\infty} = \mathbf{R}^*$, and so $\bar{I}^{\mathfrak{m}}K^*/I_f^{\mathfrak{m}}K^* = \mathbf{R}^*/\mathcal{O}^{\mathfrak{m}}$. $\square$

Given a Hecke character χ with module of definition $\mathfrak{m}$, we may now construct a *Größencharakter* mod $\mathfrak{m}$ as follows. For every $p \nmid \infty$, we choose a fixed prime element π_p of K_p and obtain a homomorphism

$$c : J^{\mathfrak{m}} \longrightarrow C(\mathfrak{m})$$

which maps a prime ideal $p \nmid \mathfrak{m}$ to the class of the idèle $\langle \pi_p \rangle = (\dots, 1, 1, \pi_p, 1, 1, \dots)$. This mapping does not depend on the choice of the prime elements, since the idèles $\langle u_p \rangle$, $u_p \in U_p$, for $p \nmid \mathfrak{m}$, lie in $I_f^{\mathfrak{m}}$. Taking the composite map

$$J^{\mathfrak{m}} \xrightarrow{c} C(\mathfrak{m}) \xrightarrow{\chi} S^1$$

yields a 1–1 correspondence between Hecke characters with module of definition $\mathfrak{m}$ and *Größencharaktere* mod $\mathfrak{m}$. The reason for this is the following

(6.13) Proposition. *There is a canonical exact sequence*

$$1 \longrightarrow K^{(\mathfrak{m})}/\mathcal{O}^{\mathfrak{m}} \xrightarrow{\delta} J^{\mathfrak{m}} \times (\mathcal{O}/\mathfrak{m})^* \times \mathbf{R}^*/\mathcal{O}^{\mathfrak{m}} \xrightarrow{f} C(\mathfrak{m}) \longrightarrow 1,$$

where δ is given by

$$\delta(a) = ((a)^{-1}, a \bmod \mathfrak{m}, a \bmod \mathcal{O}^{\mathfrak{m}}).$$

Proof: For every $a \in K^{(\mathfrak{m})}$, let $\widehat{a} \in I$ be the idèle with components $\widehat{a}_{\mathfrak{p}} = a$ for $\mathfrak{p} \nmid \mathfrak{m}\infty$ and $\widehat{a}_{\mathfrak{p}} = 1$ for $\mathfrak{p} \mid \mathfrak{m}\infty$. It is then obvious that

$$c((a)) = \widehat{a} \bmod I_f^{\mathfrak{m}} K^*.$$

Let us decompose the principal idèle a according to its components in $I = I_f \times I_{\infty}$ as a product $a = a_f a_{\infty}$, and define the homomorphisms

$$\varphi : (\mathcal{O}/\mathfrak{m})^* \longrightarrow C(\mathfrak{m}), \quad \psi : \mathbf{R}^*/\mathcal{O}^{\mathfrak{m}} \longrightarrow C(\mathfrak{m})$$

by

$$\varphi(a) = \widehat{a} a_{\infty} \bmod I_f^{\mathfrak{m}} K^*, \quad \psi(b) = b^{-1} \bmod I_f^{\mathfrak{m}} K^*,$$

where every $b \in \mathbf{R}^* = I_{\infty}$ is considered as an idèle in I . For $a \in \mathcal{O}$, $a \equiv 1 \bmod \mathfrak{m}$, we have $a_f \widehat{a}^{-1} \in I_f^{\mathfrak{m}} \subseteq I$, so we get in $C(\mathfrak{m})$ the equation $\varphi(a) = [\widehat{a} a_{\infty}] = [a_f a_{\infty}] = [a] = 1$, where $[\]$ indicates taking classes. This shows that φ is well-defined. For every $\varepsilon \in \mathcal{O}^{\mathfrak{m}}$, one has $\varepsilon_f \in I_f^{\mathfrak{m}}$, so $[\varepsilon_{\infty}] = [\varepsilon_{\infty} \varepsilon_f] = [\varepsilon] = 1$ in $C(\mathfrak{m})$, and thus $\psi(\varepsilon_{\infty}) = 1$. Consequently ψ is well-defined. We now define the homomorphism

$$f : J^{\mathfrak{m}} \times (\mathcal{O}/\mathfrak{m})^* \times \mathbf{R}^*/\mathcal{O}^{\mathfrak{m}} \longrightarrow C(\mathfrak{m})$$

by

$$f((a, a \bmod \mathfrak{m}, b \bmod \mathcal{O}^{\mathfrak{m}})) = c(a)\varphi(a)\psi(b),$$

and we show that the resulting sequence is exact. The homomorphism δ is clearly injective. For $a \in K^{(\mathfrak{m})}$ one has

$$f(\delta(a)) = c((a))^{-1} \varphi(a) \psi(a) = \widehat{a}^{-1} \widehat{a} a_{\infty} a_{\infty}^{-1} \bmod I_f^{\mathfrak{m}} K^* = 1,$$

so that $f \circ \delta = 1$. Conversely, let

$$f((a, a \bmod \mathfrak{m}, b \bmod \mathcal{O}^{\mathfrak{m}})) = c(a)\varphi(a)\psi(b) = 1,$$

and let $\mathfrak{a} = \prod_{\mathfrak{p} \nmid \mathfrak{m}\infty} \mathfrak{p}^{v_{\mathfrak{p}}}$. Then

$$c(\mathfrak{a}) = \gamma \bmod I_f^{\mathfrak{m}} K^*$$

for some idèle γ with components $\gamma_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}}$ for $\mathfrak{p} \nmid \mathfrak{m}\infty$, and $\gamma_{\mathfrak{p}} = 1$ for $\mathfrak{p} \mid \mathfrak{m}\infty$. This yields an identity

$$\gamma \widehat{a} a_{\infty} b^{-1} = \xi x \quad \text{with} \quad \xi \in I_f^{\mathfrak{m}} \quad \text{and} \quad x \in K^*.$$

For $\mathfrak{p} \nmid m\infty$ one has $(\gamma \widehat{a} a_\infty b^{-1})_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}} a = \xi_{\mathfrak{p}} x$ in $K_{\mathfrak{p}}$, and so $v_{\mathfrak{p}} = v_{\mathfrak{p}}(a^{-1}x)$. For $\mathfrak{p} \mid m$ one has $(\gamma \widehat{a} a_\infty b^{-1})_{\mathfrak{p}} = 1 = \xi_{\mathfrak{p}} x$, so that $x \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$, and also $0 = v_{\mathfrak{p}} = v_{\mathfrak{p}}(a^{-1}x)$ since a is relatively prime to m . This gives

$$a = (ax^{-1}).$$

As $x \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$, one has $x \equiv 1 \pmod{m}$, hence

$$\varphi(ax^{-1}) = \varphi(a).$$

Finally, for $\mathfrak{p} \mid \infty$ we find $(\gamma \widehat{a} a_\infty b^{-1})_{\mathfrak{p}} = ab_{\mathfrak{p}}^{-1} = x$ in $K_{\mathfrak{p}}$, so that $b = a_\infty x^{-1}$, and thus

$$\psi(ax^{-1}) = \psi(b).$$

So we have

$$(a, a \bmod m, b \bmod \mathfrak{o}^m) = ((ax^{-1}), ax^{-1} \bmod m, ax^{-1} \bmod \mathfrak{o}^m),$$

and this shows the exactness of our sequence in the middle.

The surjectivity of f is proved as follows. Let $\alpha \bmod I_{\mathfrak{f}}^m K^*$ be a class in $C(m)$. By the approximation theorem, we may modify the representing idèle α , multiplying it by a suitable $x \in K^*$, in such a way that $\alpha_{\mathfrak{p}} \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$ for $\mathfrak{p} \mid m$. Let $a = \prod_{\mathfrak{p} \nmid m\infty} \mathfrak{p}^{v_{\mathfrak{p}}(\alpha_{\mathfrak{p}})}$. Then we have

$$c(a) = \gamma \bmod I_{\mathfrak{f}}^m K^*,$$

where the idèle γ has components $\gamma_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}(\alpha_{\mathfrak{p}})} = \varepsilon_{\mathfrak{p}} \alpha_{\mathfrak{p}}$, $\varepsilon_{\mathfrak{p}} \in U_{\mathfrak{p}}$, for $\mathfrak{p} \nmid m\infty$, and $\gamma_{\mathfrak{p}} = 1$ for $\mathfrak{p} \mid m\infty$. This gives $\gamma \alpha^{-1} \alpha_\infty \in I_{\mathfrak{f}}^m$, and if we define $b = \alpha_\infty^{-1}$, then $f((a, 1 \bmod m, b \bmod \mathfrak{o}^m)) = \gamma b^{-1} \equiv \gamma \alpha_\infty \equiv \alpha \bmod I_{\mathfrak{f}}^m K^*$. $\square$

By the preceding proposition, the characters of $C(m)$ correspond 1–1 to the characters of $J^m \times (\mathfrak{o}/m)^* \times \mathbf{R}^*/\mathfrak{o}^m$ that vanish on $\delta(K^{(m)}/\mathfrak{o}^m)$, i.e., to the triples $\chi, \chi_{\mathfrak{f}}, \chi_\infty$ of characters of J^m , resp. $(\mathfrak{o}/m)^*$, resp. $\mathbf{R}^*/\mathfrak{o}^m$, such that

$$\chi((a))^{-1} \chi_{\mathfrak{f}}(a \bmod m) \chi_\infty(a \bmod \mathfrak{o}^m) = 1$$

for $a \in K^{(m)}$. This makes χ a *Größencharakter* mod m , and since $\chi_{\mathfrak{f}}$ and χ_∞ are uniquely determined by χ , we obtain the

(6.14) Corollary. *The correspondence $\chi \mapsto \chi \circ c$ is 1–1 between characters χ of $C(m)$, i.e., Hecke characters with module of definition m , and Größencharaktere mod m .*

Exercise 1. Let $\mathfrak{m} = \prod_{i=1}^r \mathfrak{m}_i$ be a decomposition of $\mathfrak{m}$ into integral ideals which are pairwise relatively prime. Then one has the decompositions

$$(\mathcal{O}/\mathfrak{m})^* \cong \prod_{i=1}^r (\mathcal{O}/\mathfrak{m}_i)^*$$

and

$$\mathfrak{m}^{-1}\mathfrak{d}^{-1}/\mathfrak{d}^{-1} \cong \bigoplus_{i=1}^r \mathfrak{m}_i^{-1}\mathfrak{d}^{-1}/\mathfrak{d}^{-1}.$$

Let χ_f be a character of $(\mathcal{O}/\mathfrak{m})^*$, and let χ_{fi} be the characters of $(\mathcal{O}/\mathfrak{m}_i)^*$ defined by χ_f . If $y \in \mathfrak{m}^{-1}\mathfrak{d}^{-1}/\mathfrak{d}^{-1}$, and if $y_i \in \mathfrak{m}_i^{-1}\mathfrak{d}^{-1}/\mathfrak{d}^{-1}$ are the components of y with respect to the above decomposition, then

$$\tau_{\mathfrak{m}}(\chi_f, y) = \prod_{i=1}^r \tau_{\mathfrak{m}_i}(\chi_{fi}, y_i).$$

Exercise 2. Prove the **Möbius inversion formula**: let $f(\mathfrak{a})$ be any function of integral ideals $\mathfrak{a}$ with values in an additive abelian group, and let

$$g(\mathfrak{a}) = \sum_{\mathfrak{b}|\mathfrak{a}} f(\mathfrak{b}).$$

Then one has

$$f(\mathfrak{a}) = \sum_{\mathfrak{b}|\mathfrak{a}} \mu\left(\frac{\mathfrak{a}}{\mathfrak{b}}\right) g(\mathfrak{b}).$$

Exercise 3. Which of the characters $\lambda(x) = N(x^p|x|^{-p+iq})$ of $\mathbf{R}^*$ are characters of $\mathbf{R}^*/\mathcal{O}^m$?

Exercise 4. The characters of the “small ray class group” $J^m/\bar{P}^m \bmod \mathfrak{m}$ are the *Größencharaktere* $\bmod \mathfrak{m}$ such that $\chi_{\infty} = 1$.

Exercise 5. Show that every pair of characters $\chi_f : (\mathcal{O}/\mathfrak{m})^* \rightarrow S^1$ and $\chi_{\infty} : \mathbf{R}^*/\mathcal{O}^m \rightarrow S^1$ such that

$$\chi_f(\varepsilon)\chi_{\infty}(\varepsilon) = 1 \quad \text{for all } \varepsilon \in \mathcal{O}^*$$

comes from a *Größencharakter* $\bmod \mathfrak{m}$.

Exercise 6. Show that the homomorphism $c : J^m \rightarrow C(\mathfrak{m})$ is injective.

§ 7. Theta Series of Algebraic Number Fields

The group P of fractional principal ideals (a) is constituted from the elements $a \in K^*$, and it sits in the exact sequence

$$1 \longrightarrow \mathcal{O}^* \longrightarrow K^* \longrightarrow P \longrightarrow 1.$$

In order to form the theta series we will need, let us now extend K^* to a group $\widehat{K}^*$ whose elements represent *all* fractional ideals $\mathfrak{a} \in J$.

(7.1) Proposition. *There is a commutative exact diagram*

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathcal{O}^* & \longrightarrow & K^* & \xrightarrow{(\cdot)} & P \longrightarrow 1 \\ & & \parallel & & \downarrow & & \downarrow \\ 1 & \longrightarrow & \mathcal{O}^* & \longrightarrow & \widehat{K}^* & \xrightarrow{(\cdot)} & J \longrightarrow 1 \end{array}$$

with a subgroup $\widehat{K}^* \subseteq \mathbf{C}^*$ containing K^* such that $|a| \in \mathbf{R}_+^*$, and

$$\mathfrak{N}((a)) = |N(a)|$$

for all $a \in \widehat{K}^*$.

Proof: Let the ideal class group J/P be given by a basis $[b_1], \dots, [b_r]$, and choose, for every one of these basic classes, an ideal $\mathfrak{b}_1, \dots, \mathfrak{b}_r$. Then every fractional ideal $\mathfrak{a} \in J$ can be written in the form

$$\mathfrak{a} = a \mathfrak{b}_1^{v_1} \cdots \mathfrak{b}_r^{v_r}$$

where $a \in K^*$ is well-determined up to a unit $\varepsilon \in \mathcal{O}^*$, and the exponents $v_i \bmod h_i$ are uniquely determined, h_i being the order of $[b_i]$ in J/P . Let $\mathfrak{b}_i^{h_i} = (b_i)$. For every $\tau \in \text{Hom}(K, \mathbb{C})$, we choose a fixed root

$$\widehat{b}_{i\tau} = \sqrt[h_i]{\tau b_i}$$

in $\mathbb{C}$ in such a way that $\widehat{b}_{i\bar{\tau}} = \overline{\widehat{b}_{i\tau}}$ whenever τ is complex. We define $\widehat{K}^*$ to be the subgroup of $\mathbf{C}^*$ generated by K^* and by the elements $\widehat{b}_i = (\widehat{b}_{i\tau}) \in \mathbf{C}^*$. Each class $[b] \in J/P$ contains a uniquely determined ideal of the form

$$\mathfrak{b} = \mathfrak{b}_1^{v_1} \cdots \mathfrak{b}_r^{v_r} \quad \text{with} \quad 0 \leq v_i < h_i,$$

and we consider the mapping

$$f : J/P \longrightarrow \widehat{K}^*/K^*, \quad f([b]) = \widehat{b}_1^{v_1} \cdots \widehat{b}_r^{v_r} \bmod K^*.$$

It is a homomorphism, for if $\mathfrak{b} = \mathfrak{b}_1^{v_1} \cdots \mathfrak{b}_r^{v_r}$ and $\mathfrak{b}' = \mathfrak{b}_1^{v'_1} \cdots \mathfrak{b}_r^{v'_r}$, and if $v_i + v'_i = \mu_i + \lambda_i h_i$, $0 \leq \mu_i < h_i$, then $\mathfrak{b}_1^{\mu_1} \cdots \mathfrak{b}_r^{\mu_r}$ is the ideal belonging to the class $[b][b']$, and

$$\begin{aligned} f([b][b']) &= \widehat{b}_1^{\mu_1} \cdots \widehat{b}_r^{\mu_r} \equiv \widehat{b}_1^{\mu_1} \cdots \widehat{b}_r^{\mu_r} \widehat{b}_1^{\lambda_1} \cdots \widehat{b}_r^{\lambda_r} \\ &= (\widehat{b}_1^{v_1} \cdots \widehat{b}_r^{v_r})(\widehat{b}_1^{v'_1} \cdots \widehat{b}_r^{v'_r}) \bmod K^* = f([b])f([b']). \end{aligned}$$

f is clearly surjective. To show the injectivity, let $\widehat{b}_1^{v_1} \cdots \widehat{b}_r^{v_r} = a \in K^*$, and let $h = h_1 \cdots h_r$ be the class number of K . Then we have for

the ideal $\mathfrak{a} = a^{-1}b_1^{v_1} \cdots b_r^{v_r} \in J$ that $\mathfrak{a}^h = a^{-h}(b_1^{v_1 h/h_1} \cdots b_r^{v_r h/h_r}) = a^{-h}(\widehat{b}_1^{v_1} \cdots \widehat{b}_r^{v_r})^h = (1)$. Since J is torsion-free, it follows that $\mathfrak{a} = (1)$, and so $b_1^{v_1} \cdots b_r^{v_r} = (a) \in P$. From this we deduce that every element $\widehat{a} \in \widehat{K}^*$ admits a unique representation

$$\widehat{a} = a\widehat{b}_1^{v_1} \cdots \widehat{b}_r^{v_r}, \quad 0 \leq v_i < h_i, \quad a \in K^*.$$

We define a map

$$(\) : \widehat{K}^* \longrightarrow J$$

by

$$\widehat{a} = a\widehat{b}_1^{v_1} \cdots \widehat{b}_r^{v_r} \longmapsto (\widehat{a}) = ab_1^{v_1} \cdots b_r^{v_r}.$$

Arguing as above, we see that this is a homomorphism. It is surjective and obviously has kernel $\mathcal{O}^*$. Finally we have that $|\widehat{b}_i| = (|\widehat{b}_{i\tau}|) \in \mathbf{R}_+^*$ and

$$\mathfrak{N}((\widehat{b}_i))^{h_i} = \mathfrak{N}(b_i^{h_i}) = |N(b_i)| = \left| \prod_{\tau} \tau b_i \right| = \left| \prod_{\tau} \widehat{b}_{i\tau}^{h_i} \right| = |N(\widehat{b}_i)|^{h_i},$$

so that $\mathfrak{N}((\widehat{b}_i)) = |N(\widehat{b}_i)|$, and thus $|a| \in \mathbf{R}_+^*$, $\mathfrak{N}((a)) = |N(a)|$ for all $a \in \widehat{K}^*$. $\square$

The elements a of $\widehat{K}^*$ used to be called **ideal numbers** — a name which is somewhat forgotten but will be used in what follows. The diagram (7.1) implies an isomorphism

$$\widehat{K}^*/K^* \cong J/P.$$

For $a, b \in \widehat{K}^*$ we write $a \sim b$ if a and b lie in the same class, i.e., if $ab^{-1} \in K^*$. We call a an **ideal integer**, or an integral ideal number, if (a) is an integral ideal. The semigroup of all ideal integers will be denoted by $\widehat{\mathcal{O}}$. Furthermore we write $a | b$ if $\frac{b}{a} \in \widehat{\mathcal{O}}$, and for every pair $a, b \in \widehat{K}^*$, we have the notion of $\gcd(a, b) \in \widehat{K}^*$ (which is lacking inside K^*). The greatest common divisor is the ideal number d (which is unique up to a unit) such that the ideal (d) is the gcd of the ideals $(a), (b)$. Observe that the ideal numbers are not defined in a canonical way. This is the reason why they have not been able to hold their own in the development of number theory. (They are treated in [46], [65].)

We now form an analogous extension of the prime residue groups $(\mathbb{Z}/m\mathbb{Z})^*$. For three ideal numbers a, b, m , the congruence

$$a \equiv b \pmod{m}$$

signifies that $a \sim b$ and $\frac{a-b}{m} \in \widehat{\mathcal{O}} \cup \{0\}$. If $\mathfrak{m} = (m)$, we also write this relation as $a \equiv b \pmod{\mathfrak{m}}$. Let $\mathfrak{m}$ be an integral ideal. The semigroup $\widehat{\mathcal{O}}^{(\mathfrak{m})}$ of all integral ideal numbers relatively prime to $\mathfrak{m}$ is partitioned by the equivalence relation $\equiv$ into classes, which we will write as $a \pmod{\mathfrak{m}}$. They are given explicitly as follows.

(7.2) Lemma. For every $a \in \widehat{\mathcal{O}}^{(\mathfrak{m})}$ one has

$$a \bmod \mathfrak{m} = a + a(a^{-1})\mathfrak{m}.$$

Proof: Let $b \in a \bmod \mathfrak{m}$, $b \neq a$, i.e., $b = a\alpha$ for some $\alpha \in K^*$, $\alpha \neq 1$, and $b - a = c\mathfrak{m}$, $c \in \widehat{\mathcal{O}}$. Then

$$a^{-1}(b - a) = \alpha - 1 \in (\alpha - 1) = (a^{-1})(c)(\mathfrak{m}) \subseteq (a^{-1})\mathfrak{m},$$

so that $b \in a + a(a^{-1})\mathfrak{m}$. Let conversely $b \in a + a(a^{-1})\mathfrak{m}$, $b \neq a$, and thus $b/a = \alpha \in 1 + (a^{-1})\mathfrak{m}$. Then one has $b \sim a$ and $(b - a) = (a)(\alpha - 1) \subseteq (a)(a^{-1})\mathfrak{m} = (\mathfrak{m})$, i.e., $\mathfrak{m} \mid b - a$ and therefore $b \equiv a \bmod \mathfrak{m}$. $\square$

We now consider the set

$$(\widehat{\mathcal{O}}/\mathfrak{m})^* := \{a \bmod \mathfrak{m} \mid a \in \widehat{\mathcal{O}}^{(\mathfrak{m})}\}$$

of all equivalence classes in the semigroup $\widehat{\mathcal{O}}^{(\mathfrak{m})}$ of ideal integers prime to $\mathfrak{m}$.

(7.3) Proposition. $(\widehat{\mathcal{O}}/\mathfrak{m})^*$ is an abelian group, and we have a canonical exact sequence

$$1 \longrightarrow (\mathcal{O}/\mathfrak{m})^* \longrightarrow (\widehat{\mathcal{O}}/\mathfrak{m})^* \longrightarrow J/P \longrightarrow 1.$$

Proof: For $a, b \in \widehat{\mathcal{O}}^{(\mathfrak{m})}$, the class $ab \bmod \mathfrak{m}$ only depends on the classes $a \bmod \mathfrak{m}$, $b \bmod \mathfrak{m}$, so we get a well-defined product in $(\widehat{\mathcal{O}}/\mathfrak{m})^*$. Every class $a \bmod \mathfrak{m}$ has an inverse. Indeed, since $(a) + \mathfrak{m} = \mathcal{O}$, we may write $1 = \alpha + \mu$, $0 \neq \alpha \in (a)$, $\mu \in \mathfrak{m}$. Consequently $a \mid \alpha$, so that $\alpha = ax$, $x \in \widehat{\mathcal{O}}^{(\mathfrak{m})}$, and since $1 \in \alpha(1 + \alpha^{-1}\mathfrak{m}) = \alpha \bmod \mathfrak{m}$, we see that $ax \bmod \mathfrak{m}$ is the unit class, i.e., $x \bmod \mathfrak{m}$ is inverse to $a \bmod \mathfrak{m}$.

The right-hand arrow in the sequence is induced by $a \mapsto (a)$. It is surjective since every class of J/P contains an integral ideal relatively prime to $\mathfrak{m}$. If the class $a \bmod \mathfrak{m} = a(1 + (a)^{-1}\mathfrak{m})$ is mapped to 1, then one has $(a) \in P$, and so $a \in \mathcal{O}$, $(a, \mathfrak{m}) = 1$. Hence $a \bmod \mathfrak{m} = a + \mathfrak{m}$ is a unit in $\mathcal{O}/\mathfrak{m}$. The injectivity of the arrow on the left is completely trivial, i.e., we have shown the exactness. $\square$

For an ideal class $\mathfrak{K} \in J/P$, we will denote by $\mathfrak{K}' \in J/P$ in what follows the class defined by

$$\mathfrak{K}\mathfrak{K}' = [\mathfrak{m}\mathcal{O}],$$

where $\mathfrak{d}$ is the different of $K|\mathbb{Q}$. Let $\mathfrak{m} = (m)$ and $\mathfrak{d} = (d)$, with some fixed ideal numbers m, d . For $\mathfrak{m} = \mathfrak{o}$ let $m = 1$. We now study characters

$$\chi : (\widehat{\mathfrak{o}}/\mathfrak{m})^* \longrightarrow \mathbb{C}^*,$$

and put $\chi(a) = 0$ for $a \in \mathfrak{o}$ such that $(a, \mathfrak{m}) \neq 1$. In the applications, χ will come from a *Größencharakter* mod $\mathfrak{m}$, but the treatment of the theta series is independent of such an origin of χ .

(7.4) Definition. Let $a \in \widehat{\mathfrak{o}}$ be an ideal integer, and let $\mathfrak{K}$ be the class of (a) . Then we define the **Gauss sum**

$$\tau(\chi, a) = \sum_{\hat{x} \bmod \mathfrak{m}} \chi(\hat{x}) e^{2\pi i \text{Tr}(\hat{x}a/md)},$$

where $\hat{x} \bmod \mathfrak{m}$ runs through the classes of $(\widehat{\mathfrak{o}}/\mathfrak{m})^*$ which are mapped to the class $\mathfrak{K}'$. In particular, we put $\tau(\chi) = \tau(\chi, 1)$.

The Gauss sum $\tau(\chi, a)$ reduces immediately to the one considered in §6,

$$\tau_{\mathfrak{m}}(\chi, y) = \sum_{\substack{x \bmod \mathfrak{m} \\ (x, \mathfrak{m})=1}} \chi(x) e^{2\pi i \text{Tr}(xy)}.$$

In fact, on the one hand we have

$$y = \hat{x}a/md \in \mathfrak{m}^{-1}\mathfrak{d}^{-1},$$

since the class of the ideal $(y) = (a)(\hat{x})(m)^{-1}(d)^{-1}$ is the principal class $\mathfrak{K}\mathfrak{K}'\mathfrak{m}^{-1}\mathfrak{d}^{-1}$, so $y \in K^*$, and one finds

$$y \in (y) = (a\hat{x})\mathfrak{m}^{-1}\mathfrak{d}^{-1} \subseteq \mathfrak{m}^{-1}\mathfrak{d}^{-1},$$

because a and $\hat{x}$ are integral. On the other hand, if $\hat{x} \bmod \mathfrak{m}$ is a fixed class of $(\widehat{\mathfrak{o}}/\mathfrak{m})^*$ which maps to $\mathfrak{K}'$, then, in view of (7.3), we get the others by $\hat{x}x \bmod \mathfrak{m}$, with $x \bmod \mathfrak{m}$ varying over the classes of $(\mathfrak{o}/\mathfrak{m})^*$. Therefore

$$\tau(\chi, a) = \chi(\hat{x})\tau_{\mathfrak{m}}(\chi, y),$$

and in particular

$$\tau(\chi) = \chi(\hat{x})\tau_{\mathfrak{m}}(\chi, y)$$

with $y = \hat{x}/md$, which satisfies $(ym\mathfrak{d}, \mathfrak{m}) = 1$ since $ym\mathfrak{d} = (\hat{x})$ and $((\hat{x}), \mathfrak{m}) = 1$. Consequently, $\tau(\chi, a)$ does not depend on the choice of representatives $\hat{x}$, and theorem (6.4) yields at once the

(7.5) Proposition. For a primitive character χ of $(\widehat{\mathfrak{o}}/\mathfrak{m})^*$, one has

$$\tau(\chi, a) = \overline{\chi}(a)\tau(\chi)$$

and $|\tau(\chi)| = \sqrt{\mathfrak{N}(\mathfrak{m})}$.

The theta series $\theta(\chi, z)$ used in § 2 in the treatment of Dirichlet L -series are attached to the field $\mathbb{Q}$. We now have to find their analogues relative to an arbitrary number field K . Given any admissible element $p \in \prod_{\tau} \mathbb{Z}$ (see § 3, p. 448) and a character χ of $(\widehat{\mathcal{O}}/\mathfrak{m})^*$, we form the **Hecke theta series**

$$\theta^p(\chi, z) = \sum_{a \in \widehat{\mathcal{O}} \cup \{0\}} \chi(a) N(a^p) e^{\pi i \langle az / |md|, a \rangle},$$

where m, d are fixed ideal numbers such that $(m) = \mathfrak{m}$ and $(d) = \mathfrak{d}$. We take $m = 1$ if $\mathfrak{m} = 1$. The case $\mathfrak{m} = 1, p = 0$ is exceptional in that the constant term of the theta series is $\chi(0)N(0^p) = 1$, whereas it is 0 in all other cases.

Let us decompose the theta series according to the ideal classes $\mathfrak{K} \in J/P$ into **partial Hecke theta series**

$$\theta^p(\mathfrak{K}, \chi, z) = \sum_{a \in (\widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}}) \cup \{0\}} \chi(a) N(a^p) e^{\pi i \langle az / |md|, a \rangle},$$

where a varies over all ideal integers in the class $\widehat{\mathfrak{K}} \in \widehat{K}^*/K^*$ which corresponds to the ideal class $\mathfrak{K}$ under the isomorphism $\widehat{K}^*/K^* \cong J/P$. For these partial theta series, we want to deduce a transformation formula, and to this end we decompose them further into theta series for which we have the general transformation formula (3.6) at our disposal.

Let $\mathfrak{a}$ be an integral ideal relatively prime to $\mathfrak{m}$ which belongs to the class $\mathfrak{K}$, and let $a \in \widehat{\mathcal{O}}^{(\mathfrak{m})}$ be an ideal number such that $(a) = \mathfrak{a}$.

(7.6) Lemma. Assume that $\mathfrak{m} \neq 1$ or $p \neq 0$. If $x \bmod \mathfrak{m}$ varies over the classes of $(\mathcal{O}/\mathfrak{m})^*$, then one has

$$\theta^p(\mathfrak{K}, \chi, z) = \chi(a) N(a^p) \sum_{x \bmod \mathfrak{m}} \chi(x) \theta_{\Gamma}^p(x, 0, z | a^2 / |md|),$$

where Γ is the lattice $\mathfrak{m}/\mathfrak{a} \subseteq \mathbf{R}$ and

$$\theta_{\Gamma}^p(x, 0, z) = \sum_{g \in \Gamma} N((x + g)^p) e^{\pi i \langle (x+g)z, x+g \rangle}.$$

Proof: In the theta series $\theta^p(\mathfrak{K}, \chi, z)$, it suffices to sum over the elements of $\widehat{\mathfrak{K}} \cap \mathcal{O}^{(\mathfrak{m})}$ because χ is zero on the others. Every class $\widehat{x} \bmod \mathfrak{m} \in (\widehat{\mathcal{O}}/\mathfrak{m})^*$ is either disjoint from $\widehat{\mathfrak{K}}$ or else it is contained in $\widehat{\mathfrak{K}}$. In view of the exact sequence (7.3)

$$1 \longrightarrow (\mathcal{O}/\mathfrak{m})^* \longrightarrow (\widehat{\mathcal{O}}/\mathfrak{m})^* \longrightarrow J/P \longrightarrow 1,$$

the classes

$$ax \bmod \mathfrak{m} = a(x + \mathfrak{a}^{-1}\mathfrak{m})$$

are the different residue classes of $(\widehat{\mathcal{O}}/\mathfrak{m})^*$ contained in $\widehat{\mathfrak{K}}$. This gives

$$\begin{aligned}\theta^P(\mathfrak{K}, \chi, z) &= \sum_{x \bmod \mathfrak{m}} \sum_{g \in \Gamma} \chi(ax) N((ax + ag)^P) e^{\pi i \langle ax + ag, z \rangle / |md|, ax + ag)} \\ &= \chi(a) N(a^P) \sum_{x \bmod \mathfrak{m}} \chi(x) \sum_{g \in \Gamma} N((x + g)^P) e^{\pi i \langle x + g, z \rangle / |a^2|md|, x + g)} \\ &= \chi(a) N(a^P) \sum_{x \bmod \mathfrak{m}} \chi(x) \theta_{\Gamma}^P(x, 0, z | a^2 / |md|). \quad \square\end{aligned}$$

For any admissible element $p = (p_{\tau})$, we will write $\bar{p}$ for the admissible element with components $\bar{p}_{\tau} = p_{\bar{\tau}}$. From the transformation formula (3.6) for the series θ_{Γ}^P and proposition (7.5) on Gauss sums, we now obtain the

(7.7) Theorem. *For a primitive character χ of $(\widehat{\mathcal{O}}/\mathfrak{m})^*$, one has the transformation formula*

$$\theta^P(\mathfrak{K}, \chi, -1/z) = W(\chi, \bar{p}) N((z/i)^{p+\frac{1}{2}}) \theta^{\bar{p}}(\mathfrak{K}', \bar{\chi}, z)$$

with the constant factor

$$W(\chi, \bar{p}) = \left[i^{\text{Tr}(\bar{p})} N\left(\left(\frac{md}{|md|}\right)^{\bar{p}}\right) \right]^{-1} \frac{\tau(\chi)}{\sqrt{\mathfrak{N}(\mathfrak{m})}}.$$

This factor has absolute value $|W(\chi, \bar{p})| = 1$.

Proof: The lattice Γ' dual to the lattice $\Gamma = \mathfrak{m}/\mathfrak{a} \subseteq \mathbf{R}$ is given, according to (5.7), by ${}^*\Gamma' = \mathfrak{a}/\mathfrak{m}\mathfrak{d}$. (Here as in §4, the asterisk signifies adjunction with respect to $\langle \ , \ \rangle$, i.e., $\langle x, \alpha y \rangle = \langle {}^* \alpha x, y \rangle$.) The volume of the fundamental mesh of Γ is by chap. I, (5.2),

$$\text{vol}(\Gamma) = \mathfrak{N}(\mathfrak{m}/\mathfrak{a}) \sqrt{|d_K|} = N(|m/a|) N(|d|)^{1/2}.$$

From (3.6) we now get

$$(1) \quad \theta_{\Gamma}^P(x, 0, -1/|md/a^2|z) = A(z) \theta_{\Gamma'}^P(0, x, z | md/a^2),$$

with the factor

$$\begin{aligned}A(z) &= \left[i^{\text{Tr}(p)} N(|m/a|) N(|d|)^{1/2} \right]^{-1} N(|md/a^2|z/i)^{p+\frac{1}{2}} \\ &= \left[i^{\text{Tr}(p)} \sqrt{\mathfrak{N}(\mathfrak{m})} \right]^{-1} N(|md/a^2|^P) N((z/i)^{p+\frac{1}{2}})\end{aligned}$$

and the series

$$(2) \quad \theta_{\Gamma'}^P(0, x, z | md/a^2) = \sum_{g' \in \Gamma'} N(g'^P) e^{2\pi i \langle x, g' \rangle} e^{\pi i \langle g', z \rangle / |md/a^2|, g'}.$$

Writing $g' = \frac{*g}{*(md/a)}$, the rules stated in §3 give

$$\langle x, g' \rangle = \text{Tr}(axg/md),$$

$$\langle g'|md/a^2|z, g' \rangle = \langle *gz|md/a^2|/|md/a|^2, *g \rangle = \langle gz/|md|, g \rangle$$

and $N((g')^p) = N(g^{\bar{p}})$. If g' varies over the lattice Γ' , then g varies over the set

$$(md/a)*\Gamma' = (md/a)\mathfrak{a}(\mathfrak{m}\mathfrak{d})^{-1} = (\widehat{\mathfrak{K}}' \cap \widehat{\mathfrak{O}}) \cup \{0\}.$$

Substituting all this into (2) yields

$$(3) \quad \theta_{\Gamma'}^p(0, x, z|md/a^2|) \\ = N\left(\left(\frac{a}{md}\right)^{\bar{p}}\right) \sum_{g \in (\widehat{\mathfrak{K}}' \cap \widehat{\mathfrak{O}}) \cup \{0\}} N(g^{\bar{p}}) e^{2\pi i \text{Tr}(axg/md)} e^{\pi i \langle gz/|md|, g \rangle}.$$

Let us now consider first the special case $\mathfrak{m} = 1$, $p = 0$ (which was essentially treated already in §5). In this case, we have $(\widehat{\mathfrak{K}} \cap \widehat{\mathfrak{O}}) \cup \{0\} = \{ag \mid g \in K, (ag) \subseteq \mathfrak{o}\} = a\mathfrak{a}^{-1} = a\Gamma$. Consequently

$$\theta^p(\mathfrak{K}, \chi, z) = \sum_{g \in \Gamma} e^{\pi i \langle agz/|d|, ag \rangle} = \sum_{g \in \Gamma} e^{\pi i \langle gz/a^2/d|, g \rangle} = \theta_{\Gamma}(z|a^2/d|), \\ \theta^{\bar{p}}(\mathfrak{K}', \bar{\chi}, z) = \sum_{g \in (\widehat{\mathfrak{K}}' \cap \widehat{\mathfrak{O}}) \cup \{0\}} e^{\pi i \langle gz/|d|, g \rangle} = \theta_{\Gamma'}(z|d/a^2|).$$

Equation (1) thus becomes

$$\theta^p(\mathfrak{K}, \chi, -1/z) = N(z/i)^{\frac{1}{2}} \theta^{\bar{p}}(\mathfrak{K}', \bar{\chi}, z).$$

Now assume $\mathfrak{m} \neq 1$ or $p \neq 0$. Then we have $\chi(0)N(0^{\bar{p}}) = 0$. Substituting (3) into (1) and (1) into formula (7.6), with $-1/z$ instead of z , we obtain

$$\theta^p(\mathfrak{K}, \chi, -1/z) = N(a^p) \sum_{x \bmod \mathfrak{m}} \chi(ax) \theta_{\Gamma'}^p(x, 0, -1/z|md/a^2|) \\ = B(z) \sum_{g \in \widehat{\mathfrak{K}}' \cap \widehat{\mathfrak{O}}} N(g^{\bar{p}}) \left(\sum_{x \bmod \mathfrak{m}} \chi(ax) e^{2\pi i \text{Tr}(axg/md)} \right) e^{\pi i \langle gz/|md|, g \rangle}$$

with the factor

$$B(z) = A(z) \frac{N(a^p)}{N((md/a)^{\bar{p}})}.$$

Now consider the sum in parentheses. If x varies over a system of representatives of $(\mathfrak{o}/\mathfrak{m})^*$, then ax varies over a system of representatives of those classes of $(\mathfrak{o}/\mathfrak{m})^*$ which are mapped under $(\widehat{\mathfrak{O}}/\mathfrak{m})^* \rightarrow J/P$ to the class $\mathfrak{K}$. Furthermore, (g) is an integral ideal in the class $\mathfrak{K}'$, and since $\mathfrak{K}'$

bears the same relation $\mathfrak{K}'\mathfrak{K} = [\mathfrak{m}\mathfrak{d}]$ to $\mathfrak{K}$ as $\mathfrak{K}$ does to $\mathfrak{K}'$, we recognize the sum in question as the *Gauss sum*

$$\tau(\chi, g) = \sum_{x \bmod \mathfrak{m}} \chi(ax) e^{2\pi i \operatorname{Tr}(axg/md)}.$$

Substituting in now the result (7.5),

$$\tau(\chi, g) = \bar{\chi}(g)\tau(\chi),$$

we finally arrive at the identity

$$(4) \quad \theta^p(\mathfrak{K}, \chi, -1/z) = W(\chi, \bar{p}) N((z/i)^{p+\frac{1}{2}}) \theta^{\bar{p}}(\mathfrak{K}', \bar{\chi}, z)$$

with the factor

$$\begin{aligned} W(\chi, \bar{p}) &= [i^{\operatorname{Tr}(p)} \sqrt{\mathfrak{N}(\mathfrak{m})}]^{-1} N(|md/a^2|^p) \frac{N(a^p)\tau(\chi)}{N((md/a)^{\bar{p}})} \\ &= \frac{\tau(\chi)}{i^{\operatorname{Tr}(p)} \sqrt{\mathfrak{N}(\mathfrak{m})}} N\left(\left(\frac{|md|}{md}\right)^{\bar{p}}\right) N\left(\frac{a^p a^p}{|a|^{2p}}\right) \\ &= \frac{\tau(\chi)}{\sqrt{\mathfrak{N}(\mathfrak{m})}} \left[i^{\operatorname{Tr}(\bar{p})} N\left(\left(\frac{md}{|md|}\right)^{\bar{p}}\right) \right]^{-1}, \end{aligned}$$

where one has to observe that $\operatorname{Tr}(p) = \operatorname{Tr}(\bar{p})$, $a^{\bar{p}} = {}^*a^p$, $a^*a = |a|^2$, and $|md|^p = ({}^*|md|)^p = |md|^{\bar{p}}$ because $|md| \in \mathbf{R}_+^*$. Since $|\tau(\chi)| = \sqrt{\mathfrak{N}(\mathfrak{m})}$, we have $|W(\chi, \bar{p})| = 1$. $\square$

If $\mathfrak{m} \neq 1$ or $p \neq 0$, we find for the special theta series:

$$\theta^p(\chi, z) = \sum_{a \in \hat{\mathcal{O}}} \chi(a) N(a^p) e^{\pi i \langle az/|md|, a \rangle} = \sum_{\mathfrak{K}} \theta^p(\mathfrak{K}, \chi, z),$$

and (7.7) yields the

$$(7.8) \text{ Corollary. } \theta^p(\chi, -1/z) = W(\chi, \bar{p}) N((z/i)^{p+\frac{1}{2}}) \theta^{\bar{p}}(\bar{\chi}, z).$$

We recommend that the reader who has studied the above proof allow himself a moment of contemplation. Looking back, he will realize the peculiar way in which almost all fundamental arithmetic properties of the number field K have been used. First they served to break up the theta series, then these constituents were reshuffled by the analytic transformation law, but in the end they are reassembled to form a new theta series. Having contemplated this, the reader should reflect upon the admirable simplicity of the theta formula which encapsulates all these aspects of the arithmetic of the number field.

There is however one important fundamental law of number theory which does not enter into this formula, that is, **Dirichlet's unit theorem**. This will play an essential rôle when we now pass from theta series to L -series in the next section.

Exercise 1. Define ideal prime numbers and show that unique prime factorization holds in $\widehat{K}^*$.

Exercise 2. Let $\widehat{\mathcal{O}}$ be the semigroup of all ideal integers. If $d = (a, b)$ is the gcd of a, b , then there exist elements $x, y \in \widehat{\mathcal{O}} \cup \{0\}$ such that

$$d = xa + yb.$$

Furthermore, we have $x \sim d/a$, resp. $y \sim d/b$, unless $x = 0$, resp. $y = 0$. Here the notation $\alpha \sim \beta$ means $\alpha\beta^{-1} \in K^*$.

Exercise 3. The congruence $ax \equiv b \pmod{m}$ has a solution in $\widehat{\mathcal{O}}$ with integral x if and only if $(a, m) | b$. This solution is unique $\pmod{m}$, provided $(a, m) = 1$.

Exercise 4. A system of finitely many congruences with pairwise relatively prime moduli is simultaneously solvable if every congruence is solvable individually in such a way that the individual solutions are equivalent (with respect to $\sim$).

Exercise 5. If $a, m \in \widehat{\mathcal{O}}$, then there exists in every residue class $\pmod{m}$ prime to m , an ideal integer prime to a .

Exercise 6. For the factor group $J^{\mathfrak{m}}/\bar{P}^{\mathfrak{m}}$ by the group $\bar{P}^{\mathfrak{m}}$ of all principal ideals (a) such that $a \equiv 1 \pmod{\mathfrak{m}}$, one has the exact sequence

$$1 \rightarrow \mathcal{O}^*/\mathcal{O}^{\mathfrak{m}} \rightarrow (\widehat{\mathcal{O}}/\mathfrak{m})^* \rightarrow J^{\mathfrak{m}}/\bar{P}^{\mathfrak{m}} \rightarrow 1,$$

where $\mathcal{O}^{\mathfrak{m}} = \{\varepsilon \in \mathcal{O}^* \mid \varepsilon \equiv 1 \pmod{\mathfrak{m}}\}$.

Exercise 7. Let $\widehat{K}^{(\mathfrak{m})}$ be the preimage of $J^{\mathfrak{m}}$ under $\widehat{K}^* \rightarrow J$, and let $K^{\mathfrak{m}} = \{a \in K^* \mid a \equiv 1 \pmod{\mathfrak{m}}\}$. Then one has $(\widehat{\mathcal{O}}/\mathfrak{m})^* = \widehat{K}^{(\mathfrak{m})}/K^{\mathfrak{m}}$.

§ 8. Hecke L -series

Let $\mathfrak{m}$ be again an integral ideal of the number field K and let

$$\chi : J^{\mathfrak{m}} \longrightarrow S^1$$

be a character of the group of ideals relatively prime to $\mathfrak{m}$. With respect to this character, we form the L -series

$$L(\chi, s) = \sum_{\mathfrak{a}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s},$$

where $\mathfrak{a}$ varies over the integral ideals of K and we put $\chi(\mathfrak{a}) = 0$ whenever $(\mathfrak{a}, \mathfrak{m}) \neq 1$. Then the following proposition holds in complete generality.

(8.1) Proposition. *The L -series $L(\chi, s)$ converges absolutely and uniformly in the domain $\operatorname{Re}(s) \geq 1 + \delta$, for all $\delta > 0$, and one has*

$$L(\chi, s) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p})\mathfrak{N}(\mathfrak{p})^{-s}},$$

where $\mathfrak{p}$ varies over the prime ideals of K .

Proof: Taking formally the logarithm of the product

$$E(s) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p})\mathfrak{N}(\mathfrak{p})^{-s}}$$

gives the series

$$\log E(s) = \sum_{\mathfrak{p}} \sum_{n=1}^{\infty} \frac{\chi(\mathfrak{p})^n}{n\mathfrak{N}(\mathfrak{p})^{ns}}.$$

It converges absolutely and uniformly for $\operatorname{Re}(s) = \sigma \geq 1 + \delta$. In fact, since $|\chi(\mathfrak{p})| \leq 1$, and $|\mathfrak{N}(\mathfrak{p})^s| = |\mathfrak{N}(\mathfrak{p})|^\sigma \geq p^{f_{\mathfrak{p}}(1+\delta)} \geq p^{1+\delta}$, and since $\#\{\mathfrak{p}|p\} \leq d = [K : \mathbb{Q}]$, it admits the following convergent upper bound which is independent of s :

$$\sum_{p, n} \frac{d}{np^{n(1+\delta)}} = d \log \zeta(1 + \delta).$$

This shows that the product

$$E(s) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p})\mathfrak{N}(\mathfrak{p})^{-s}} = \exp \left(\sum_{\mathfrak{p}} \left(\sum_{n=1}^{\infty} \frac{\chi(\mathfrak{p})^n}{n\mathfrak{N}(\mathfrak{p})^s} \right) \right)$$

is absolutely and uniformly convergent for $\operatorname{Re}(s) \geq 1 + \delta$. Now develop in this product the factors

$$\frac{1}{1 - \chi(\mathfrak{p})\mathfrak{N}(\mathfrak{p})^{-s}} = 1 + \frac{\chi(\mathfrak{p})}{\mathfrak{N}(\mathfrak{p})^s} + \frac{\chi(\mathfrak{p})^2}{\mathfrak{N}(\mathfrak{p})^{2s}} + \cdots$$

for the finitely many prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ such that $\mathfrak{N}(\mathfrak{p}_i) \leq N$, and multiply them. This yields the equation

$$\begin{aligned} (*) \quad \prod_{i=1}^r \frac{1}{1 - \chi(\mathfrak{p}_i)\mathfrak{N}(\mathfrak{p}_i)^{-s}} &= \sum_{v_1, \dots, v_r=0}^{\infty} \frac{\chi(\mathfrak{p}_1)^{v_1} \cdots \chi(\mathfrak{p}_r)^{v_r}}{(\mathfrak{N}(\mathfrak{p}_1)^{v_1} \cdots \mathfrak{N}(\mathfrak{p}_r)^{v_r})^s} \\ &= \sum'_{\mathfrak{a}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s}, \end{aligned}$$

where $\sum'$ denotes the sum over all integral ideals $\mathfrak{a}$ which are divisible at most by the prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. Since the sum $\sum'$ contains in particular the terms such that $\mathfrak{N}(\mathfrak{a}) \leq N$, we may also write

$$\prod_{i=1}^r \frac{1}{1 - \chi(\mathfrak{p}_i)\mathfrak{N}(\mathfrak{p}_i)^{-s}} = \sum_{\mathfrak{N}(\mathfrak{a}) \leq N} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s} + \sum'_{\mathfrak{N}(\mathfrak{a}) > N} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s}.$$

Comparing now in (*) the sum $\sum'$ with the series $L(\chi, s)$, we get

$$\begin{aligned} \left| \prod_{i=1}^r \frac{1}{1 - \chi(\mathfrak{p}_i) \mathfrak{N}(\mathfrak{p}_i)^{-s}} - L(\chi, s) \right| &\leq \left| \sum_{\substack{\mathfrak{N}(\mathfrak{a}) > N \\ \mathfrak{p}_i \nmid \mathfrak{a}}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s} \right| \\ &\leq \sum_{\mathfrak{N}(\mathfrak{a}) > N} \frac{1}{\mathfrak{N}(\mathfrak{a})^{1+\delta}}. \end{aligned}$$

For $N \rightarrow \infty$ the right-hand side tends to zero, as it is the remainder term of a convergent series, since the sequence $\left(\sum_{\mathfrak{N}(\mathfrak{a}) \leq N} \frac{1}{\mathfrak{N}(\mathfrak{a})^{1+\delta}} \right)_{N \in \mathbb{N}}$ is monotone increasing and bounded from above. Indeed, with the previous notations we find

$$\begin{aligned} \sum_{\mathfrak{N}(\mathfrak{a}) \leq N} \frac{1}{\mathfrak{N}(\mathfrak{a})^{1+\delta}} &\leq \sum_{\mathfrak{a}}' \frac{1}{\mathfrak{N}(\mathfrak{a})^{1+\delta}} \\ &= \prod_{i=1}^r (1 - \mathfrak{N}(\mathfrak{p}_i)^{-(1+\delta)})^{-1}, \end{aligned}$$

and

$$\begin{aligned} \log \left(\prod_{i=1}^r (1 - \mathfrak{N}(\mathfrak{p}_i)^{-(1+\delta)})^{-1} \right) &= \sum_{i=1}^r \log \left((1 - \mathfrak{N}(\mathfrak{p}_i)^{-(1+\delta)})^{-1} \right) \\ &= \sum_{i=1}^r \sum_{n=1}^{\infty} \frac{1}{n \mathfrak{N}(\mathfrak{p}_i)^{(1+\delta)n}} \\ &\leq \sum_{\mathfrak{p}} \sum_{n=1}^{\infty} \frac{1}{n \mathfrak{N}(\mathfrak{p})^{(1+\delta)n}} \\ &\leq \sum_{p, n} d \frac{1}{np^{n(1+\delta)}} \\ &= d \log(\zeta(1+\delta)). \end{aligned} \quad \square$$

We now face the task of analytically continuing the L -series $L(\chi, s)$ attached to a **Größencharakter** $\chi \bmod \mathfrak{m}$, and setting up a suitable functional equation for it at the same time. So we are given a character

$$\chi : J^{\mathfrak{m}} \longrightarrow S^1,$$

such that

$$(*) \quad \chi((a)) = \chi_f(a) \chi_{\infty}(a)$$

for all integers $a \in \mathcal{O}$ relatively prime to $\mathfrak{m}$, and there are two associated characters

$$\chi_f : (\mathcal{O}/\mathfrak{m})^* \longrightarrow S^1 \quad \text{and} \quad \chi_{\infty} : \mathbf{R}^* \longrightarrow S^1.$$

The character χ_f extends in a unique way to a character

$$\chi_f : (\widehat{\mathcal{O}}/\mathfrak{m})^* \longrightarrow S^1$$

such that the identity $(*)$ holds for all integral ideal numbers $a \in \widehat{\mathcal{O}}^{(\mathfrak{m})}$ prime to $\mathfrak{m}$. Indeed, the restriction of the function $\chi_f(a) := \chi((a))\chi_\infty(a)^{-1}$ of $\widehat{\mathcal{O}}^{(\mathfrak{m})}$ to $\mathcal{O}^{(\mathfrak{m})}$ is given by the original character χ_f of $(\mathcal{O}/\mathfrak{m})^*$, so it is in particular trivial on $1 + \mathfrak{m}$ and thus yields a character of $(\widehat{\mathcal{O}}/\mathfrak{m})^*$.

The L -series of a *Größencharacter* of $J^{\mathfrak{m}}$ is called a **Hecke L -series**. If χ is a (generalized) Dirichlet character mod $\mathfrak{m}$, i.e., a character of the ray class group $J^{\mathfrak{m}}/P^{\mathfrak{m}}$, then we call it a (generalized) **Dirichlet L -series**. The proof of the functional equation of the Hecke L -series proceeds in exactly the same way as for the Dedekind zeta function, except that it is based on the theta transformation formula (7.7).

We decompose the Hecke L -series according to the classes $\mathfrak{K}$ of the ideal class group J/P as a sum

$$L(\chi, s) = \sum_{\mathfrak{K}} L(\mathfrak{K}, \chi, s)$$

of the **partial L -series**

$$L(\mathfrak{K}, \chi, s) = \sum_{\substack{\mathfrak{a} \in \mathfrak{K} \\ \text{integral}}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s}$$

and deduce a functional equation for those. If all one wants is the functional equation of the L -series $L(\chi, s)$, this decomposition is unnecessary; it may also be derived directly using the transformation formula (7.8), because we know how to represent any ideal $\mathfrak{a}$ by an ideal number (this was not yet the case when we were treating the Dedekind zeta function). However, we prefer to establish the finer result for the partial L -series.

By (7.1), we have a bijective mapping

$$(\widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}})/\mathcal{O}^* \xrightarrow{\sim} \{ \mathfrak{a} \in \mathfrak{K} \mid \mathfrak{a} \text{ integral} \}, \quad a \longmapsto (a),$$

where $\widehat{\mathfrak{K}} \in \widehat{K}^*/K^*$ corresponds to the class $\mathfrak{K} \in J/P$ with respect to the isomorphism $\widehat{K}^*/K^* \cong J/P$. Therefore we get

$$L(\mathfrak{K}, \chi, s) = \sum_{a \in \mathfrak{R}} \frac{\chi((a))}{|N(a)|^s},$$

where $\mathfrak{R}$ is a system of representatives of $(\widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}})/\mathcal{O}^*$. We want to write this function as a Mellin transform. To this end, we recall from §4 the L -function

$$L_X(s) = N(\pi^{-s/2})\Gamma_X(s/2) = N(\pi^{-s/2}) \int_{\mathbf{R}_+^*} N(e^{-y}y^{s/2}) \frac{dy}{y},$$

which has been attached to the $G(\mathbb{C}|\mathbb{R})$ -set $X = \text{Hom}(K, \mathbb{C})$. The character χ_∞ of $\mathbf{R}^*$ corresponding to χ is given by (6.7) as

$$\chi_\infty(x) = N(x^p |x|^{-p+iq}),$$

for an admissible $p \in \prod_r \mathbb{Z}$ and a $q \in \mathbf{R}_\pm$. We put $\mathbf{s} = \mathbf{s}\mathbf{1} + p - iq$, where $s \in \mathbb{C}$ is a single complex variable, and

$$L_\infty(\chi, s) = L_X(\mathbf{s}) = L_X(\mathbf{s}\mathbf{1} + p - iq).$$

In the integral

$$\Gamma_X(\mathbf{s}/2) = \int_{\mathbf{R}_+^*} N(e^{-y} y^{s/2}) \frac{dy}{y},$$

we make the substitution

$$y \mapsto \pi |a|^2 y / |md| \quad (a \in \mathfrak{R}),$$

where $m, d \in \widehat{\mathcal{O}}$ are fixed ideal numbers such that $(m) = \mathfrak{m}$ and $(d) = \mathfrak{d}$ is the different of $K|\mathbb{Q}$. We then obtain

$$\Gamma_X(\mathbf{s}/2) = N\left(\left(\frac{\pi}{|md|}\right)^{s/2}\right) N(|a|^s) \int_{\mathbf{R}_+^*} e^{-\pi \langle ay/|md|, a \rangle} N(y^{s/2}) \frac{dy}{y}$$

and, since $N(|md|^{\mathbf{1}s/2}) = (|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2}$,

$$(|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2} L_\infty(\chi, s) \frac{1}{N(|a|^s)} = c(\chi) \int_{\mathbf{R}_+^*} e^{-\pi \langle ay/|md|, a \rangle} N(y^{s/2}) \frac{dy}{y}$$

where $c(\chi) = N(|md|^{-p+iq})^{1/2}$. Multiplying this by $\chi_{\mathfrak{f}}(a)N(a^p)$ and summing over $a \in \mathfrak{R}$ yields, in view of

$$\frac{\chi_{\mathfrak{f}}(a)N(a^p)}{N(|a|^s)} = \frac{\chi_{\mathfrak{f}}(a)N(a^p |a|^{-p+iq})}{N(|a|^s)} = \frac{\chi((a))}{|N(a)|^s},$$

the equation

$$(|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2} L_\infty(\chi, s) L(\mathfrak{K}, \chi, s) = c(\chi) \int_{\mathbf{R}_+^*} g(y) N(y^{s/2}) \frac{dy}{y}$$

with the series

$$g(y) = \sum_{a \in \mathfrak{R}} \chi_{\mathfrak{f}}(a) N(a^p) e^{-\pi \langle ay/|md|, a \rangle}.$$

We now consider the completed L -series

$$\Lambda(\mathfrak{K}, \chi, s) = (|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2} L_\infty(\chi, s) L(\mathfrak{K}, \chi, s).$$

Then we get

$$\Lambda(\mathfrak{K}, \chi, s) = c(\chi) \int_{\mathbf{R}_+^*} g(y) N(y^{s/2}) \frac{dy}{y}.$$

We now want to write this function as an integral over the series

$$\theta(\mathfrak{K}, \chi, z) := \theta^p(\mathfrak{K}, \chi_f, z) = \varepsilon(\chi) + \sum_{a \in \widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}}} \chi_f(a) N(a^p) e^{\pi i \langle az / |md|, a \rangle},$$

where the summation is extended not only – as in the case of $g(y)$ – over a system of representatives $\mathfrak{R}$ of $(\widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}}) / \mathcal{O}^*$, but over all $a \in \widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}}$. We have $\varepsilon(\chi) = 1$ if $\mathfrak{m} = 1$ and $p = 0$, and $\varepsilon(\chi) = 0$ otherwise. We will proceed in the same way as with the Dedekind zeta function (see (5.5)). Just as we did there, using

$$y = xt^{1/n}, \quad x = \frac{y}{N(y)^{1/n}}, \quad t = N(y),$$

with $n = [K : \mathbb{Q}]$, we decompose

$$\mathbf{R}_+^* = \mathbf{S} \times \mathbb{R}_+^*, \quad \frac{dy}{y} = d^*x \times \frac{dt}{t}.$$

Then, observing that

$$N(y^{s/2}) = N(x^{s/2}) N(t^{s/2n}) = N(x^{(p-iq)/2}) t^{\frac{1}{2}(s + \text{Tr}(p-iq)/n)},$$

we obtain the identity

$$(*) \quad \Lambda(\mathfrak{K}, \chi, s) = c(\chi) \int_0^\infty \int_{\mathbf{S}} N(x^{(p-iq)/2}) g(xt^{1/n}) d^*x t^{s'} \frac{dt}{t}$$

with $s' = \frac{1}{2}(s + \text{Tr}(p - iq)/n)$. The function under the second integral will be denoted by

$$g_{\mathfrak{R}}(x, t) = N(x^{(p-iq)/2}) \sum_{a \in \mathfrak{R}} \chi_f(a) N(a^p) e^{-\pi \langle ax t^{1/n} / |md|, a \rangle}.$$

From it, the theta series $\theta(\mathfrak{K}, \chi, ixt^{1/n})$ is constructed as follows.

(8.2) Lemma. $N(x^{(p-iq)/2}) (\theta(\mathfrak{K}, \chi, ixt^{1/n}) - \varepsilon(\chi)) = \sum_{\varepsilon \in \mathcal{O}^*} g_{\mathfrak{R}}(|\varepsilon|^2 x, t).$

Proof: For every unit $\varepsilon \in \mathcal{O}^*$, one has $\chi_\infty(\varepsilon) \chi_f(\varepsilon) = \chi((\varepsilon)) = 1$, so that we get

$$N(|\varepsilon|^{p-iq}) = \overline{\chi}_\infty(\varepsilon) N(\varepsilon^p) = \chi_f(\varepsilon) N(\varepsilon^p).$$

We put for short $\xi = xt^{1/n}/|md|$ and obtain

$$g_{\mathfrak{R}}(|\varepsilon|^2 x, t) = N(x^{(p-iq)/2}) \sum_{a \in \mathfrak{R}} \chi_{\mathfrak{f}}(\varepsilon a) N((\varepsilon a)^p) e^{-\pi \langle \varepsilon a \xi, \varepsilon a \rangle} = g_{\varepsilon \mathfrak{R}}(x, t).$$

Since $\widehat{\mathfrak{K}} \cap \widehat{\mathcal{O}} = \bigcup_{\varepsilon \in \mathcal{O}^*} \varepsilon \mathfrak{R}$, we get

$$\begin{aligned} N(x^{(p-iq)/2}) (\theta(\mathfrak{K}, \chi, ixt^{1/n}) - \varepsilon(\chi)) &= \\ &= \sum_{\varepsilon \in \mathcal{O}^*} \sum_{a \in \varepsilon \mathfrak{R}} N(x^{(p-iq)/2}) \chi_{\mathfrak{f}}(\varepsilon a) N((\varepsilon a)^p) e^{-\pi \langle \varepsilon a \xi, \varepsilon a \rangle} \\ &= \sum_{\varepsilon \in \mathcal{O}^*} g_{\varepsilon \mathfrak{R}}(x, t) = \sum_{\varepsilon \in \mathcal{O}^*} g_{\mathfrak{R}}(|\varepsilon|^2 x, t). \end{aligned} \quad \square$$

From this lemma we now obtain the desired integral representation of the function $\Lambda(\mathfrak{K}, \chi, s)$. We choose as in § 5 a fundamental domain F of $\mathbf{S}$ for the action of the group $|\mathcal{O}^*|^2$. F is mapped by $\log : \mathbf{R}_+^* \xrightarrow{\sim} \mathbf{R}_{\pm}$ to a fundamental mesh of the lattice $2 \log |\mathcal{O}^*|$. This means that we have

$$\mathbf{S} = \bigcup_{\eta \in |\mathcal{O}^*|} \eta^2 F.$$

(8.3) Proposition. *The function*

$$\Lambda(\mathfrak{K}, \chi, s) = (|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2} L_{\infty}(\chi, s) L(\mathfrak{K}, \chi, s)$$

is the Mellin transform

$$\Lambda(\mathfrak{K}, \chi, s) = L(f, s')$$

of the function

$$f(t) = f_F(\mathfrak{K}, \chi, t) = \frac{c(\chi)}{w} \int_F N(x^{(p-iq)/2}) \theta(\mathfrak{K}, \chi, ixt^{1/n}) d^*x$$

at $s' = \frac{1}{2}(s + \text{Tr}(p - iq)/n)$. Here we have set $n = [K : \mathbb{Q}]$, $c(\chi) = N(|md|^{-p+iq})^{1/2}$, and w denotes the number of roots of unity in K .

Proof: One has

$$f(\infty) = \frac{c(\chi) \varepsilon(\chi)}{w} \int_F N(x^{(p-iq)/2}) d^*x.$$

We have seen before that

$$(*) \quad \Lambda(\mathfrak{R}, \chi, s) = \int_0^\infty f_0(t) t^{s'} \frac{dt}{t} = L(f, s')$$

where

$$f_0(t) = c(\chi) \int_{\mathbf{S}} g_{\mathfrak{R}}(x, t) d^*x.$$

Since $\mathbf{S} = \bigcup_{\eta \in |\mathcal{O}^*|} \eta^2 F$, one has

$$f_0(t) = c(\chi) \sum_{\eta \in |\mathcal{O}^*|} \int_{\eta^2 F} g_{\mathfrak{R}}(x, t) d^*x.$$

In each one of the integrals on the right, we make the transformation $F \rightarrow \eta^2 F$, $x \mapsto \eta^2 x$, and obtain

$$f_0(t) = c(\chi) \int_F \sum_{\eta \in |\mathcal{O}^*|} g_{\mathfrak{R}}(\eta^2 x, t) d^*x.$$

The fact that we may swap summation and integration is justified in exactly the same way as for the case of Dirichlet L -series in §2, p.436. In view of the exact sequence

$$1 \longrightarrow \mu(K) \longrightarrow \mathcal{O}^* \longrightarrow |\mathcal{O}^*| \longrightarrow 1,$$

where $\mu(K)$ denotes the group of roots of unity in K , one has $\#\{\varepsilon \in \mathcal{O}^* \mid |\varepsilon| = \eta\} = w$, so that we get

$$\sum_{|\varepsilon|=\eta} g_{\mathfrak{R}}(|\varepsilon|^2 x, t) = w g_{\mathfrak{R}}(\eta^2 x, t).$$

Using (8.2), this gives

$$\begin{aligned} f_0(t) &= \frac{c(\chi)}{w} \int_F \sum_{\varepsilon \in \mathcal{O}^*} g_{\mathfrak{R}}(|\varepsilon|^2 x, t) d^*x \\ &= \frac{c(\chi)}{w} \int_F N(x^{(p-iq)/2}) (\theta(\mathfrak{R}, \chi, ixt^{1/n}) - \varepsilon(\chi)) d^*x \\ &= f(t) - f(\infty). \end{aligned}$$

This together with (*) yields the claim of the proposition. $\square$

It is now the transformation formula (7.7) for the theta series $\theta(\mathfrak{R}, \chi, z) = \theta^p(\mathfrak{R}, \chi_f, z)$ which guarantees that the functions $f(t) = f_F(\mathfrak{R}, \chi, t)$ satisfy the hypotheses of the Mellin principle.

(8.4) Proposition. We have $f_F(\mathfrak{R}, \chi, t) = a_0 + O(e^{-ct^{1/n}})$ for some $c > 0$, and

$$a_0 = \frac{N(|d|^{iq/2})}{w} \int_F N(x^{-iq/2}) d^*x$$

if $\mathfrak{m} = 1$ and $p = 0$, and $a_0 = 0$ otherwise. Furthermore we have

$$f_F\left(\mathfrak{R}, \chi, \frac{1}{t}\right) = W(\chi) t^{\frac{1}{2} + \text{Tr}(p)/n} f_{F^{-1}}(\mathfrak{R}', \bar{\chi}, t)$$

where $\mathfrak{R}\mathfrak{R}' = [\mathfrak{m}d]$, and the constant factor is given by

$$W(\chi) = \left[i^{\text{Tr}(\bar{p})} N\left(\left(\frac{md}{|md|}\right)^{\bar{p}}\right) \right]^{-1} \frac{\tau(\chi_f)}{\sqrt{\mathfrak{N}(\mathfrak{m})}}.$$

Proof: The first statement follows exactly as in the proof of (5.8). For the second, we make use of formula (7.7). It gives us

$$\begin{aligned} \theta(\mathfrak{R}, \chi, -1/z) &= \theta^p(\mathfrak{R}, \chi_f, -1/z) = W(\chi) N\left((z/i)^{p+\frac{1}{2}}\right) \theta^{\bar{p}}(\mathfrak{R}', \bar{\chi}_f, z) \\ &= W(\chi) N\left((z/i)^{p+\frac{1}{2}}\right) \theta(\mathfrak{R}', \bar{\chi}, z), \end{aligned}$$

because $\bar{\chi}_\infty(x) = \overline{N(x^p|x|^{-p+iq})} = N((x^p)^{\bar{p}}|x|^{-p-iq}) = N(x^{\bar{p}}|x|^{-\bar{p}-iq})$. Observing the fact that the transformation $x \mapsto x^{-1}$ leaves the Haar measure d^*x invariant and takes the fundamental domain F to the fundamental domain F^{-1} , (7.7) yields for $z = ixt^{1/n}$:

$$\begin{aligned} f_F\left(\mathfrak{R}, \chi, \frac{1}{t}\right) &= \frac{c(\chi)}{w} \int_F N(x^{(p-iq)/2}) \theta(\mathfrak{R}, \chi, ixt^{1/n}) d^*x \\ &= \frac{c(\chi)}{w} \int_{F^{-1}} N(x^{-(p-iq)/2}) \theta(\mathfrak{R}, \chi, -1/ixt^{1/n}) d^*x \\ &= \frac{c(\chi)W(\chi)}{w} \int_{F^{-1}} N(x^{-\frac{p-iq}{2}+p+\frac{1}{2}}) N(t^{(p+\frac{1}{2})/n}) \theta(\mathfrak{R}', \bar{\chi}, ixt^{1/n}) d^*x \\ &= \frac{c(\chi)W(\chi)}{w} \int_{F^{-1}} N(x^{(\bar{p}+iq)/2}) t^{1/2+\text{Tr}(p)/n} \theta(\mathfrak{R}', \bar{\chi}, ixt^{1/n}) d^*x \\ &= W(\chi) t^{\frac{1}{2} + \text{Tr}(p)/n} f_{F^{-1}}(\mathfrak{R}', \bar{\chi}, t). \end{aligned}$$

We have used in this calculation that $N(x^{1/2}) = N(x)^{1/2} = 1$ and $N(x^p) = N((x^p)^{\bar{p}}) = N(x^{\bar{p}})$, and that the character $\bar{\chi}_\infty$, the complex conjugate of χ_∞ , is given by

$$\bar{\chi}_\infty(x) = N(x^{\bar{p}}|x|^{-\bar{p}-iq}).$$

□

From this proposition and (1.4), we now finally get our main result. We may assume that χ is a primitive *Größencharacter* mod $\mathfrak{m}$, i.e., that the corresponding character χ_f of $(\mathcal{O}/\mathfrak{m})^*$ is primitive (see §6, p.472). The L -series of an arbitrary character differs from the L -series of the corresponding primitive character only by finitely many Euler factors. So analytic continuation and functional equation of one follow from those of the other.

(8.5) Theorem. *Let χ be a primitive Größencharacter mod $\mathfrak{m}$. Then the function*

$$\Lambda(\mathfrak{K}, \chi, s) = (|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2} L_\infty(\chi, s) L(\mathfrak{K}, \chi, s), \quad \operatorname{Re}(s) > 1,$$

has a meromorphic continuation to the complex plane $\mathbb{C}$ and satisfies the functional equation

$$\Lambda(\mathfrak{K}, \chi, s) = W(\chi) \Lambda(\mathfrak{K}', \bar{\chi}, 1-s)$$

where $\mathfrak{K}\mathfrak{K}' = [\mathfrak{m}\mathfrak{d}]$, and the constant factor is given by

$$W(\chi) = \left[i^{\operatorname{Tr}(\bar{p})} N \left(\left(\frac{md}{|md|} \right)^{\bar{p}} \right) \right]^{-1} \frac{\tau(\chi_f)}{\sqrt{\mathfrak{N}(\mathfrak{m})}}.$$

It has absolute value $|W(\chi)| = 1$.

$\Lambda(\mathfrak{K}, \chi, s)$ is holomorphic except for poles of order at most one at $s = \operatorname{Tr}(-p + iq)/n$ and $s = 1 + \operatorname{Tr}(p + iq)/n$. In the case $\mathfrak{m} \neq 1$ or $p \neq 0$, $\Lambda(\mathfrak{K}, \chi, s)$ is holomorphic on all of $\mathbb{C}$.

Proof: Let $f(t) = f_F(\mathfrak{K}, \chi, t)$ and $g(t) = f_{F^{-1}}(\mathfrak{K}', \bar{\chi}, t)$. From $f(t) = a_0 + O(e^{-ct^{1/n}})$, $g(t) = b_0 + O(e^{-ct^{1/n}})$ and

$$f\left(\frac{1}{t}\right) = W(\chi) t^{\frac{1}{2} + \operatorname{Tr}(p)/n} g(t),$$

it follows by (1.4) that the Mellin transforms $L(f, s)$ and $L(g, s)$ can be meromorphically continued, and from (8.3) we get

$$\begin{aligned} \Lambda(\mathfrak{K}, \chi, s) &= L\left(f, \frac{1}{2}(s + \operatorname{Tr}(p - iq)/n)\right) \\ &= W(\chi) L\left(g, \frac{1}{2} + \operatorname{Tr}(p)/n - \frac{1}{2}(s + \operatorname{Tr}(p - iq)/n)\right) \\ &= W(\chi) L\left(g, \frac{1}{2}(1 - s + \operatorname{Tr}(\bar{p} + iq)/n)\right) \\ &= W(\chi) \Lambda(\mathfrak{K}', \bar{\chi}, 1-s), \end{aligned}$$

where we have to take into account again that $\bar{\chi}_\infty(x) = N(x^{\bar{p}} |x|^{-\bar{p}-iq})$.

According to (1.4), in the case $a_0 \neq 0$, $L(f, s)$ has a simple pole at $s = 0$ and $s = \frac{1}{2} + \text{Tr}(p)/n$, i.e., $\Lambda(\mathfrak{K}, \chi, s) = L(f, \frac{1}{2}(s + \text{Tr}(p - iq)/n))$ has a simple pole at $s = \text{Tr}(-p + iq)/n$ and $s = 1 + \text{Tr}(p + iq)/n$. If $\mathfrak{m} \neq 1$ or $p \neq 0$, then $a_0 = 0$, i.e., $\Lambda(\mathfrak{K}, \chi, s)$ is holomorphic on all of $\mathbb{C}$. $\square$

For the **completed Hecke L -series**

$$\Lambda(\chi, s) = (|d_K| \mathfrak{N}(\mathfrak{m}))^{s/2} L_\infty(\chi, s) L(\chi, s) = \sum_{\mathfrak{K}} \Lambda(\mathfrak{K}, \chi, s)$$

we derive immediately from the theorem the

(8.6) Corollary. *The L -series $\Lambda(\chi, s)$ admits a holomorphic continuation to*

$$\mathbb{C} \setminus \{ \text{Tr}(-p + iq)/n, 1 + \text{Tr}(p + iq)/n \}$$

and satisfies the functional equation

$$\Lambda(\chi, s) = W(\chi) \Lambda(\overline{\chi}, 1 - s).$$

It is holomorphic on all of $\mathbb{C}$, if $\mathfrak{m} \neq 1$ or $p \neq 0$.

Remark 1: For a *Dirichlet character* $\chi \bmod \mathfrak{m}$, the functional equation can be proved without using ideal numbers, by splitting the ray class group $J^{\mathfrak{m}}/P^{\mathfrak{m}}$ into its classes $\mathfrak{K}$, and then proceeding exactly as for the Dedekind zeta function. The Gauss sums to be used then are those treated by *HASSE* in [52]. On the other hand, one may prove the functional equation for the Dedekind zeta function by using ideal numbers, imitating the above proof, without decomposing the ideal group at all.

Remark 2: There is an important alternative approach to the results of this section. It starts from a character of the idèle class group and from the representation (8.1) of the corresponding L -series as an Euler product. The proof of the functional equation is then based on the local-to-global principle of algebraic number theory and on the Fourier analysis of p -adic number fields and their idèle class group. This theory was developed by the American mathematician *JOHN TATE*, and is commonly known for short as **Tate's thesis**. Even though it does meet the goal of this book of presenting modern conceptual approaches, we still decided not to include it here. The reason for this is the clarity and conciseness of Tate's original paper [24], which cannot be improved upon. In addition *SERGE LANG*'s account of the theory [94] provides an illustrative complement.

Thus instead of idly copying this theory, we have chosen to provide a conceptual framework and a modern treatment of Hecke's original proof which is somewhat difficult to fathom. It turns out that Hecke's approach continues to have a relevance of its own, and can even claim a number of advantages over Tate's theory. For the functional equation of the Riemann zeta function and the Dirichlet L -series, for example, it would be out of proportion to develop Tate's formalism with all its p -adic expense, since they can be settled at a beginner's level with the method used here. Also, L -series, and the very theory of theta series has to be seen as an important arithmetic accomplishment in its own right.

It was for pedagogical reasons that we have proved the analytic continuation and functional equation of L -series four times over: for the Riemann zeta function, for the Dirichlet L -series, for the Dedekind zeta function, and finally for general Hecke L -series. This explains the number of pages needed. Attacking the general case directly would shrink the exposé to little more than the size of Tate's thesis. Still, it has to be said that Tate's theory has acquired fundamental importance for number theory at large through its far reaching generalizations.

§ 9. Values of Dirichlet L -series at Integer Points

The results of § 1 and § 2 on the values $\zeta(1-k)$ and $L(\chi, 1-k)$ of the Riemann zeta function and the Dirichlet L -series will now be extended to generalized Dirichlet L -series over a totally real number field. We do this using a method devised by the Japanese mathematician *TAKURO SHINTANI* (who died an early and tragic death) (see [127], [128]).

We first prove a new kind of unit theorem for which we need the following notions from linear algebra. Let V be an n -dimensional $\mathbb{R}$ -vector space, k a subfield of $\mathbb{R}$, and V_k a fixed k -structure of V , i.e., a k -subspace such that $V = V_k \otimes_k \mathbb{R}$. By definition, an (open) **k -rational simplicial cone** of dimension d is a subset of the form

$$C(v_1, \dots, v_d) = \{t_1 v_1 + \dots + t_d v_d \mid t_\ell \in \mathbb{R}_+^*\}$$

where $v_1, \dots, v_d$ are linearly independent vectors in V_k . A finite disjoint union of k -rational simplicial cones is called a **k -rational polyhedric cone**. We call a linear form L on V *k -rational* if its coefficients with respect to a k -basis of V_k lie in k .

(9.1) Lemma. *Every nonempty subset different from $\{0\}$ of the form*

$$P = \{x \in V \mid L_i(x) \geq 0, \quad 0 < i \leq \ell, \quad M_j(x) > 0, \quad 0 < j \leq m\}$$

with nonzero k -rational linear forms L_i, M_j ($\ell = 0$ or $m = 0$ is allowed) is a disjoint union of finitely many k -rational cones, and possibly the origin.

Proof: First let $P = \{x \in V \mid L_i(x) \geq 0, \quad i = 1, \dots, \ell\}$, with k -rational linear forms $L_1, \dots, L_\ell \neq 0$. For $n = 1$ and $n = 2$ the lemma is obvious. We assume it is established for all $\mathbb{R}$ -vector spaces of dimension smaller than n . If P has no inner point, then there is a linear form L among the $L_1, \dots, L_\ell$ such that P is contained in the hyperplane $L = 0$. In this case the lemma follows from the induction hypothesis. So let $u \in P$ be an inner point, i.e., $L_1(u) > 0, \dots, L_\ell(u) > 0$. Since V_k is dense in V , we may assume $u \in V_k$. For every $i = 1, \dots, \ell$, let $\partial_i P = \{x \in P \mid L_i(x) = 0\}$. If $\partial_i P \neq \{0\}$, then $\partial_i P \setminus \{0\}$ is by the induction hypothesis a disjoint union of a finite number of k -rational simplicial cones of dimension $< n$. If a simplicial cone in $\partial_i P$ has a nonempty intersection with some $\partial_j P$, then it is clearly contained in $\partial_i P \cap \partial_j P$. Therefore $\partial_1 P \cup \dots \cup \partial_\ell P \setminus \{0\}$ is a disjoint union of k -rational simplicial cones of dimension $< n$, so that

$$\partial_1 P \cup \dots \cup \partial_\ell P \setminus \{0\} = \bigcup_{j \in J} C_j,$$

where $C_j = C(v_1, \dots, v_{d_j})$, $v_1, \dots, v_{d_j} \in V_k$, $d_j < n$. For every $j \in J$ we put $C_j(u) = C(v_1, \dots, v_{d_j}, u)$. This is a $(d_j + 1)$ -dimensional k -rational simplicial cone. We claim that

$$P \setminus \{0\} = \bigcup_{j \in J} C_j \cup \bigcup_{j \in J} C_j(u) \cup \mathbb{R}_+^* u.$$

Indeed, if the point $x \in P \setminus \{0\}$ lies on the boundary of P , then it belongs to some $\partial_i P$, hence to $\bigcup_{j \in J} C_j$. On the other hand, if x belongs to the interior of P , then $L_i(x) > 0$ for all i . If x is a scalar multiple of u , then we have $x \in \mathbb{R}_+^* u$. Assume this is not the case, and let s be the minimum of the numbers $L_1(x)/L_1(u), \dots, L_\ell(x)/L_\ell(u)$. Then $s > 0$ and $x - su$ lies on the boundary of P . Since $x - su \neq 0$, there is a unique $j \in J$ such that $x - su \in C_j$, and thus there is a unique $j \in J$ such that $x \in C_j(u)$. This proves the claim.

Now let

$$P = \{x \in V \mid L_i(x) \geq 0, \quad 0 < i \leq \ell, \quad M_j(x) > 0, \quad j = 1, \dots, m\}.$$

Then

$$\bar{P} = \{x \in V \mid L_i(x) \geq 0, \quad M_j(x) \geq 0\}$$

is a disjoint union of a finite number of k -rational simplicial cones and $\{0\}$. For every $j = 1, \dots, m$, let $\partial_j \bar{P} = \{x \in \bar{P} \mid M_j(x) = 0\}$. If a simplicial cone in $\bar{P}$ has nonempty intersection with $\partial_j \bar{P}$, then it is contained in $\partial_j \bar{P}$. As $P = \bar{P} \setminus \bigcup_{j=1}^m \partial_j \bar{P}$, we see that since $\bar{P} \setminus \{0\}$ is a disjoint union of finitely many k -rational simplicial cones, then so is P . $\square$

(9.2) Corollary. *If C and C' are k -rational polyhedric cones, then $C \setminus C'$ is also a k -rational polyhedric cone.*

Proof: We may assume without loss of generality that C and C' are k -rational cones. Let d be the dimension of C' . Then there are n k -rational linear forms $L_1, \dots, L_{n-d}, M_1, \dots, M_d$ such that

$$C' = \{x \in V \mid L_1(x) = \dots = L_{n-d}(x) = 0, M_1(x) > 0, \dots, M_d(x) > 0\}.$$

If we define, for each $i = 1, \dots, n-d$,

$$C_i^\pm = \{x \in C \mid L_1(x) = \dots = L_{i-1}(x) = 0, \pm L_i(x) > 0\},$$

and for each $j = 1, \dots, d$,

$$C_j = \left\{x \in C \mid \begin{array}{l} L_1(x) = \dots = L_{n-d}(x) = 0, \\ M_1(x) > 0, \dots, M_{j-1}(x) > 0, M_j(x) \leq 0 \end{array} \right\},$$

then we find, as can be checked immediately, that $C \setminus C'$ is the disjoint union of the sets $C_1^+, \dots, C_{n-d}^+, C_1^-, \dots, C_{n-d}^-, C_1, \dots, C_d$. By (9.1), these are either empty or k -rational polyhedric cones. Therefore $C \setminus C'$ is also. $\square$

It is a rare and special event if a new substantial insight is added to the foundations of algebraic number theory. The following theorem, proved by SHINTANI in 1979, falls into this category. Let K be a number field of degree $n = [K : \mathbb{Q}]$, and let $\mathbf{R} = \left[\prod_{\tau} \mathbb{C} \right]^+$ be the corresponding Minkowski space ($\tau \in \text{Hom}(K, \mathbb{C})$). Define

$$\mathbf{R}_{(+)}^* = \{(x_{\tau}) \in \mathbf{R}^* \mid x_{\tau} > 0 \text{ for all real } \tau\}.$$

(Observe that one has $\mathbf{R}_{(+)}^* = \mathbf{R}_+^*$ only in the case where K is totally real.) Since $\mathbf{R} = K \otimes_{\mathbb{Q}} \mathbb{R}$, the field K is a $\mathbb{Q}$ -structure of $\mathbf{R}$. The group

$$\mathcal{O}_+^* = \mathcal{O}^* \cap \mathbf{R}_{(+)}^*$$

of totally positive units acts on $\mathbf{R}_{(+)}^*$ via multiplication, and we will show that this action has a fundamental domain which is a $\mathbb{Q}$ -rational polyhedric cone:

(9.3) Shintani's Unit Theorem. *If E is a subgroup of finite index in $\mathcal{O}_+^*$, then there exists a $\mathbb{Q}$ -rational polyhedric cone P such that*

$$\mathbf{R}_{(+)}^* = \bigcup_{\varepsilon \in E} \varepsilon P \quad (\text{disjoint union}).$$

Proof: We consider in $\mathbf{R}_{(+)}^*$ the norm-one hypersurface

$$S = \{x \in \mathbf{R}_{(+)}^* \mid |N(x)| = 1\}.$$

Every $x \in \mathbf{R}_{(+)}^*$ is in a unique way the product of an element of S and of a positive scalar element. Indeed, $x = |N(x)|^{1/n} (x/|N(x)|^{1/n})$. By Dirichlet's unit theorem, E (being a subgroup of finite index in $\mathcal{O}^*$) is mapped by the mapping

$$\ell : S \longrightarrow \left[\prod_{\tau} \mathbb{R} \right]^+, \quad (x_{\tau}) \longmapsto (\log |x_{\tau}|),$$

onto a complete lattice Γ of the trace-zero space $H = \{x \in [\prod_{\tau} \mathbb{R}]^+ \mid \text{Tr}(x) = 0\}$. Let Φ be a fundamental mesh of Γ , let $\overline{\Phi}$ be the closure of Φ in H , and put $F = \ell^{-1}(\overline{\Phi})$. Since $\overline{\Phi}$ is bounded and closed, so is F . It is therefore compact, and we have

$$(1) \quad S = \bigcup_{\varepsilon \in E} \varepsilon F.$$

Let $x \in F$ and $U_{\delta}(x) = \{y \in \mathbf{R} \mid \|x - y\| < \delta\} \subseteq \mathbf{R}_{(+)}^*$, $\delta > 0$. Then there is clearly a basis $v_1, \dots, v_n \in U_{\delta}(x)$ of $\mathbf{R}$ such that $x = t_1 v_1 + \dots + t_n v_n$ with $t_i > 0$. Since K is dense in $\mathbf{R}$ by the approximation theorem, we may even choose the v_i to lie in $K \cap U_{\delta}(x)$. Then $C_{\delta} = C(v_1, \dots, v_n)$ is a $\mathbb{Q}$ -rational simplicial cone in $\mathbf{R}_{(+)}^*$ with $x \in C_{\delta}$, and every $y \in C_{\delta}$ is of the form $y = \lambda z$ with $\lambda \in \mathbb{R}_{+}^*$ and $z \in U_{\delta}(x)$. We may now choose δ sufficiently small so that

$$C_{\delta} \cap \varepsilon C_{\delta} = \emptyset \quad \text{for all } \varepsilon \in E, \varepsilon \neq 1.$$

If not, then we would find sequences $\lambda_v z_v, \lambda'_v z'_v \in C_{1/v}$, $\lambda_v, \lambda'_v \in \mathbb{R}_{+}^*$, $z_v, z'_v \in U_{1/v}(x)$, and $\varepsilon_v \in E$, $\varepsilon_v \neq 1$, such that $\lambda_v z_v = \varepsilon_v \lambda'_v z'_v$, and thus $\rho_v z_v = \varepsilon_v z'_v$, $\rho_v = \lambda_v / \lambda'_v$. z_v and z'_v would converge to x ; now ρ_v would converge to 1 as $\rho_v^n N(z_v) = N(z'_v)$, i.e., $x = (\lim \varepsilon_v)x$. This would mean that $\lim \varepsilon_v = 1$, which is impossible, since E is discrete in $\mathbf{R}$.

F being compact, we thus find a finite number of $\mathbb{Q}$ -rational cones $C_1, \dots, C_m$ in $\mathbf{R}_{(+)}^*$ such that

$$(2) \quad F = \bigcup_{i=1}^m (C_i \cap F)$$

and $C_i \cap \varepsilon C_i = \emptyset$ for all $\varepsilon \in E$, $\varepsilon \neq 1$, and all $i = 1, \dots, m$. From (1) and (2), we deduce that

$$\mathbf{R}_{(+)}^* = \bigcup_{i=1}^m \bigcup_{\varepsilon \in E} \varepsilon C_i.$$

In order to turn this union into a disjoint one, we put $C_1^{(1)} = C_1$ and

$$C_i^{(1)} = C_i \setminus \bigcup_{\varepsilon \in E} \varepsilon C_1, \quad i = 2, \dots, m.$$

εC_1 and C_i are disjoint for almost all $\varepsilon \in E$. Hence, by (9.2), $C_i^{(1)}$ is a $\mathbb{Q}$ -rational polyhedric cone. Observing that $C_i \cap \varepsilon C_i = \emptyset$ for $\varepsilon \in E$, $\varepsilon \neq 1$, we obtain

$$\mathbf{R}_{(+)}^* = \bigcup_{i=1}^m \bigcup_{\varepsilon \in E} \varepsilon C_i^{(1)}$$

and $\varepsilon C_1^{(1)} \cap C_i^{(1)} = \emptyset$ for all $\varepsilon \in E$ and $i = 2, \dots, m$.

We now assume by induction that we have found a finite system of $\mathbb{Q}$ -rational polyhedric cones $C_1^{(\nu)}, \dots, C_m^{(\nu)}$, $\nu = 1, \dots, m-2$ satisfying the following properties:

- (i) $C_i^{(\nu)} \subseteq C_i$,
- (ii) $\mathbf{R}_{(+)}^* = \bigcup_{i=1}^m \bigcup_{\varepsilon \in E} \varepsilon C_i^{(\nu)}$,
- (iii) $\varepsilon C_i^{(\nu)} \cap C_j = \emptyset$ for all $\varepsilon \in E$, if $i \leq \nu$ and $i \neq j$.

We put $C_i^{(\nu+1)} = C_i^{(\nu)}$ for $i \leq \nu+1$, and

$$C_i^{(\nu+1)} = C_i^{(\nu)} \setminus \bigcup_{\varepsilon \in E} \varepsilon C_{\nu+1}^{(\nu)} \quad \text{for } i \geq \nu+2.$$

Then $C_1^{(\nu+1)}, \dots, C_m^{(\nu+1)}$ is a finite system of $\mathbb{Q}$ -rational polyhedric cones which enjoys properties (i), (ii), and (iii) with $\nu+1$ instead of ν . Consequently, $C_1^{(m-1)}, \dots, C_m^{(m-1)}$ is a system of $\mathbb{Q}$ -rational polyhedric cones such that

$$\mathbf{R}_{(+)}^* = \bigcup_{i=1}^m \bigcup_{\varepsilon \in E} \varepsilon C_i^{(m-1)} \quad (\text{disjoint union}). \quad \square$$

Based on Shintani's unit theorem, we now obtain the following description of Dirichlet's L -series. Let $\mathfrak{m}$ be an integral ideal, $J^{\mathfrak{m}}/P^{\mathfrak{m}}$ the ray class group mod $\mathfrak{m}$. Let $\chi : J^{\mathfrak{m}}/P^{\mathfrak{m}} \rightarrow \mathbb{C}^*$ be a Dirichlet character mod $\mathfrak{m}$, and

$$L(\chi, s) = \sum_{\mathfrak{a}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s}$$

the associated Dirichlet L -series. If $\mathfrak{K}$ varies over the classes of $J^{\mathfrak{m}}/P^{\mathfrak{m}}$, then we have

$$L(\chi, s) = \sum_{\mathfrak{K}} \chi(\mathfrak{K}) \zeta(\mathfrak{K}, s)$$

with the partial zeta functions

$$\zeta(\mathfrak{K}, s) = \sum_{\substack{\mathfrak{a} \in \mathfrak{K} \\ \mathfrak{a} \text{ integral}}} \frac{1}{\mathfrak{N}(\mathfrak{a})^s}.$$

Let $\mathfrak{K}$ be a fixed class, and $\mathfrak{a}$ an integral ideal in $\mathfrak{K}$. Furthermore let $(1 + \mathfrak{a}^{-1}\mathfrak{m})_+ = (1 + \mathfrak{a}^{-1}\mathfrak{m}) \cap \mathbf{R}_{(+)}^*$ be the set of all totally positive elements in $1 + \mathfrak{a}^{-1}\mathfrak{m}$. The group

$$E = \mathcal{O}_+^{\mathfrak{m}} = \{ \varepsilon \in \mathcal{O}^* \mid \varepsilon \equiv 1 \pmod{\mathfrak{m}}, \varepsilon \in \mathbf{R}_{(+)}^* \}$$

acts on $(1 + \mathfrak{a}^{-1}\mathfrak{m})_+$, and we have the

(9.4) Lemma. *There is a bijection*

$$(1 + \mathfrak{a}^{-1}\mathfrak{m})_+/E \xrightarrow{\sim} \mathfrak{K}_{\text{int}}, \quad \bar{a} \mapsto a\mathfrak{a},$$

onto the set $\mathfrak{K}_{\text{int}}$ of integral ideals in $\mathfrak{K}$.

Proof: Let $a \in (1 + \mathfrak{a}^{-1}\mathfrak{m})_+$. Then we have $(a - 1)\mathfrak{a} \subseteq \mathfrak{m}$, and since $\mathfrak{a}$ and $\mathfrak{m}$ are relatively prime, we get $a - 1 \in \mathfrak{m}$, i.e., $(a) \in P^{(\mathfrak{m})}$. Hence $a\mathfrak{a}$ lies in $\mathfrak{K}$. Furthermore, we have $a\mathfrak{a} \subseteq \mathfrak{a}(1 + \mathfrak{a}^{-1}\mathfrak{m}) = \mathfrak{a} + \mathfrak{m} = \mathcal{O}$, so that $a\mathfrak{a}$ is integral. Therefore $a \mapsto a\mathfrak{a}$ gives us a mapping

$$(1 + \mathfrak{a}^{-1}\mathfrak{m})_+ \rightarrow \mathfrak{K}_{\text{int}}.$$

It is surjective, for if $a\mathfrak{a}$, $a \in P^{\mathfrak{m}}$, is an integral ideal in $\mathfrak{K}$, then $(a - 1)\mathfrak{a} \subseteq \mathfrak{m}\mathfrak{a} \subseteq \mathfrak{m}$, so that $a \in 1 + \mathfrak{a}^{-1}\mathfrak{m}$, and also $a \in \mathbf{R}_{(+)}^*$, and so $a \in (1 + \mathfrak{a}^{-1}\mathfrak{m})_+$. For $a, b \in (1 + \mathfrak{a}^{-1}\mathfrak{m})_+$, we have $a\mathfrak{a} = b\mathfrak{a}$ if and only if $(a) = (b)$, so that $a = b\varepsilon$ with $\varepsilon \in \mathcal{O}^*$. Since $\varepsilon \in (1 + \mathfrak{a}^{-1}\mathfrak{m})_+$, it follows that $\varepsilon \in E$, i.e., a and b have exactly the same image if and only if they belong to the same class under the action of E . $\square$

The lemma implies the following formula for the partial zeta function $\zeta(\mathfrak{K}, s)$:

$$\zeta(\mathfrak{K}, s) = \frac{1}{\mathfrak{N}(\mathfrak{a})^s} \sum_{a \in \mathfrak{K}} \frac{1}{|N(a)|^s},$$

where $\mathfrak{R}$ runs through a system of representatives of $(1 + \mathfrak{a}^{-1}\mathfrak{m})_+/E$. To this we now apply Shintani's unit theorem. Let

$$\mathbf{R}_{(+)}^* = \bigcup_{i=1}^m \bigcup_{\varepsilon \in E} \varepsilon C_i$$

be a disjoint decomposition of $\mathbf{R}_{(+)}^*$ into finitely many $\mathbb{Q}$ -rational simplicial cones C_i . For every $i = 1, \dots, m$, let $v_{i1}, \dots, v_{id_i}$ be a linearly independent system of generators of C_i . Multiplying if necessary by a convenient totally positive integer, we may assume that all $v_{i\ell}$ lie in $\mathfrak{m}$. Let

$$C_i^1 = \{t_1 v_{i1} + \dots + t_{d_i} v_{id_i} \mid 0 < t_\ell \leq 1\},$$

and

$$R(\mathfrak{R}, C_i) = (1 + \mathfrak{a}^{-1}\mathfrak{m})_+ \cap C_i^1.$$

Then we have the

(9.5) Proposition. *The sets $R(\mathfrak{R}, C_i)$ are finite, and one has*

$$\zeta(\mathfrak{R}, s) = \frac{1}{\mathfrak{N}(\mathfrak{a})^s} \sum_{i=1}^m \sum_{x \in R(\mathfrak{R}, C_i)} \zeta(C_i, x, s)$$

with the zeta functions

$$\zeta(C_i, x, s) = \sum_{\mathbf{z}} |N(x + z_1 v_{i1} + \dots + z_{d_i} v_{id_i})|^{-s},$$

where $\mathbf{z} = (z_1, \dots, z_{d_i})$ varies over all d_i -tuples of nonnegative integers.

Proof: $R(\mathfrak{R}, C_i)$ is a bounded subset of the lattice $\mathfrak{a}^{-1}\mathfrak{m}$ in $\mathbf{R}$, translated by 1. It is therefore finite. Since $C_i \subseteq \mathbf{R}_{(+)}^*$ is the simplicial cone generated by $v_{i1}, \dots, v_{id_i} \in \mathfrak{m}$, every $a \in (1 + \mathfrak{a}^{-1}\mathfrak{m}) \cap C_i$ can be written uniquely as

$$a = \sum_{\ell=1}^{d_i} y_\ell v_{i\ell}$$

with rational numbers $y_\ell > 0$. Putting

$$y_\ell = x_\ell + z_\ell, \quad 0 < x_\ell \leq 1, \quad 0 \leq z_\ell \in \mathbb{Z},$$

we have $\sum x_\ell v_{i\ell} \in 1 + \mathfrak{a}^{-1}\mathfrak{m}$ because $\sum z_\ell v_{i\ell} \in \mathfrak{m} \subseteq \mathfrak{a}^{-1}\mathfrak{m}$. In other words, every $a \in (1 + \mathfrak{a}^{-1}\mathfrak{m}) \cap C_i$ can be written uniquely in the form

$$a = x + \sum_{\ell=1}^{d_i} z_\ell v_{i\ell}$$

with $x = \sum x_\ell v_{i\ell} \in R(\mathfrak{K}, C_i)$. Since

$$(1 + \mathfrak{a}^{-1}\mathfrak{m})_+ = \bigcup_{i=1}^m \bigcup_{\varepsilon \in E} (1 + \mathfrak{a}^{-1}\mathfrak{m}) \cap \varepsilon C_i,$$

$a = x + \sum z_\ell v_{i\ell}$ runs through a system $\mathfrak{R}$ of representatives of $(1 + \mathfrak{a}^{-1}\mathfrak{m})_+/E$ if i runs through the numbers $1, \dots, m$, x through the elements of $R(\mathfrak{K}, C_i)$, and $\mathbf{z} = (z_1, \dots, z_{d_i})$ through integer tuples with $z_\ell \geq 0$. Thus we indeed find that

$$\zeta(\mathfrak{K}, s) = \frac{1}{\mathfrak{N}(\mathfrak{a})^s} \sum_{i=1}^m \sum_{x \in R(\mathfrak{K}, C_i)} \zeta(C_i, x, s). \quad \square$$

(9.6) Corollary. *For the Dirichlet L -series attached to the Dirichlet character $\chi : J^{\mathfrak{m}}/P^{\mathfrak{m}} \rightarrow \mathbb{C}^*$, we have the decomposition*

$$L(\chi, s) = \sum_{\mathfrak{K}} \frac{\chi(\mathfrak{a})}{\mathfrak{N}(\mathfrak{a})^s} \sum_{i=1}^m \sum_{x \in R(\mathfrak{K}, C_i)} \zeta(C_i, x, s),$$

where $\mathfrak{K}$ runs through the classes $J^{\mathfrak{m}}/P^{\mathfrak{m}}$, and $\mathfrak{a}$ denotes an integral ideal in $\mathfrak{K}$, one for each class.

The relation between zeta functions and Bernoulli numbers hinges on a purely analytic fact which is independent of number theory. This is what we will describe now.

Let A be a real $r \times n$ -matrix, $r \leq n$, with *positive* entries a_{ji} , $1 \leq j \leq r$, $1 \leq i \leq n$. From this matrix we construct the linear forms

$$L_j(t_1, \dots, t_n) = \sum_{i=1}^n a_{ji} t_i \quad \text{and} \quad L_i^*(z_1, \dots, z_r) = \sum_{j=1}^r a_{ji} z_j.$$

For an r -tuple $x = (x_1, \dots, x_r)$ of *positive* real numbers, we write the following series

$$\zeta(A, x, s) = \sum_{z_1, \dots, z_r=0}^{\infty} \prod_{i=1}^n L_i^*(z + x)^{-s}.$$

On the other hand we define the generalized **Bernoulli polynomials** $B_k(A, x)$ by

$$B_k(A, x) = \frac{1}{n} \sum_{i=1}^n B_k(A, x)^{(i)},$$

where $B_k(A, x)^{(i)}/(k!)^n$ is the coefficient of

$$u^{(k-1)n} (t_1 \cdots t_{i-1} t_{i+1} \cdots t_n)^{k-1}$$

in the Laurent expansion at 0 of the function

$$\prod_{j=1}^r \frac{\exp(ux_j L_j(t))}{\exp(uL_j(t)) - 1} \Big|_{t_i=1}$$

in the variables $u, t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_n$. For $r = n = 1$ and $A = a$, we have $B_k(a, x) = a^{k-1} B_k(x)$, with the usual Bernoulli polynomial $B_k(x)$ (see § 1, exercise 2). The equation

$$B_k(A, 1-x) = (-1)^{n(k-1)+r} B_k(A, x),$$

where $1-x$ signifies $(1-x_1, \dots, 1-x_r)$, is easily proved.

(9.7) Proposition. *The series $\zeta(A, x, s)$ is absolutely convergent for $\operatorname{Re}(s) > r/n$, and it can be meromorphically continued to the whole complex plane. Its values at the points $s = 1-k$, $k = 1, 2, \dots$, are given by*

$$\zeta(A, x, 1-k) = (-1)^r \frac{B_k(A, x)}{k^n}.$$

Proof: The absolute convergence for $\operatorname{Re}(s) > r/n$ is deduced from the convergence of a series $\sum_{n=1}^{\infty} \frac{1}{n^{1+\delta}}$ by the same arguments that we have used repeatedly. It will be left to the reader. The remainder of the proof is similar to that of (1.8). In the gamma function

$$\Gamma(s)^n = \int_0^{\infty} \cdots \int_0^{\infty} \prod_{i=1}^n e^{-t_i} (t_1 \cdots t_n)^{s-1} dt_1 \cdots dt_n,$$

we substitute

$$t_i \longmapsto L_i^*(z+x)t_i,$$

and obtain

$$\begin{aligned} & \Gamma(s)^n \prod_{i=1}^n L_i^*(z+x)^{-s} \\ &= \int_0^{\infty} \cdots \int_0^{\infty} \exp \left[- \sum_{i=1}^n t_i L_i^*(z+x) \right] (t_1 \cdots t_n)^{s-1} dt_1 \cdots dt_n. \end{aligned}$$

Summing this over all $z = (z_1, \dots, z_r)$, $z_i \in \mathbb{Z}$, $z_i \geq 0$, and observing that

$$\sum_{i=1}^n t_i L_i^*(z+x) = \sum_{j=1}^r (z_j + x_j) L_j(t),$$

yields the equation

$$\Gamma(s)^n \zeta(A, x, s) = \int_0^\infty \cdots \int_0^\infty g(t)(t_1 \cdots t_n)^{s-1} dt_1 \cdots dt_n$$

with the function

$$g(t) = g(t_1, \dots, t_n) = \prod_{j=1}^r \frac{\exp((1-x_j)L_j(t))}{\exp(L_j(t)) - 1}.$$

We cut up the space $\mathbb{R}^n$ into the subsets

$$D_i = \{t \in \mathbb{R}^n \mid 0 \leq t_\ell \leq t_i, \ell = 1, \dots, i-1, i+1, \dots, n\}$$

for $i = 1, \dots, n$, and get

$$(1) \quad \zeta(A, x, s) = \Gamma(s)^{-n} \sum_{i=1}^n \int_{D_i} g(t)(t_1 \cdots t_n)^{s-1} dt_1 \cdots dt_n.$$

In D_i we make the transformation of variables

$$t = uy = u(y_1, \dots, y_n),$$

where $0 < u, 0 \leq y_\ell \leq 1$ for $\ell \neq i$ and $y_i = 1$. This gives

$$\begin{aligned} & \Gamma(s)^{-n} \int_{D_i} g(t)(t_1 \cdots t_n)^{s-1} dt_1 \cdots dt_n \\ &= \Gamma(s)^{-n} \int_0^\infty \left[\int_0^1 \cdots \int_0^1 g(uy) \left(\prod_{\ell \neq i} y_\ell \right)^{s-1} \prod_{\ell \neq i} dy_\ell \right] u^{ns-1} du. \end{aligned}$$

For $0 < \varepsilon < 1$, let now $I_\varepsilon(1)$, resp. $I_\varepsilon(+\infty)$, denote the path in $\mathbb{C}$ consisting of the interval $[1, \varepsilon]$, resp. $[+\infty, \varepsilon]$, followed by a circle around 0 of radius ε in the positive direction, and the interval $[\varepsilon, 1]$, resp. $[\varepsilon, +\infty]$. For ε sufficiently small, the right-hand side of the last equation following (1.9) becomes

$$(2) \quad A(s) \int_{I_\varepsilon(+\infty)} \int_{I_\varepsilon(1)}^{n-1} \left[g(uy) u^{ns-1} \left(\prod_{\ell \neq i} y_\ell \right)^{s-1} \prod_{\ell \neq i} dy_\ell \right] du,$$

with the factor

$$A(s) = \frac{\Gamma(s)^{-n}}{(e^{2\pi i n s} - 1)(e^{2\pi i s} - 1)^{n-1}},$$

where one has to observe that the linear forms $L_1, \dots, L_r$ have positive coefficients. It is easy to check that the above expression, as a function of the

variable s , is meromorphic on all of $\mathbb{C}$. As for the factor $A(s)$, (1.2) implies that

$$A(s) = \frac{1}{(2\pi i)^n} \frac{\Gamma(1-s)^n}{(e^{2\pi i n s} - 1)(e^{2\pi i s} - 1)^{-1} e^{n\pi i s}}.$$

Let us now put $s = 1 - k$. The function $e^{n\pi i s}(e^{2\pi i n s} - 1)/(e^{2\pi i s} - 1)$ takes the value $(-1)^{n(k-1)}n$ at $s = 1 - k$. Thus expression (2) turns into

$$(-1)^{n(k-1)} \frac{\Gamma(k)^n}{n} \cdot \frac{1}{(2\pi i)^n} \int_{K_\varepsilon} \int_{K_\varepsilon^{n-1}} \left[g(uy) u^{n(1-k)-1} \left(\prod_{\ell \neq i} y_\ell \right)^{-k} \prod_{\ell \neq i} dy_\ell \right] du,$$

where K_ε denotes the positively oriented circumference of the circle of radius ε , and where we have to observe that the integrals over $(\infty, \varepsilon]$ and $[\varepsilon, \infty)$, resp. over $[1, \varepsilon]$ and $[\varepsilon, 1]$, kill each other in (2) if $s = 1 - k$. This value is obviously $((-1)^{n(k-1)} \Gamma(k)^n / n)$ times the coefficient of $u^{n(k-1)} (\prod_{\ell \neq i} y_\ell)^{k-1}$ in the Laurent expansion of the function

$$g(uy_1, \dots, uy_{i-1}, u, uy_{i+1}, \dots, uy_n) = \prod_{j=1}^r \frac{\exp(u(1-x_j)L_j(y))}{\exp(uL_j(y)) - 1} \Big|_{y_i=1},$$

which is a holomorphic function of $u, t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_n$ in the direct product of n copies of the punctured disc of radius ε . Therefore the value of (2) at $s = 1 - k$ equals $(-1)^{n(k-1)} k^{-n} B_k(A, 1-x)^{(i)} / n$. Inserting this into (1) gives

$$\begin{aligned} \zeta(A, x, 1-k) &= (-1)^{n(k-1)} k^{-n} \frac{1}{n} \sum_{i=1}^n B_k(A, 1-x)^{(i)} \\ &= (-1)^{n(k-1)} \frac{B_k(A, 1-x)}{k^n}. \end{aligned}$$

Together with the equation $B_k(A, 1-x) = (-1)^{n(k-1)+r} B_k(A, x)$ mentioned above, this gives the desired result. $\square$

Theorems (9.5) and (9.6) now imply our main result concerning the values of Dirichlet L -series $L(\chi, s)$ at integer points $s = 1 - k$, $k = 1, 2, \dots$. If K is not totally real, then these values are all zero (except if χ is the trivial character, for which $s = 0$ is not a zero). This can be read off immediately from the functional equation (8.6) and (5.11).

So we let K be a totally real number field of degree n . Numbering the embeddings $\tau : K \rightarrow \mathbb{R}$ identifies the Minkowski space $\mathbf{R}$ with $\mathbb{R}^n$, and $\mathbf{R}_{(+)}^* = \mathbf{R}_+^n$ with the set $\mathbb{R}_+^n$ of vectors $(x_1, \dots, x_n)$ with positive coefficients x_i . Given the $\mathbb{Q}$ -rational simplicial cone $C_i \subseteq \mathbb{R}_+^n$ generated by $v_{i1}, \dots, v_{id_i}$, we again consider the zeta functions

$$\zeta(C_i, x, s) = \sum_{\mathbf{z}} \left| N(x + z_1 v_{i1} + \dots + z_{d_i} v_{id_i}) \right|^{-s}.$$

If

$$v_{ij} = (a_{j1}^{(i)}, \dots, a_{jn}^{(i)}), \quad j = 1, \dots, d_i,$$

then $A_i = (a_{jk}^{(i)})$ is a $(d_i \times n)$ -matrix with positive entries, and the k -th component of $z_1 v_{i1} + \dots + z_{d_i} v_{id_i}$ becomes

$$L_k^*(z_1, \dots, z_{d_i}) = \sum_{j=1}^{d_i} a_{jk}^{(i)} z_j.$$

For $x \in \mathbb{R}_+^*$, we therefore get

$$\zeta(C_i, x, s) = \sum_{\mathbf{z}} \prod_{k=1}^n L_k^*(z_1, \dots, z_{d_i})^{-s} = \zeta(A_i, x, s),$$

and, from (9.5) and (9.6), we obtain by putting $s = 1 - k$ the

(9.8) Theorem. *The values of the partial zeta function $\zeta(\mathfrak{R}, s)$ at the integral points $s = 1 - k$, $k = 1, 2, 3, \dots$, are given by*

$$\zeta(\mathfrak{R}, 1 - k) = \mathfrak{N}(\mathfrak{a})^{k-1} \sum_{i=1}^m \left[(-1)^{d_i} \sum_{x \in R(\mathfrak{R}, C_i)} \frac{B_k(A_i, x)}{k^n} \right],$$

and the values of the Dirichlet L -series $L(\chi, s)$ are given by

$$L(\chi, 1 - k) = \sum_{\mathfrak{R}} \chi(\mathfrak{a}) \mathfrak{N}(\mathfrak{a})^{k-1} \sum_{i=1}^m \left[(-1)^{d_i} \sum_{x \in R(\mathfrak{R}, C_i)} \frac{B_k(A_i, x)}{k^n} \right].$$

Here $\mathfrak{a}$ is an integral ideal in the class $\mathfrak{R}$ of J^m/P^m .

This result about the Dirichlet L -series $L(\chi, s)$ also covers the Dedekind zeta function $\zeta_K(s)$. The theorem says in particular that the values $L(\chi, 1 - k)$, for $k \geq 1$, are algebraic numbers which all lie in the cyclotomic field $\mathbb{Q}(\chi_f)$ generated by the values of the character χ_f . The values $\zeta_K(1 - k)$ are even rational numbers. From the functional equation (5.11),

$$\zeta_K(1 - s) = |d_K|^{s-1/2} \left(\cos \frac{\pi s}{2} \right)^{r_1+r_2} \left(\sin \frac{\pi s}{2} \right)^{r_2} \Gamma_{\mathbb{C}}(s)^n \zeta_K(s),$$

we deduce that $\zeta_K(1 - k) = 0$ for odd $k > 1$, and it is $\neq 0$ for even $k > 1$. If the number field K is not totally real, then we have $\zeta_K(s) = 0$ for all $s = -1, -2, -3, \dots$

(9.9) Corollary (SIEGEL-KLINGEN). *The values of the partial zeta function $\zeta(\mathfrak{R}, s)$ at the points $s = 0, -1, -2, \dots$ are rational numbers.*

Proof: Let $a_1, \dots, a_r$ be nonzero numbers in K , and let A be the $(r \times n)$ -matrix (a_{ji}) , where a_{ji} is the i -th component of a_j , after identifying $\mathbf{R} = \mathbb{R}^n$ according to the chosen numbering of the embeddings $\tau : K \rightarrow \mathbb{R}$. It is enough to show that $B_k(A, x)$ is a rational number for every r -tuple of rational numbers $x = (x_1, \dots, x_r)$. To see this, let $L|\mathbb{Q}$ be the normal closure of $K|\mathbb{Q}$ and $\sigma \in G(L|\mathbb{Q})$. Then σ induces a permutation of the indices $\{1, 2, \dots, n\}$ so that

$$\sigma a_{ji} = a_{j\sigma(i)} \quad (1 \leq j \leq r, i = 1, \dots, n).$$

Now we had $B_k(A, x) = \frac{1}{n} \sum_{i=1}^n B_k(A, x)^{(i)}$, where $B_k(A, x)^{(i)}$ was the coefficient of $u^{n(k-1)+r} (t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_n)^{k-1}$ in the Taylor expansion of the function

$$u^r \prod_{j=1}^r \frac{\exp(x_j u L_j(t))}{\exp(u L_j(t) - 1)} \Big|_{t_i=1},$$

with $L_j(t) = a_{j1}t_1 + \dots + a_{jn}t_n$. This makes it clear that $B_k(A, x)^{(i)}$ lies in L and that $\sigma B_k(A, x)^{(i)} = B_k(A, x)^{(\sigma(i))}$. Therefore $B_k(A, x)$ is invariant under the action of the Galois group $G(L|\mathbb{Q})$, and thus belongs to $\mathbb{Q}$. $\square$

The nature of the special values of L -series at integer points has recently found increasing interest. Like in the class number formula, which expresses the behaviour of the Dedekind zeta function at the point $s = 0$, the properties of all the special values indicate a deep arithmetic law which appears to extend to an extremely wide class of L -series, the L -series attached to “motives”. According to a conjecture of the American mathematician *STEPHEN LICHENBAUM*, the significance of these L -values can be explained by a strikingly simple geometric interpretation: they appear according to the **Lichtenbaum conjecture** as Euler characteristics in étale cohomology (see [99], [12]). The proof of this conjecture is a great, if still remote, goal of number theory. On the way towards it, the insights into the nature of L -series which we have encountered may prove to be important.

Finally we want to mention that the French mathematicians *DANIEL BARSKY* and *PIERETTE CASSOU-NOGUÈS* have used *SHINTANI*’s result to prove the existence of p -adic L -series. These play a major rôle in *Iwasawa theory*, which we have mentioned before. The p -adic zeta function of a totally real number field K is a continuous function

$$\zeta_p : \mathbb{Z}_p \setminus \{1\} \longrightarrow \mathbb{Q}_p,$$

which is related to the ordinary Dedekind zeta function $\zeta_K(s)$ by

$$\zeta_p(-n) = \zeta_K(-n) \prod_{\mathfrak{p}|p} (1 - \mathfrak{N}(\mathfrak{p})^n)$$

for all $n \in \mathbb{N}$ such that $-n \equiv 1 \pmod{d}$, where $d = [K(\mu_{2p}) : K]$ denotes the degree of the field $K(\mu_{2p})$ of $2p$ -th roots of unity over K . The p -adic zeta function is uniquely determined by this relation. Its existence hinges on the fact that the rational values $\zeta_K(-n)$ are subjected to severe congruences with respect to p .

§ 10. Artin L -series

So far, all L -series we have considered were associated to an individual number field K . With the *Artin L -series*, a new type of L -series enters the stage; these are derived from representations of the Galois group $G(L|K)$ of a Galois extension $L|K$. This new kind of L -series is intimately related to the old ones via the main theorem of class field theory. In this way they appear as far-reaching generalizations of the old L -series. Let us explain this for the case of a Dirichlet L -series

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \chi(p)p^{-s}}$$

attached to a Dirichlet character

$$\chi : (\mathbb{Z}/m\mathbb{Z})^* \longrightarrow \mathbb{C}^*.$$

Let $G = G(\mathbb{Q}(\mu_m)|\mathbb{Q})$ be the Galois group of the field $\mathbb{Q}(\mu_m)$ of m -th roots of unity. The main theorem of class field theory in this particular case simply describes the familiar isomorphism

$$(\mathbb{Z}/m\mathbb{Z})^* \xrightarrow{\sim} G,$$

which sends the residue class $p \pmod{m}$ of a prime number $p \nmid m$ to the *Frobenius automorphism* φ_p , which in turn is defined by

$$\varphi_p \zeta = \zeta^p \quad \text{for } \zeta \in \mu_m.$$

Using this isomorphism we may interpret χ as a character of the Galois group G , or in other words, as a 1-dimensional *representation* of G , i.e., a homomorphism

$$\chi : G \longrightarrow GL_1(\mathbb{C}).$$

This interpretation describes the Dirichlet L -series in a purely Galois-theoretic fashion,

$$L(\chi, s) = \prod_{p \nmid m} \frac{1}{1 - \chi(\varphi_p)p^{-s}},$$

and allows us the following generalization.

Let $L|K$ be a Galois extension of finite algebraic number fields with Galois group $G = G(L|K)$. A **representation** of G is an action of G on a finite dimensional $\mathbb{C}$ -vector space V , i.e., a homomorphism

$$\rho : G \rightarrow GL(V) = \text{Aut}_{\mathbb{C}}(V).$$

Our shorthand notation for the action of $\sigma \in G$ on $v \in V$ is σv , instead of the complete expression $\rho(\sigma)v$. Let $\mathfrak{p}$ be a prime ideal of K , and let $\mathfrak{P}|\mathfrak{p}$ be a prime ideal of L lying above $\mathfrak{p}$. Let $G_{\mathfrak{P}}$ be the decomposition group and $I_{\mathfrak{P}}$ the inertia group of $\mathfrak{P}$ over $\mathfrak{p}$. Then we have a canonical isomorphism

$$G_{\mathfrak{P}}/I_{\mathfrak{P}} \xrightarrow{\sim} G(\kappa(\mathfrak{P})|\kappa(\mathfrak{p}))$$

onto the Galois group of the residue field extension $\kappa(\mathfrak{P})|\kappa(\mathfrak{p})$ (see chap. I, (9.5)). The factor group $G_{\mathfrak{P}}/I_{\mathfrak{P}}$ is therefore generated by the **Frobenius automorphism** $\varphi_{\mathfrak{P}}$ whose image in $G(\kappa(\mathfrak{P})|\kappa(\mathfrak{p}))$ is the q -th power map $x \mapsto x^q$, where $q = \mathfrak{N}(\mathfrak{p})$. $\varphi_{\mathfrak{P}}$ is an endomorphism of the module $V^{I_{\mathfrak{P}}}$ of invariants. The **characteristic polynomial**

$$\det(1 - \varphi_{\mathfrak{P}}t; V^{I_{\mathfrak{P}}})$$

only depends on the prime ideal $\mathfrak{p}$, not on the choice of the prime ideal $\mathfrak{P}$ above $\mathfrak{p}$. In fact, a different choice $\mathfrak{P}'|\mathfrak{p}$ yields an endomorphism conjugate to $\varphi_{\mathfrak{P}}$, as the decomposition groups $G_{\mathfrak{P}}$ and $G_{\mathfrak{P}'}$, the inertia groups $I_{\mathfrak{P}}$ and $I_{\mathfrak{P}'}$, and the Frobenius automorphisms $\varphi_{\mathfrak{P}}$ and $\varphi_{\mathfrak{P}'}$ are simultaneous conjugates. We thus arrive at the following

(10.1) Definition. Let $L|K$ be a Galois extension of algebraic number fields with Galois group G , and let (ρ, V) be a representation of G . Then the **Artin L -series** attached to ρ is defined to be

$$\mathcal{L}(L|K, \rho, s) = \prod_{\mathfrak{p}} \frac{1}{\det(1 - \varphi_{\mathfrak{P}}\mathfrak{N}(\mathfrak{p})^{-s}; V^{I_{\mathfrak{P}}})},$$

where $\mathfrak{p}$ runs through all prime ideals of K .

The Artin L -series converges absolutely and uniformly in the half-plane $\text{Re}(s) \geq 1 + \delta$, for any $\delta > 0$. It thus defines an analytic function on the half-plane $\text{Re}(s) > 1$. This is shown in the same way as for the Hecke L -series (see (8.1)), observing that the ε_i in the factorization

$$\det(1 - \varphi_{\mathfrak{P}}\mathfrak{N}(\mathfrak{p})^{-s}; V^{I_{\mathfrak{P}}}) = \prod_{i=1}^d (1 - \varepsilon_i \mathfrak{N}(\mathfrak{p})^{-s})$$

are roots of unity because the endomorphism $\varphi_{\mathfrak{P}}$ of $V^{I_{\mathfrak{P}}}$ has finite order.

For the trivial representation $(\rho, \mathbb{C})$, $\rho(\sigma) \equiv 1$, the Artin L -series is simply the Dedekind zeta function $\zeta_K(s)$. An additive expression analogous to the expansion

$$\zeta_K(s) = \sum_{\mathfrak{a}} \frac{1}{\mathfrak{N}(\mathfrak{a})^s}$$

does not exist for general Artin L -series. But they exhibit a perfectly regular behaviour under change of extensions $L|K$ and representations ρ . This allows to deduce many of their excellent properties. As a preparation for this study, we first collect basic facts from representation theory of finite groups. For their proofs we refer to [125].

The **degree** of a representation (ρ, V) of a finite group G is the dimension of V . The representation is called **irreducible** if the G -module V does not admit any proper G -invariant subspace. An irreducible representation of an *abelian* group is simply a character

$$\rho : G \longrightarrow \mathbb{C}^* = GL_1(\mathbb{C}).$$

Two representations (ρ, V) and (ρ', V') are called **equivalent** if the G -modules V and V' are isomorphic. Every representation (ρ, V) factors into a direct sum

$$V = V_1 \oplus \cdots \oplus V_s$$

of *irreducible* representations. If an irreducible representation (ρ_α, V_α) is equivalent to precisely r_α among the representations in this decomposition, then r_α is called the **multiplicity** of ρ_α in ρ , and one writes

$$\rho \sim \sum_{\alpha} r_{\alpha} \rho_{\alpha},$$

where ρ_α varies over all non-equivalent irreducible representations of G .

The **character** of a representation (ρ, V) is by definition the function

$$\chi_{\rho} : G \longrightarrow \mathbb{C}, \quad \chi_{\rho}(\sigma) = \text{trace } \rho(\sigma).$$

One has $\chi_{\rho}(1) = \dim V = \text{degree}(\rho)$, and $\chi_{\rho}(\sigma\tau\sigma^{-1}) = \chi_{\rho}(\tau)$ for all $\sigma, \tau \in G$. In general, a function $f : G \rightarrow \mathbb{C}$ with the property that $f(\sigma\tau\sigma^{-1}) = f(\tau)$ is called a **central** function (or **class function**). The special importance of characters comes from the following fact:

Two representations are equivalent if and only if their characters are equal. If $\rho \sim \sum_{\alpha} r_{\alpha} \rho_{\alpha}$, then

$$\chi_{\rho} = \sum_{\alpha} r_{\alpha} \chi_{\rho_{\alpha}}.$$

The character of the *trivial representation* $\rho : G \rightarrow GL(V)$, $\dim V = 1$, $\rho(\sigma) = 1$ for all $\sigma \in G$, is the constant function of value 1, and is denoted by $\mathbf{1}_G$, or simply $\mathbf{1}$. The *regular representation* is given by the G -module

$$V = \mathbb{C}[G] = \left\{ \sum_{\tau \in G} x_\tau \tau \mid x_\tau \in \mathbb{C} \right\},$$

on which the $\sigma \in G$ act via multiplication on the left. It decomposes into the direct sum of the trivial representation $V_0 = \mathbb{C} \sum_{\sigma \in G} \sigma$, and the *augmentation representation* $\left\{ \sum_{\sigma \in G} x_\sigma \sigma \mid \sum_{\sigma} x_\sigma = 0 \right\}$. The character associated with the regular, resp. the augmentation representation, is denoted by r_G , resp. u_G . We thus have $r_G = u_G + \mathbf{1}_G$, and explicitly: $r_G(\sigma) = 0$ for $\sigma \neq 1$, $r_G(1) = g = \#G$.

A character χ is called *irreducible* if it belongs to an irreducible representation. Every central function φ can be written uniquely as a linear combination

$$\varphi = \sum c_\chi \chi, \quad c_\chi \in \mathbb{C},$$

of irreducible characters. φ is a character of a representation of G if and only if the c_χ are rational integers ≥ 0 . For instance, for the character r_G of the regular representation we find

$$r_G = \sum \chi(1)\chi,$$

where χ varies over all irreducible characters of G . Given any two central functions φ and ψ of G , we put

$$(\varphi, \psi) = \frac{1}{g} \sum_{\sigma \in G} \varphi(\sigma) \overline{\psi(\sigma)}, \quad g = \#G,$$

where $\overline{\psi}$ is the function which is the complex conjugate of ψ . For two irreducible characters χ and χ' , this gives

$$(\chi, \chi') = \begin{cases} 1, & \text{if } \chi = \chi', \\ 0, & \text{if } \chi \neq \chi'. \end{cases}$$

In other words, $(,)$ is a hermitian scalar product on the space of all central functions on G , and the irreducible characters form an orthonormal basis of this hermitian space.

For the representations itself, this scalar product has the following meaning. Let

$$V = V_1 \oplus \cdots \oplus V_r$$

be the decomposition of a representation V with character χ into the direct sum of irreducible representations V_i . If V' is an irreducible representation with character χ' , then (χ, χ') is the number of times that V' occurs

among the V_i , up to isomorphism. For if χ_i is the character of V_i , then $\chi = \chi_1 + \cdots + \chi_r$, so that

$$(\chi, \chi') = (\chi_1, \chi') + \cdots + (\chi_r, \chi'),$$

and we have $(\chi_i, \chi') = 1$ or 0 , depending whether V_i is or is not isomorphic to V' . Applying this to the trivial representation $V' = \mathbb{C}$, we obtain in particular that

$$\dim V^G = \frac{1}{g} \sum_{\sigma \in G} \chi(\sigma), \quad g = \#G.$$

Now let $h : H \rightarrow G$ be a homomorphism of finite groups. If φ is a central function on G , then $h^*(\varphi) = \varphi \circ h$ is a central function on H . Conversely, one has the following proposition.

(10.2) Frobenius Reciprocity. *For every central function ψ on H there is one and only one central function $h_*(\psi)$ on G such that one has*

$$(\varphi, h_*(\psi)) = (h^*(\varphi), \psi)$$

for all central functions φ on G .

This will be applied chiefly to the following two special cases.

a) *H is a subgroup of G and h is inclusion.*

In this case we write $\varphi|H$ or simply φ instead of $h^*(\varphi)$, and ψ_* instead of $h_*(\psi)$ (the **induced function**). If φ is the character of a representation (ρ, V) of G , then $\varphi|H$ is the character of the representation $(\rho|H, V)$. If ψ is the character of a representation (ρ, V) of H , then ψ_* is the character of the representation $(\text{ind}(\rho), \text{Ind}_G^H(V))$ given by the **induced** G -module

$$\text{Ind}_G^H(V) = \{ f : G \rightarrow V \mid f(\tau x) = \tau f(x) \text{ for all } \tau \in H \},$$

on which $\sigma \in G$ acts by $(\sigma f)(x) = f(x\sigma)$ (see chap. IV, § 7). One has

$$\psi_*(\sigma) = \sum_{\tau} \psi(\tau \sigma \tau^{-1}),$$

where τ varies over a system of representatives on the right of G/H , and we put $\psi(\tau \sigma \tau^{-1}) = 0$ if $\tau \sigma \tau^{-1} \notin H$.

b) *G is a quotient group H/N of H and h is the projection.*

We then write φ instead of $h^*(\varphi)$, and $\psi_{\mathfrak{h}}$ instead of $h_*(\psi)$. One has

$$\psi_{\mathfrak{h}}(\sigma) = \frac{1}{\#N} \sum_{\tau \mapsto \sigma} \psi(\tau).$$

If φ is the character of a representation (ρ, V) of G , then $h^*(\varphi)$ is the character of the representation $(\rho \circ h, V)$.

The following result is of great importance.

(10.3) Brauer's Theorem. Every character χ of a finite group G is a $\mathbb{Z}$ -linear combination of characters χ_{i*} induced from characters χ_i of degree 1 associated to subgroups H_i of G .

Note that a character of degree 1 of a group H is simply a homomorphism $\chi : H \rightarrow \mathbb{C}^*$.

After this brief survey of representation theory for finite groups, we now return to Artin L -series. Since two representations (ρ, V) and (ρ', V') are equivalent if and only if their characters χ and χ' coincide, we will henceforth write

$$\mathcal{L}(L|K, \chi, s) = \prod_{\mathfrak{p}} \frac{1}{\det(1 - \rho(\varphi_{\mathfrak{p}}) \mathfrak{N}(\mathfrak{p})^{-s}; V^{I_{\mathfrak{p}}})}$$

instead of $\mathcal{L}(L|K, \rho, s)$. These L -series exhibit the following functorial behaviour.

(10.4) Proposition. (i) For the principal character $\chi = \mathbf{1}$, one has

$$\mathcal{L}(L|K, \mathbf{1}, s) = \zeta_K(s).$$

(ii) If χ, χ' are two characters of $G(L|K)$, then

$$\mathcal{L}(L|K, \chi + \chi', s) = \mathcal{L}(L|K, \chi, s) \mathcal{L}(L|K, \chi', s).$$

(iii) For a bigger Galois extension $L'|K$, $L' \supseteq L \supseteq K$, and a character χ of $G(L|K)$ one has

$$\mathcal{L}(L'|K, \chi, s) = \mathcal{L}(L|K, \chi, s).$$

(iv) If M is an intermediate field, $L \supseteq M \supseteq K$, and χ is a character of $G(L|M)$, then

$$\mathcal{L}(L|M, \chi, s) = \mathcal{L}(L|K, \chi_*, s).$$

Proof: We have already noted (i) earlier. (ii) If (ρ, V) , (ρ', V') are representations of $G(L|K)$ with characters χ, χ' , then the direct sum $(\rho \oplus \rho', V \oplus V')$ is a representation with character $\chi + \chi'$, and

$$\det(1 - \varphi_{\mathfrak{p}} t; (V \oplus V')^{I_{\mathfrak{p}}}) = \det(1 - \varphi_{\mathfrak{p}} t; V^{I_{\mathfrak{p}}}) \det(1 - \varphi_{\mathfrak{p}} t; V'^{I_{\mathfrak{p}}}).$$

This yields (ii).

(iii) Let $\mathfrak{P}'|\mathfrak{P}|\mathfrak{p}$ be prime ideals of $L'|L|K$, each lying above the next. Let χ be the character belonging to the $G(L|K)$ -module V . $G(L'|K)$ acts on V via the projection $G(L'|K) \rightarrow G(L|K)$. It induces surjective homomorphisms

$$G_{\mathfrak{P}'} \longrightarrow G_{\mathfrak{P}}, \quad I_{\mathfrak{P}'} \longrightarrow I_{\mathfrak{P}}, \quad G_{\mathfrak{P}'} / I_{\mathfrak{P}'} \longrightarrow G_{\mathfrak{P}} / I_{\mathfrak{P}}$$

of the decomposition and inertia groups. The latter maps the Frobenius automorphism $\varphi_{\mathfrak{P}'}$ to the Frobenius automorphism $\varphi_{\mathfrak{P}}$ so that $(\varphi_{\mathfrak{P}'}, V^{I_{\mathfrak{P}'}}) = (\varphi_{\mathfrak{P}}, V^{I_{\mathfrak{P}}})$, i.e.,

$$\det(1 - \varphi_{\mathfrak{P}'} t; V^{I_{\mathfrak{P}'}}) = \det(1 - \varphi_{\mathfrak{P}} t, V^{I_{\mathfrak{P}}}).$$

This yields (iii).

(iv) Let $G = G(L|K)$ and $H = G(L|M)$. Let $\mathfrak{p}$ be a prime ideal of K , $\mathfrak{q}_1, \dots, \mathfrak{q}_r$ the various prime ideals of M above $\mathfrak{p}$, and $\mathfrak{P}_i$ a prime ideal of L above $\mathfrak{q}_i$, $i = 1, \dots, r$. Let G_i , resp. I_i , be the decomposition, resp. inertia, group of $\mathfrak{P}_i$ over $\mathfrak{p}$. Then $H_i = G_i \cap H$, resp. $I'_i = I_i \cap H$, are the decomposition, resp. inertia, groups of $\mathfrak{P}_i$ over $\mathfrak{q}_i$. The degree of $\mathfrak{q}_i$ over $\mathfrak{p}$ is $f_i = (G_i : H_i I_i)$, i.e.,

$$\mathfrak{N}(\mathfrak{q}_i) = \mathfrak{N}(\mathfrak{p})^{f_i}.$$

We choose elements $\tau_i \in G$ such that $\mathfrak{P}_i = \mathfrak{P}_1^{\tau_i}$. Then $G_i = \tau_i^{-1} G_1 \tau_i$, and $I_i = \tau_i^{-1} I_1 \tau_i$. Let $\varphi \in G_1$ be an element which is mapped to the Frobenius $\varphi_{\mathfrak{P}_1} \in G_1/I_1$. Then $\varphi_i = \tau_i^{-1} \varphi \tau_i \in G_i$ is mapped to the Frobenius $\varphi_{\mathfrak{P}_i} \in G_i/I_i$, and the image of $\varphi_i^{f_i}$ in H_i/I'_i is the Frobenius of $\mathfrak{P}_i$ over $\mathfrak{q}_i$.

Now let $\rho : H \rightarrow GL(W)$ be a representation of H with character χ . Then χ_* is the character of the induced representation $\text{ind}(\rho) : G \rightarrow GL(V)$, $V = \text{Ind}_H^G(W)$. Clearly, what we have to show is that

$$\det(1 - \varphi t; V^{I_1}) = \prod_{i=1}^r \det(1 - \varphi_i^{f_i} t^{f_i}; W^{I'_i}).$$

We reduce the problem to the case $G_1 = G$, i.e., $r = 1$. Conjugating by τ_i , we obtain

$$\det(1 - \varphi_i^{f_i} t^{f_i}; W^{I'_i}) = \det(1 - \varphi^{f_i} t^{f_i}; (\tau_i W)^{I_1 \cap \tau_i H \tau_i^{-1}})$$

and $f_i = (G_1 : (G_1 \cap \tau_i H \tau_i^{-1}) I_1)$. For every i we choose a system of representatives on the left, σ_{ij} , of $G_1 \bmod G_1 \cap \tau_i H \tau_i^{-1}$. One checks immediately that then $\{\sigma_{ij} \tau_i\}$ is a system of representatives on the left of $G \bmod H$. We thus have (see chap. IV, § 5, p. 297) the decomposition

$$V = \bigoplus_{i,j} \sigma_{ij} \tau_i W.$$

Putting $V_i = \bigoplus_j \sigma_{ij} \tau_i W$, we obtain a decomposition $V = \bigoplus_i V_i$ of V as a G_1 -module. Hence

$$\det(1 - \varphi t; V^{I_1}) = \prod_{i=1}^r \det(1 - \varphi t; V_i^{I_1}).$$

It is therefore sufficient to prove that

$$\det(1 - \varphi t; V_i^{I_1}) = \det(1 - \varphi^{f_i} t^{f_i}; (\tau_i W)^{I_1 \cap \tau_i H \tau_i^{-1}}).$$

We simplify the notation by replacing G_1 by G , I_1 by I , $G_1 \cap \tau_i H \tau_i^{-1}$ by H , f_i by $f = (G : HI)$, V_i by V , and $\tau_i W$ by W . Then we have still $V = \text{Ind}_G^H(W)$, i.e., we are reduced to the case $r = 1$, $G_1 = G$.

We may further assume that $I = 1$. For if we put $\bar{G} = G/I$, $\bar{H} = H/I \cap H$, then $V^I = \text{Ind}_{\bar{G}}^{\bar{H}}(W^{I \cap H})$. Indeed, a function $f : G \rightarrow W$ in V is invariant under I if and only if one has $f(x\tau) = f(x)$ for all $\tau \in I$, i.e., if and only if it is constant on the right (and therefore also on the left) cosets of $G \bmod I$, i.e., if and only if it is a function on $\bar{G}$. It then automatically takes values in $W^{I \cap H}$, because $\tau f(x) = f(\tau x) = f(x)$ for $\tau \in I \cap H$.

So let $I = 1$. Then G is generated by φ , $f = (G : H)$, and thus

$$V = \bigoplus_{i=0}^{f-1} \varphi^i W.$$

Let A be the matrix of φ^f with respect to a basis $w_1, \dots, w_d$ of W . If E denotes the $(d \times d)$ unit matrix, then

$$\begin{pmatrix} 0 & E & \cdots & 0 \\ 0 & 0 & \cdots & E \\ A & 0 & \cdots & 0 \end{pmatrix}$$

is the matrix of φ with respect to the basis $\{\varphi^i w_j\}$ of V . This gives

$$\det(1 - \varphi t; V) = \det \begin{pmatrix} E & -tE & \cdots & 0 \\ 0 & 0 & \cdots & -tE \\ -tA & 0 & \cdots & E \end{pmatrix} = \det(1 - \varphi^f t^f; W)$$

as desired. The last identity is obtained by first multiplying the first column by t and adding it to the second, and then multiplying the second column by t and adding it to the third, etc. $\square$

The character $\mathbf{1}_*$ induced from the trivial character $\mathbf{1}$ of the subgroup $\{1\} \subseteq G(L|K)$ is the character $r_G = \sum_{\chi} \chi(1)\chi$ of the regular representation of $G(L|K)$. We therefore deduce from (10.4) the

(10.5) Corollary. *One has*

$$\zeta_L(s) = \zeta_K(s) \prod_{\chi \neq \mathbf{1}} \mathcal{L}(L|K, \chi, s)^{\chi(1)},$$

where χ varies over the nontrivial irreducible characters of $G(L|K)$.

The starting point of Artin's investigations on L -series had been the question whether, for a Galois extension $L|K$, the quotient $\zeta_L(s)/\zeta_K(s)$ is an entire function, i.e., a holomorphic function on the whole complex plane. Corollary (10.5) shows that this could be deduced from the famous

Artin Conjecture: For every irreducible character $\chi \neq \mathbf{1}$, the Artin L -series $\mathcal{L}(L|K, \chi, s)$ defines an **entire** function.

We will see presently that this conjecture holds for *abelian* extensions. In general it is not known. In view of its momentous consequences, it constitutes one of the big challenges in number theory.

We will show next that the Artin L -series in the case of *abelian* extensions $L|K$ coincide with certain Hecke L -series, more precisely, with generalized Dirichlet L -series. This means that the properties of Hecke's series, and in particular their functional equation, transfer to Artin series in the abelian case. Via functoriality (10.4) they may then be extended to the non-abelian case.

The link between Artin and Hecke L -series is provided by class field theory. Let $L|K$ be an abelian extension, and let $\mathfrak{f}$ be the **conductor** of $L|K$, i.e., the smallest module

$$\mathfrak{f} = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{n_{\mathfrak{p}}}$$

such that $L|K$ lies in the ray class field $K^{\mathfrak{f}}|K$ (see chap. VI, (6.2)). The **Artin symbol** $\left(\frac{L|K}{a}\right)$ then gives us a surjective homomorphism

$$J^{\mathfrak{f}}/P^{\mathfrak{f}} \longrightarrow G(L|K), \quad a \bmod P^{\mathfrak{f}} \longmapsto \left(\frac{L|K}{a}\right),$$

from the **ray class group** $J^{\mathfrak{f}}/P^{\mathfrak{f}}$. Here $J^{\mathfrak{f}}$ is the group of fractional ideals prime to $\mathfrak{f}$, and $P^{\mathfrak{f}}$ is the group of principal ideals (a) such that $a \equiv 1 \bmod \mathfrak{f}$ and a is positive in $K_{\mathfrak{p}} = \mathbb{R}$ if $\mathfrak{p}$ is real.

Now let χ be an *irreducible* character of the abelian group $G(L|K)$, i.e., a homomorphism

$$\chi : G(L|K) \rightarrow \mathbb{C}^*.$$

Composing with the Artin symbol $\left(\frac{L|K}{a}\right)$, this gives a character of the ray class group $J^{\mathfrak{f}}/P^{\mathfrak{f}}$, i.e., a Dirichlet character $\bmod \mathfrak{f}$. It induces a character on $J^{\mathfrak{f}}$, which we denote by

$$\tilde{\chi} : J^{\mathfrak{f}} \rightarrow \mathbb{C}^*.$$

By (6.9), this character on ideals is a **Größencharacter** $\bmod \mathfrak{f}$ of type $(p, 0)$, and we have the

(10.6) Theorem. Let $L|K$ be an abelian extension, let $\mathfrak{f}$ be the conductor of $L|K$, let $\chi \neq 1$ be an irreducible character of $G(L|K)$, and $\tilde{\chi}$ the associated Größencharakter mod $\mathfrak{f}$.

Then the Artin L -series for the character χ and the Hecke L -series for the Größencharakter $\tilde{\chi}$ satisfy the identity

$$\mathcal{L}(L|K, \chi, s) = \prod_{\mathfrak{p} \in S} \frac{1}{1 - \chi(\varphi_{\mathfrak{p}})\mathfrak{N}(\mathfrak{p})^{-s}} L(\tilde{\chi}, s),$$

where $S = \{\mathfrak{p} | \mathfrak{f} \mid \chi(I_{\mathfrak{p}}) \neq 1\}$.

Proof: The representation of $G(L|K)$ associated to the character χ is given by a 1-dimensional vector space $V = \mathbb{C}$ on which $G(L|K)$ acts via multiplication by χ , i.e., $\sigma v = \chi(\sigma)v$. Since $\mathfrak{f}$ is the conductor of $L|K$, we find by chap. VI, (6.6), that

$$\mathfrak{p} | \mathfrak{f} \iff \mathfrak{p} \text{ is ramified} \iff I_{\mathfrak{p}} \neq 1.$$

If $\chi(I_{\mathfrak{p}}) \neq 1$, then $V^{I_{\mathfrak{p}}} = \{0\}$, and the corresponding Euler factor does not occur in the Artin L -series. If on the other hand $\chi(I_{\mathfrak{p}}) = 1$, then $V^{I_{\mathfrak{p}}} = \mathbb{C}$, so that

$$\det(1 - \varphi_{\mathfrak{p}}\mathfrak{N}(\mathfrak{p})^{-s}; V^{I_{\mathfrak{p}}}) = 1 - \chi(\varphi_{\mathfrak{p}})\mathfrak{N}(\mathfrak{p})^{-s}.$$

We thus have

$$\mathcal{L}(L|K, \chi, s) = \prod_{\mathfrak{p} \nmid \mathfrak{f}} \frac{1}{1 - \chi(\varphi_{\mathfrak{p}})\mathfrak{N}(\mathfrak{p})^{-s}} \prod_{\mathfrak{p} \in S} \frac{1}{1 - \chi(\varphi_{\mathfrak{p}})\mathfrak{N}(\mathfrak{p})^{-s}}$$

and

$$L(\tilde{\chi}, s) = \prod_{\mathfrak{p} \nmid \mathfrak{f}} \frac{1}{1 - \tilde{\chi}(\mathfrak{p})\mathfrak{N}(\mathfrak{p})^{-s}}.$$

For $\mathfrak{p} \nmid \mathfrak{f}$, one has $(\frac{L|K}{\mathfrak{p}}) = \varphi_{\mathfrak{p}}$, and so $\tilde{\chi}(\mathfrak{p}) = \chi(\varphi_{\mathfrak{p}})$. This proves the claim. $\square$

Remark: If the character $\chi : G(L|K) \rightarrow \mathbb{C}^*$ is *injective*, then $S = \emptyset$, and one has complete equality

$$\mathcal{L}(L|K, \chi, s) = L(\tilde{\chi}, s).$$

In this case $\tilde{\chi}$ is a *primitive Größencharakter* mod $\mathfrak{f}$.

If on the other hand χ is the trivial character 1_G , then $\tilde{\chi}$ is the trivial Dirichlet character mod $\mathfrak{f}$, and we have

$$\zeta_K(s) = \prod_{\mathfrak{p} \nmid \mathfrak{f}} \frac{1}{1 - \mathfrak{N}(\mathfrak{p})^{-s}} L(\tilde{\chi}, s).$$

The theorem implies that the **Artin conjecture** holds for all Artin L -series $\mathcal{L}(L|K, \chi, s)$ which correspond to nontrivial irreducible characters χ of *abelian* Galois groups $G(L|K)$. For if L_χ is the fixed field of the kernel of χ and $\tilde{\chi}$ is the *Größencharakter* associated with $\chi : G(L_\chi|K) \hookrightarrow \mathbb{C}^*$, then the above remark shows that $\mathcal{L}(L|K, \chi, s) = \mathcal{L}(L_\chi|K, \chi, s) = L(\tilde{\chi}, s)$. Hence $\mathcal{L}(L|K, \chi, s)$ is holomorphic on all of $\mathbb{C}$, because the same is true for $L(\tilde{\chi}, s)$, as was shown in (8.5). This also settles the Artin conjecture for every solvable extension $L|K$.

Our goal now is to prove a functional equation for Artin L -series. The basis for this will be the above theorem and the functional equation we have already established for Hecke L -series. We however have to complete the Artin L -series by the right “Euler factors” at the infinite places. In looking for these Euler factors, the first natural guideline is provided by the case of Hecke L -series. But in order to go the whole way, we need an additional Galois-theoretic complement which will be dealt with in the next section.

§ 11. The Artin Conductor

The discriminant $\mathfrak{d} = \mathfrak{d}_{L|K}$ of a Galois extension $L|K$ of algebraic number fields admits a fine structure based on group theory. It is expressed by a product decomposition

$$\mathfrak{d} = \prod \mathfrak{f}(\chi)^{\chi(1)},$$

where χ varies over the irreducible characters of the Galois group $G = G(L|K)$. The ideals $\mathfrak{f}(\chi)$ are given by

$$\mathfrak{f}(\chi) = \prod_{\mathfrak{p} \nmid \infty} \mathfrak{p}^{f_{\mathfrak{p}}(\chi)}$$

with

$$f_{\mathfrak{p}}(\chi) = \sum_{i \geq 0} \frac{g_i}{g_0} \text{codim } V^{G_i},$$

where V is a representation with character χ , G_i is the i -th ramification group of $L_{\mathfrak{p}}|K_{\mathfrak{p}}$, and g_i denotes its order. This discovery goes back to *EMIL ARTIN* and *HELMUT HASSE*. The ideals $\mathfrak{f}(\chi)$ are called **Artin conductors**. They play an important rôle in the functional equation of the Artin L -series, which we are going to prove in the next section. Here we collect the properties needed for this, following essentially the treatment given by *J.-P. SERRE* in [122].

First let us consider a Galois extension $L|K$ of **local fields**, with Galois group $G = G(L|K)$. Let $f = f_{L|K} = [\lambda : \kappa]$ be the inertia degree of $L|K$. In chap. II, § 10, we defined, for any $\sigma \in G$,

$$i_G(\sigma) = v_L(\sigma x - x),$$

where x is an element such that $\mathcal{O}_L = \mathcal{O}_K[x]$, and v_L is the normalized valuation of L . With this notation we can write the i -th ramification group as

$$G_i = \{ \sigma \in G \mid i_G(\sigma) \geq i + 1 \}.$$

One has $i_G(\tau\sigma\tau^{-1}) = i_G(\sigma)$, and $i_H(\sigma) = i_G(\sigma)$ for every subgroup $H \subseteq G$. If $L|K$ is unramified, then $i_G(\sigma) = 0$ for all $\sigma \in G$, $\sigma \neq 1$. We put

$$a_G(\sigma) = \begin{cases} -fi_G(\sigma) & \text{for } \sigma \neq 1, \\ f \sum_{\tau \neq 1} i_G(\tau) & \text{for } \sigma = 1. \end{cases}$$

a_G is a central function on G , and we have

$$(a_G, \mathbf{1}_G) = \frac{1}{\#G} \sum_{\sigma \in G} a_G(\sigma) = 0.$$

We may therefore write

$$a_G = \sum_{\chi} f(\chi) \chi, \quad f(\chi) \in \mathbb{C},$$

with χ varying over the irreducible characters of G . Our chief problem is to prove that the coefficients $f(\chi)$ are rational integers ≥ 0 . Once we have shown this, we may form the ideal $\mathfrak{f}_{\mathfrak{p}}(\chi) = \mathfrak{p}^{f(\chi)}$, which will be the $\mathfrak{p}$ -component of the global Artin conductor that we want. First we prove that the function a_G satisfies the following properties (we use the notation of the preceding section).

(11.1) Proposition. (i) *If H is a normal subgroup of G , then*

$$a_{G/H} = (a_G)_{\mathfrak{H}}.$$

(ii) *If H is any subgroup of G , and if K' is the fixed field with discriminant $\mathfrak{d}_{K'|K} = \mathfrak{p}^v$, then*

$$a_G|H = v r_H + f_{K'|K} a_H.$$

(iii) *Let G_i be the i -th ramification group of G , u_i the augmentation character of G_i , and $(u_i)_*$ the character of G induced from u_i . Then one has*

$$a_G = \sum_{i=0}^{\infty} \frac{1}{(G_0 : G_i)} (u_i)_*.$$

Proof: (i) follows immediately from chap. II, (10.5).

(ii) Let $\sigma \in H$, $\sigma \neq 1$. Then

$$a_G(\sigma) = -f_{L|K} i_G(\sigma), \quad a_H(\sigma) = -f_{L|K'} i_H(\sigma), \quad r_H(\sigma) = 0.$$

Since $i_G(\sigma) = i_H(\sigma)$ and $f_{L|K} = f_{L|K'} f_{K'|K}$, this implies

$$a_G(\sigma) = v r_H(\sigma) + f_{K'|K} a_H(\sigma).$$

Now let $\sigma = 1$, and let $\mathfrak{D}_{L|K}$ be the *different* of $L|K$. Let $\mathcal{O}_L = \mathcal{O}_K[x]$ and $g(X)$ be the minimal polynomial of x over K . By chap. III, (2.4), $\mathfrak{D}_{L|K}$ is then generated by $g'(x) = \prod_{\sigma \neq 1} (\sigma x - x)$. Consequently,

$$v_L(\mathfrak{D}_{L|K}) = v_L(g'(x)) = \sum_{\sigma \neq 1} i_G(\sigma) = \frac{1}{f_{L|K}} a_G(1).$$

By chap. III, (2.9), we know, on the other hand, that $\mathfrak{d}_{L|K} = N_{L|K}(\mathfrak{D}_{L|K})$, so $v_K \circ N_{L|K} = f_{L|K} v_L$ gives the identity

$$a_G(1) = f_{L|K} v_L(\mathfrak{D}_{L|K}) = v_K(\mathfrak{d}_{L|K}),$$

and in the same way $a_H(1) = v_{K'}(\mathfrak{d}_{L|K'})$. From chap. III, (2.10), we get furthermore that

$$\mathfrak{d}_{L|K} = (\mathfrak{d}_{K'|K})^{[L:K']} N_{K'|K}(\mathfrak{d}_{L|K'}).$$

Thus $r_H(1) = [L : K']$ and $v = v_K(\mathfrak{d}_{K'|K})$ yields the formula

$$a_G(1) = [L : K'] v_K(\mathfrak{d}_{K'|K}) + f_{K'|K} v_{K'}(\mathfrak{d}_{L|K'}) = v r_H(1) + f_{K'|K} a_H(1).$$

(iii) Let $g_i = \#G_i$, $g = \#G$. Since G_i is invariant in G , we have $(u_i)_*(\sigma) = 0$ if $\sigma \notin G_i$, and $(u_i)_*(\sigma) = -g/g_i = -f \cdot g_0/g_i$ if $\sigma \in G_i$, $\sigma \neq 1$, and $\sum_{\sigma \in G} (u_i)_*(\sigma) = 0$. For $\sigma \in G_k \setminus G_{k+1}$, we thus find

$$a_G(\sigma) = -f(k+1) = \sum \frac{1}{(G_0 : G_i)} (u_i)_*(\sigma).$$

This implies the identity for the case $\sigma = 1$ as well, since both sides are orthogonal to $\mathbf{1}_G$. $\square$

For the coefficients $f(\chi)$ in the linear combination

$$a_G = \sum f(\chi) \chi,$$

we have, in view of $a_G(\sigma^{-1}) = a_G(\sigma)$, that

$$f(\chi) = (a_G, \chi) = \frac{1}{g} \sum_{\sigma \in G} a_G(\sigma) \chi(\sigma^{-1}) = \frac{1}{g} \sum_{\sigma \in G} a_G(\sigma^{-1}) \chi(\sigma) = (\chi, a_G),$$

$g = \#G$. For any central function φ of G , we put

$$f(\varphi) = (\varphi, a_G)$$

and

$$\varphi(G_i) = \frac{1}{g_i} \sum_{\sigma \in G_i} \varphi(\sigma), \quad g_i = \#G_i.$$

(11.2) Proposition. (i) If φ is a central function on the quotient group G/H , and φ' is the corresponding central function on G , then

$$f(\varphi) = f(\varphi').$$

(ii) If φ is a central function on a subgroup H of G , and φ_* is the central function induced by φ on G , then

$$f(\varphi_*) = v_K(\mathfrak{d}_{K'|K})\varphi(1) + f_{K'|K}f(\varphi).$$

(iii) For a central function φ on G , one has

$$f(\varphi) = \sum_{i \geq 0} \frac{g_i}{g_0} (\varphi(1) - \varphi(G_i)).$$

Proof: (i) $f(\varphi) = (\varphi, a_{G/H}) = (\varphi, (a_G)_H) = (\varphi', a_G) = f(\varphi')$.

(ii) $f(\varphi_*) = (\varphi_*, a_G) = (\varphi, a_G|H) = v(\varphi, r_H) + f_{K'|K}(\varphi, a_H) = v\varphi(1) + f_{K'|K}f(\varphi)$ with $v = v_K(\mathfrak{d}_{K'|K})$.

(iii) We have $(\varphi, (u_i)_*) = (\varphi|G_i, u_i) = \varphi(1) - \varphi(G_i)$, so the formula follows from (11.1), (iii). $\square$

If χ is the character of a representation (ρ, V) of G , then $\chi(1) = \dim V$ and $\chi(G_i) = \dim V^{G_i}$, hence

$$f(\chi) = \sum_{i \geq 0} \frac{g_i}{g_0} \operatorname{codim} V^{G_i}.$$

Now consider the function

$$\eta_{L|K}(s) = \int_0^s \frac{dx}{(G_0 : G_x)},$$

which was introduced in chap. II, § 10. For integers $m \geq -1$, it is given by $\eta_{L|K}(-1) = -1$, $\eta_{L|K}(0) = 0$, and

$$\eta_{L|K}(m) = \sum_{i=1}^m \frac{g_i}{g_0} \quad \text{for } m \geq 1.$$

The theorem of *HASSE-ARF* (see chap. V, (6.3)) now gives us the following integrality statement for the number $f(\chi)$ in the case of a character χ of degree 1.

(11.3) Proposition. Let χ be a character of G of degree 1. Let j be the biggest integer such that $\chi|G_j \neq \mathbf{1}_{G_j}$ (when $\chi = \mathbf{1}_G$ we put $j = -1$). Then we have

$$f(\chi) = \eta_{L|K}(j) + 1,$$

and this is a rational integer ≥ 0 .

Proof: If $i \leq j$, then $\chi(G_i) = 0$, so that $\chi(1) - \chi(G_i) = 1$. If $i > j$, then $\chi(G_i) = 1$, and so $\chi(1) - \chi(G_i) = 0$. From (11.2), (iii), it thus follows that

$$f(\chi) = \sum_{i=0}^j \frac{g_i}{g_0} = \eta_{L|K}(j) + 1,$$

provided $j \geq 0$. If $j = -1$, we have $\chi(1) - \chi(G_i) = 0$ for all $i \geq 0$, and hence by (11.2), (iii), $f(\chi) = 0 = \eta_{L|K}(-1) + 1$.

Let H be the kernel of χ and L' the fixed field of H . By Herbrand's theorem (chap. II, (10.7)) one has

$$G_j(L|K)H/H = G_{j'}(L'|K) \quad \text{with} \quad j' = \eta_{L|L'}(j).$$

In terms of the upper numbering of the ramification groups, this translates into

$$G^t(L|K)H/H = G^t(L'|K),$$

where $t = \eta_{L|K}(j) = \eta_{L'|K}(\eta_{L|L'}(j)) = \eta_{L'|K}(j')$ (see chap. II, (10.8)). But $\chi(G_j(L|K)H/H) \neq 1$, and $\chi(G_{j+\delta}(L|K)H/H) = \chi(G_{j+1}(L|K)H/H) = 1$ for all $\delta > 0$, and in particular $G_j(L|K)H/H \neq G_{j+\delta}(L|K)H/H$ for all $\delta > 0$. Since $\eta_{L|K}(s)$ is continuous and strictly increasing, it follows that

$$G^t(L'|K) = G^t(L|K)H/H \neq G^{t+\varepsilon}(L|K)H/H = G^{t+\varepsilon}(L'|K)$$

for all $\varepsilon > 0$, i.e., t is a *jump* in the ramification filtration of $L'|K$. The extension $L'|K$ is abelian and therefore $t = \eta_{L|K}(j)$ is an integer, by the theorem of HASSE and ARF. $\square$

Now let χ be an arbitrary character of the Galois group $G = G(L|K)$. By Brauer's theorem (10.3), we then have

$$\chi = \sum n_i \chi_{i*}, \quad n_i \in \mathbb{Z},$$

where χ_{i*} is the character induced from a character χ_i of degree 1 of a subgroup H_i . By (11.2), (ii), we have

$$f(\chi) = \sum n_i f(\chi_{i*}) = \sum n_i (v_K(\mathfrak{d}_{K_i|K}) \chi_i(1) + f_{K_i|K} f(\chi_i)),$$

where K_i is the fixed field of H_i . Therefore $f(\chi)$ is a rational integer. On the other hand, (11.1), (iii) shows that $g_0 a_G$ is the character of a representation of G , so $g_0 f(\chi) = (\chi, g_0 a_G) \geq 0$. We have thus established the

(11.4) Theorem. *If χ is a character of the Galois group $G = G(L|K)$, then $f(\chi)$ is a rational integer ≥ 0 .*

(11.5) Definition. We define the (local) **Artin conductor** of the character χ of $G = G(L|K)$ to be the ideal

$$\mathfrak{f}_{\mathfrak{p}}(\chi) = \mathfrak{p}^{f(\chi)}.$$

In chap. V, (1.6), we defined the **conductor** of an *abelian* extension $L|K$ of local fields to be the smallest power of $\mathfrak{p}$, $\mathfrak{f} = \mathfrak{p}^n$, such that the n -th higher unit group $U_K^{(n)}$ is contained in the norm group $N_{L|K}L^*$. The latter is the kernel of the norm residue symbol

$$(\ , L|K) : K^* \longrightarrow G(L|K),$$

which maps $U_K^{(i)}$ to the higher ramification group $G^i(L|K) = G_j(L|K)$ with $i = \eta_{L|K}(j)$ — see V, (6.2). The conductor $\mathfrak{f} = \mathfrak{p}^n$ is therefore given by the smallest integer $n \geq 0$ such that $G^n(L|K) = 1$. From (11.3) we thus obtain the following result.

(11.6) Proposition. Let $L|K$ be a Galois extension of local fields, and let χ be a character of $G(L|K)$ of degree 1. Let L_χ be the fixed field of the kernel of χ , and $\mathfrak{f}$ the conductor of $L_\chi|K$. Then one has

$$\mathfrak{f} = \mathfrak{f}_{\mathfrak{p}}(\chi).$$

Proof: By (11.3), we have $f(\chi) = \eta_{L|K}(j) + 1$, where j is the largest integer such that $G_j(L|K) \not\subseteq G(L|L_\chi) =: H$. Let $t = \eta_{L|K}(j)$. Then one has

$$G^t(L_\chi|K) = G^t(L|K)H/H = G_j(L|K)H/H,$$

and $G^{t+\varepsilon}(L_\chi|K) \subseteq G_{j+1}(L|K)H/H = 1$ for all $\varepsilon > 0$. Hence t is the largest number such that $G^t(L_\chi|K) \neq 1$. By the theorem of *HASSE-ARF*, t is an integer, and we conclude that $f(\chi) = t + 1$ is the smallest integer such that $G^{f(\chi)}(L_\chi|K) = 1$, i.e., $f(\chi) = n$. $\square$

We now leave the local situation, and suppose that $L|K$ is a Galois extension of **global** fields. Let $\mathfrak{p}$ be a prime ideal of K , $\mathfrak{P}|\mathfrak{p}$ a prime ideal of L lying above $\mathfrak{p}$. Let $L_{\mathfrak{P}}|K_{\mathfrak{p}}$ be the completion of $L|K$, and $G_{\mathfrak{P}} = G(L_{\mathfrak{P}}|K_{\mathfrak{p}})$ the decomposition group of $\mathfrak{P}$ over K . We denote the function $a_{G_{\mathfrak{P}}}$ on $G_{\mathfrak{P}}$ by $a_{\mathfrak{P}}$, and extend it to $G = G(L|K)$ by zero. The central function

$$a_{\mathfrak{p}} = \sum_{\mathfrak{P}|\mathfrak{p}} a_{\mathfrak{P}}$$

immediately turns out to be the function $(a_{\mathfrak{p}})_*$ induced by $a_{\mathfrak{p}}|G_{\mathfrak{p}}$. It is therefore the character of a representation of G . If now χ is a character of G , then we put

$$f(\chi, \mathfrak{p}) = (\chi, a_{\mathfrak{p}}) = f(\chi|G_{\mathfrak{p}}).$$

Then $f_{\mathfrak{p}}(\chi) = \mathfrak{p}^{f(\chi, \mathfrak{p})}$ is the Artin conductor of the restriction of χ to $G_{\mathfrak{p}} = G(L_{\mathfrak{p}}|K_{\mathfrak{p}})$. In particular, we have $f_{\mathfrak{p}}(\chi) = 1$ if $\mathfrak{p}$ is unramified. We define the (global) **Artin conductor** of χ to be the product

$$f(\chi) = \prod_{\mathfrak{p}|\infty} f_{\mathfrak{p}}(\chi).$$

Whenever precision is called for, we write $f(L|K, \chi)$ instead of $f(\chi)$. The properties (11.2) of the numbers $f(\chi, \mathfrak{p})$ transfer immediately to the Artin conductor $f(\chi)$, and we obtain the

(11.7) Proposition. (i) $f(\chi + \chi') = f(\chi)f(\chi')$, $f(1) = (1)$.

(ii) If $L'|K$ is a Galois subextension of $L|K$, and χ is a character of $G(L'|K)$, then

$$f(L|K, \chi) = f(L'|K, \chi).$$

(iii) If H is a subgroup of G with fixed field K' , and if χ is a character of H , then

$$f(L|K, \chi_*) = \mathfrak{d}_{K'|K}^{\chi(1)} N_{K'|K} (f(L|K', \chi)).$$

Proof: (i) and (ii) are trivial. To prove (iii), we choose a fixed prime ideal $\mathfrak{P}$ of L , put

$$G = G(L|K), \quad H = G(L|K'), \quad G_{\mathfrak{P}} = G(L_{\mathfrak{P}}|K_{\mathfrak{P}}),$$

with $\mathfrak{p} = \mathfrak{P} \cap K$, and consider the decomposition

$$G = \bigcup_{\tau} G_{\mathfrak{P}} \tau H$$

into double cosets. Then representation theory yields the following formula for the character χ of H :

$$(*) \quad \chi_*|G_{\mathfrak{P}} = \sum_{\tau} \chi_*^{\tau},$$

where χ^{τ} is the character $\chi^{\tau}(\sigma) = \chi(\tau^{-1}\sigma\tau)$ of $G_{\mathfrak{P}} \cap \tau H \tau^{-1}$, and χ_*^{τ} is the character of $G_{\mathfrak{P}}$ induced by χ^{τ} (see [119], chap. 7, prop. 22). Furthermore $\mathfrak{P}'_{\tau} = \mathfrak{P}^{\tau} \cap K'$ are the different prime ideals of K' above $\mathfrak{p}$ (see chap. I, § 9, p. 55), and we have

$$G_{\mathfrak{P}'_{\tau}} = \tau^{-1} G_{\mathfrak{P}} \tau = G(L_{\mathfrak{P}^{\tau}}|K_{\mathfrak{p}}), \quad H_{\mathfrak{P}'_{\tau}} = G_{\mathfrak{P}'_{\tau}} \cap H = G(L_{\mathfrak{P}'_{\tau}}|K'_{\mathfrak{p}'_{\tau}}).$$

Now let $\mathfrak{d}_{\mathfrak{P}'_\tau} = \mathfrak{p}^{v_{\mathfrak{P}'_\tau}}$ be the discriminant ideal of $K'_{\mathfrak{P}'_\tau} | K_{\mathfrak{p}}$, and let $f_{\mathfrak{P}'_\tau}$ be the degree of $\mathfrak{P}'_\tau$ over K . Thus $N_{K'|K}(\mathfrak{P}'_\tau) = \mathfrak{p}^{f_{\mathfrak{P}'_\tau}}$. Since

$$\mathfrak{f}_{\mathfrak{p}}(L|K, \chi_*) = \mathfrak{p}^{f(\chi_*|G_{\mathfrak{p}})} \quad \text{and} \quad \mathfrak{f}_{\mathfrak{P}'_\tau}(L|K', \chi) = \mathfrak{P}'_\tau^{f(\chi|H_{\mathfrak{P}'_\tau})},$$

we have to show that

$$f(\chi_*|G_{\mathfrak{p}}) = \sum_{\tau} v_{\mathfrak{P}'_\tau} \chi(1) + f_{\mathfrak{P}'_\tau} f(\chi|H_{\mathfrak{P}'_\tau}),$$

or, in view of (11.2), (ii), that

$$(**) \quad f(\chi_*|G_{\mathfrak{p}}) = \sum_{\tau} f((\chi|H_{\mathfrak{P}'_\tau})_*).$$

But $H_{\mathfrak{P}'_\tau} = \tau^{-1}(G_{\mathfrak{p}} \cap \tau H \tau^{-1})\tau$, and $\chi|H_{\mathfrak{P}'_\tau}$, resp. $(\chi|H_{\mathfrak{P}'_\tau})_*$, arises by conjugation $\sigma \mapsto \tau \sigma \tau^{-1}$ from χ^τ , resp. χ_*^τ . Therefore $f((\chi|H_{\mathfrak{P}'_\tau})_*) = f(\chi_*^\tau)$, and (**) follows from (*). $\square$

We apply (iii) to the case $\chi = \mathbf{1}_H$, and denote the induced character χ_* by $s_{G/H}$. Since $\mathfrak{f}(\chi) = 1$, we obtain the

(11.8) Corollary. $\mathfrak{d}_{K'|K} = \mathfrak{f}(L|K, s_{G/H})$.

If in particular $H = \{1\}$, then $s_{G/H}$ is the character r_G of the regular representation. Its decomposition into irreducible characters χ is given by

$$r_G = \sum_{\chi} \chi(1) \chi.$$

This yields the

(11.9) Conductor-Discriminant-Formula. *For an arbitrary Galois extension $L|K$ of global fields, one has*

$$\mathfrak{d}_{L|K} = \prod_{\chi} \mathfrak{f}(\chi)^{\chi(1)},$$

where χ varies over the irreducible characters of $G(L|K)$.

For an *abelian* extension $L|K$ of global fields, we defined the conductor $\mathfrak{f}$ in VI, (6.4). By chap. VI, (6.5), it is the product

$$\mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}}$$

of the conductors $\mathfrak{f}_{\mathfrak{p}}$ of the local extensions $L_{\mathfrak{p}}|K_{\mathfrak{p}}$. (11.6) now gives rise to the following

(11.10) Proposition. *Let $L|K$ be a Galois extension of global fields, χ a character of $G(L|K)$ of degree 1, L_χ the fixed field of the kernel of χ , and $\mathfrak{f}$ the conductor of $L_\chi|K$. Then one has*

$$\mathfrak{f} = \mathfrak{f}(\chi).$$

Now let $L|K$ be a Galois extension of algebraic number fields. We form the ideal

$$\mathfrak{c}(L|K, \chi) = \mathfrak{d}_{K|\mathbb{Q}}^{\chi(1)} N_{K|\mathbb{Q}}(\mathfrak{f}(L|K, \chi))$$

of $\mathbb{Z}$. The positive generator of this ideal is the integer

$$c(L|K, \chi) = |d_K|^{\chi(1)} \mathfrak{N}(\mathfrak{f}(L|K, \chi)).$$

Applying (11.7) and observing the transitivity of the discriminant (chap. III, (2.10)), we get the

(11.11) Proposition. (i) $c(L|K, \chi + \chi') = c(L|K, \chi)c(L|K, \chi')$, $c(L|K, \mathbf{1}) = |d_K|$,

(ii) $c(L|K, \chi) = c(L'|K, \chi)$,

(iii) $c(L|K, \chi_*) = c(L|K', \chi)$.

Here the notation is that of (11.7).

§ 12. The Functional Equation of Artin L -series

The first task is to complete the Artin L -series

$$\mathcal{L}(L|K, \chi, s) = \prod_{\mathfrak{p} \nmid \infty} \frac{1}{\det(1 - \varphi_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{-s}; V^I_{\mathfrak{p}})},$$

for the character χ of $G = G(L|K)$, by the appropriate gamma factors. For every infinite place $\mathfrak{p}$ of K we put

$$\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s) = \begin{cases} L_{\mathbb{C}}(s)^{\chi(1)}, & \text{if } \mathfrak{p} \text{ is complex,} \\ L_{\mathbb{R}}(s)^{n^+} L_{\mathbb{R}}(s+1)^{n^-}, & \text{if } \mathfrak{p} \text{ is real,} \end{cases}$$

with the exponents $n^+ = \frac{\chi(1) + \chi(\varphi_{\mathfrak{p}})}{2}$, $n^- = \frac{\chi(1) - \chi(\varphi_{\mathfrak{p}})}{2}$. Here $\varphi_{\mathfrak{p}}$ is the distinguished generator of $G(L_{\mathfrak{p}}|K_{\mathfrak{p}})$, and

$$L_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2), \quad L_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s)$$

(see §4). For $\mathfrak{p}$ real, the exponents n^+ , n^- in $\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s)$ have the following meaning.

The involution $\varphi_{\mathfrak{p}}$ on V induces an eigenspace decomposition $V = V^+ \oplus V^-$, where

$$V^+ = \{x \in V \mid \varphi_{\mathfrak{p}}x = x\}, \quad V^- = \{x \in V \mid \varphi_{\mathfrak{p}}x = -x\},$$

and it follows from the remark in § 10, p. 521, that

$$\dim V^+ = \frac{1}{2}(\chi(1) + \chi(\varphi_{\mathfrak{p}})), \quad \dim V^- = \frac{1}{2}(\chi(1) - \chi(\varphi_{\mathfrak{p}})).$$

The functions $\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s)$ exhibit the same behaviour under change of fields and characters as the L -series and the Artin conductor.

(12.1) Proposition. (i) $\mathcal{L}_{\mathfrak{p}}(L|K, \chi + \chi', s) = \mathcal{L}_{\mathfrak{p}}(L|K, \chi, s)\mathcal{L}_{\mathfrak{p}}(L|K, \chi', s)$.

(ii) If $L'|K$ is a Galois subextension of $L|K$ and χ a character of $G(L'|K)$, then

$$\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s) = \mathcal{L}_{\mathfrak{p}}(L'|K, \chi, s).$$

(iii) If K' is an intermediate field of $L|K$ and χ a character of $G(L|K')$, then

$$\mathcal{L}_{\mathfrak{p}}(L|K, \chi_*, s) = \prod_{\mathfrak{q}|\mathfrak{p}} \mathcal{L}_{\mathfrak{q}}(L|K', \chi, s),$$

where $\mathfrak{q}$ varies over the places of K' lying above $\mathfrak{p}$.

Proof: (i) is trivial.

(ii) If $\mathfrak{P}|\mathfrak{P}'|\mathfrak{p}$ are places of $L \supseteq L' \supseteq K$, each lying above the next, then $\varphi_{\mathfrak{p}}$ is mapped under the projection $G(L|K) \rightarrow G(L'|K)$ to $\varphi_{\mathfrak{P}'}$. So $\chi(\varphi_{\mathfrak{p}}) = \chi(\varphi_{\mathfrak{P}'})$.

(iii) If $\mathfrak{p}$ is complex, then there are precisely $m = [K' : K]$ places $\mathfrak{q}$ above $\mathfrak{p}$. They are also complex, and the claim follows from $\chi_*(1) = m\chi(1)$.

Suppose $\mathfrak{p}$ is real. Let $G = G(L|K)$, $H = G(L|K')$, and let $H \backslash G / G_{\mathfrak{p}}$ be the set of double cosets $H\tau G_{\mathfrak{p}}$ with a fixed place $\mathfrak{P}$ of L above $\mathfrak{p}$. Then we have a bijection

$$H \backslash G / G_{\mathfrak{p}} \longrightarrow \{\mathfrak{q} \text{ place of } K' \text{ above } \mathfrak{p}\}, \quad H\tau G_{\mathfrak{p}} \longmapsto \mathfrak{q}_{\tau} = \tau\mathfrak{P}|_{K'}$$

(see chap. I, § 9, p. 55). $\mathfrak{q}_{\tau}$ is real if and only if $\varphi_{\tau\mathfrak{p}} = \tau\varphi_{\mathfrak{p}}\tau^{-1} \in H$, i.e., $G_{\tau\mathfrak{p}} = \tau G_{\mathfrak{p}}\tau^{-1} \subseteq H$. The latter inclusion holds if and only if the double coset $H\tau G_{\mathfrak{p}}$ consists of only one coset mod H :

$$H\tau G_{\mathfrak{p}} = (H\tau G_{\mathfrak{p}}\tau^{-1})\tau = H\tau.$$

We thus obtain the real places among the $\mathfrak{q}_{\tau}$ by letting τ run through a system of representatives of the cosets $H\tau$ of $H \backslash G$ such that $\tau\varphi_{\mathfrak{p}}\tau^{-1} \in H$. But, for such a system, one has

$$\chi_*(\varphi_{\mathfrak{p}}) = \sum_{\tau} \chi(\tau\varphi_{\mathfrak{p}}\tau^{-1}) = \sum_{\tau} \chi(\varphi_{\tau\mathfrak{p}}).$$

Putting $\Omega = \tau \mathfrak{P}$, makes $q = \Omega|_{K'}$ run through the real places of K' above p , i.e.,

$$\chi_*(\varphi_{\mathfrak{P}}) = \sum_{\substack{q|p \\ \text{real}}} \chi(\varphi_{\Omega}).$$

On the other hand we have

$$\chi_*(1) = \sum_{q \text{ complex}} 2\chi(1) + \sum_{q \text{ real}} \chi(1).$$

Legendre's duplication formula $L_{\mathbb{R}}(s)L_{\mathbb{R}}(s+1) = L_{\mathbb{C}}(s)$ (see (4.3)) turns this into

$$\begin{aligned} \mathcal{L}_p(L|K, \chi_*, s) &= \\ \prod_{q \text{ complex}} L_{\mathbb{C}}(s)^{\chi(1)} \prod_{q \text{ real}} L_{\mathbb{R}}(s)^{\frac{\chi(1)+\chi(\varphi_{\Omega})}{2}} \prod_{q \text{ real}} L_{\mathbb{R}}(s+1)^{\frac{\chi(1)-\chi(\varphi_{\Omega})}{2}} &= \\ = \prod_{q|p} \mathcal{L}_q(L|K', \chi, s). \end{aligned} \quad \square$$

We finally put

$$\mathcal{L}_{\infty}(L|K, \chi, s) = \prod_{p|\infty} \mathcal{L}_p(L|K, \chi, s),$$

and obtain immediately from the above proposition the equations

$$\begin{aligned} \mathcal{L}_{\infty}(L|K, \chi + \chi', s) &= \mathcal{L}_{\infty}(L|K, \chi, s) \mathcal{L}_{\infty}(L|K, \chi', s), \\ \mathcal{L}_{\infty}(L|K, \chi, s) &= \mathcal{L}_{\infty}(L'|K, \chi, s), \\ \mathcal{L}_{\infty}(L|K, \chi_*, s) &= \mathcal{L}_{\infty}(L|K', \chi, s). \end{aligned}$$

(12.2) Definition. The **completed Artin L -series** for the character χ of $G(L|K)$ is defined to be

$$\Lambda(L|K, \chi, s) = c(L|K, \chi)^{s/2} \mathcal{L}_{\infty}(L|K, \chi, s) \mathcal{L}(L|K, \chi, s),$$

where

$$c(L|K, \chi) = |d_K|^{\chi(1)} \mathfrak{N}(\mathfrak{f}(L|K, \chi)).$$

The behaviour of the factors $c(L|K, \chi)$, $\mathcal{L}_{\infty}(L|K, \chi, s)$, $\mathcal{L}(L|K, \chi, s)$ on the right-hand side, which we studied in (10.4), (11.11), and above, carries over to the function $\Lambda(L|K, \chi, s)$, i.e., we have the

(12.3) Proposition. (i) $\Lambda(L|K, \chi + \chi', s) = \Lambda(L|K, \chi, s)\Lambda(L|K, \chi', s)$.

(ii) If $L'|K$ is a Galois subextension of $L|K$ and χ a character of $G(L'|K)$, then

$$\Lambda(L|K, \chi, s) = \Lambda(L'|K, \chi, s).$$

(iii) If K' is an intermediate field of $L|K$ and χ a character of $G(L|K')$, then

$$\Lambda(L|K, \chi_*, s) = \Lambda(L|K', \chi, s).$$

For a character χ of degree 1, the completed Artin L -series $\Lambda(L|K, \chi, s)$ coincides with a completed Hecke L -series. To see this, let $L_\chi|K$ be the fixed field of the kernel of χ , and let $\mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$ be the conductor of $L_\chi|K$. By (11.10), we then have

$$\mathfrak{f} = \mathfrak{f}(\chi).$$

Via the Artin symbol

$$J^{\mathfrak{f}}/P^{\mathfrak{f}} \longrightarrow G(L_\chi|K), \quad \mathfrak{a} \longmapsto \left(\frac{L_\chi|K}{\mathfrak{a}} \right),$$

χ becomes a Dirichlet character of conductor $\mathfrak{f}$, i.e., by (6.9), a primitive *Größencharakter* mod $\mathfrak{f}(\chi)$ with exponent $p = (p_\tau)$, so that $p_\tau = 0$ if τ is complex. This *Größencharakter* will be denoted $\tilde{\chi}$.

We put $p_{\mathfrak{p}} = p_\tau$ if $\mathfrak{p}$ is the place corresponding to the embedding $\tau : K \rightarrow \mathbb{C}$. The numbers $p_{\mathfrak{p}}$ have the following Galois-theoretical meaning.

(12.4) Lemma. For every real place $\mathfrak{p}$ of K one has

$$p_{\mathfrak{p}} = [L_{\chi\mathfrak{p}} : K_{\mathfrak{p}}] - 1.$$

Proof: We consider the isomorphism

$$I/I^{\mathfrak{f}}K^* \xrightarrow{\sim} J^{\mathfrak{f}}/P^{\mathfrak{f}},$$

where $I^{\mathfrak{f}} = \prod_{\mathfrak{p}} U_{\mathfrak{p}}^{(n_{\mathfrak{p}})}$ is the congruence subgroup mod $\mathfrak{f}$ of the idèle group $I = \prod_{\mathfrak{p}} K_{\mathfrak{p}}^*$ (see chap. VI, (1.9)), and consider the composite map

$$I/I^{\mathfrak{f}}K^* \longrightarrow J^{\mathfrak{f}}/P^{\mathfrak{f}} \longrightarrow G(L_\chi|K) \xhookrightarrow{\chi} \mathbb{C}^*.$$

Let $\mathfrak{p}$ be a real place of K , and let $\alpha \in I$ be the idèle with components $\alpha_{\mathfrak{p}} = -1$ and $\alpha_{\mathfrak{q}} = 1$ for all places $\mathfrak{q}$ different from $\mathfrak{p}$. By chap. VI, (5.6), the image $\varphi_{\mathfrak{p}} = (\alpha, L_\chi|K) = (-1, L_{\chi\mathfrak{p}}|K_{\mathfrak{p}})$ in $G(L_\chi|K)$ is a generator of

the decomposition group $G_{\mathfrak{p}} = G(L_{\chi \mathfrak{p}} | K_{\mathfrak{p}})$. By the approximation theorem, we may choose an $a \in K^*$ such that $a \equiv 1 \pmod{\mathfrak{f}}$, $a < 0$ in $K_{\mathfrak{p}}$, and $a > 0$ in $K_{\mathfrak{q}}$, for all real places $\mathfrak{q} \neq \mathfrak{p}$. Then

$$\beta = \alpha a \in I^{(\mathfrak{f})} = \{x \in I \mid x_{\mathfrak{p}} \in U_{\mathfrak{p}}^{(n_{\mathfrak{p}})} \text{ for } \mathfrak{p} \mid \mathfrak{f}\infty\}, \quad \text{if } \mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}.$$

As explained in the proof of chap. VI, (1.9), the image of $\alpha \pmod{I^{\mathfrak{f}} K^*}$ in $J^{\mathfrak{f}}/P^{\mathfrak{f}}$ is the class of $(\beta) = (a)$, which therefore maps to $\varphi_{\mathfrak{p}}$. Consequently,

$$\chi((a)) = \chi_{\mathfrak{f}}(a) \chi_{\infty}(a) = \chi(\varphi_{\mathfrak{p}}).$$

Since $a \equiv 1 \pmod{\mathfrak{f}}$, we have $\chi_{\mathfrak{f}}(a) = 1$ and $\chi_{\infty}(a) = N\left(\left(\frac{a}{|a|}\right)^p\right) = \left(\frac{a}{|a|_{\mathfrak{p}}}\right)^{p_{\mathfrak{p}}} = (-1)^{p_{\mathfrak{p}}}$, i.e., $\chi(\varphi_{\mathfrak{p}}) = (-1)^{p_{\mathfrak{p}}}$, so that $\varphi_{\mathfrak{p}} = 1$ for $p_{\mathfrak{p}} = 0$, and $\varphi_{\mathfrak{p}} \neq 1$ for $p_{\mathfrak{p}} = 1$. But this is the statement of the lemma. $\square$

(12.5) Proposition. *The completed Artin L -series for the character χ of degree 1 and the completed Hecke L -series for the Größencharakter $\tilde{\chi}$ coincide:*

$$\Lambda(L|K, \chi, s) = \Lambda(\tilde{\chi}, s).$$

Proof: The completed Hecke L -series is given, according to § 8, by

$$\Lambda(\tilde{\chi}, s) = (|d_K| \mathfrak{N}(\mathfrak{f}(\tilde{\chi})))^{s/2} L_{\infty}(\tilde{\chi}, s) L(\tilde{\chi}, s)$$

with

$$L_{\infty}(\tilde{\chi}, s) = L_X(s),$$

and $s = s_1 + p$, where

$$L_X(s) = \prod_{\mathfrak{p} \mid \infty} L_{\mathfrak{p}}(s_{\mathfrak{p}})$$

is the L -function of the $G(\mathbb{C}|\mathbb{R})$ -set $X = \text{Hom}(K, \mathbb{C})$ defined in § 4. The factors $L_{\mathfrak{p}}(s_{\mathfrak{p}})$ are given explicitly by

$$(*) \quad L_{\mathfrak{p}}(s_{\mathfrak{p}}) = \begin{cases} L_{\mathbb{C}}(s), & \text{if } \mathfrak{p} \text{ complex,} \\ L_{\mathbb{R}}(s + p_{\mathfrak{p}}), & \text{if } \mathfrak{p} \text{ real,} \end{cases}$$

(see p.454). On the other hand we have

$$\Lambda(L|K, \chi, s) = c(L|K, \chi)^{s/2} \mathcal{L}_{\infty}(L|K, \chi, s) \mathcal{L}(L|K, \chi, s)$$

with

$$c(L|K, \chi) = |d_K| \mathfrak{N}(\mathfrak{f}(L|K, \chi))$$

and

$$\mathcal{L}_{\infty}(L|K, \chi, s) = \prod_{\mathfrak{p} \mid \infty} \mathcal{L}_{\mathfrak{p}}(L|K, \chi, s).$$

Let L_χ be the fixed field of the kernel of χ . By (11.11), (ii), and the remark preceding lemma (12.4), one has

$$c(L|K, \chi) = c(L_\chi|K, \chi) = |d_K| \mathfrak{N}(\mathfrak{f}(\tilde{\chi})),$$

and by (10.4), (ii), and (10.6), and the subsequent remark, one has

$$\mathcal{L}(L|K, \chi, s) = \mathcal{L}(L_\chi|K, \chi, s) = L(\tilde{\chi}, s).$$

We are thus reduced to proving

$$\mathcal{L}_p(L|K, \chi, s) = L_p(s_p)$$

for $p|\infty$ and $s = s_1 + p$. Firstly, we have $\mathcal{L}_p(L|K, \chi, s) = \mathcal{L}_p(L_\chi|K, \chi, s)$ (see p. 537). Let $\varphi_{\mathfrak{p}}$ be the generator of $G(L_{\chi\mathfrak{p}}|K_p)$. Since χ is injective on $G(L_\chi|K)$, we get $\chi(\varphi_{\mathfrak{p}}) = -1$ if $\varphi_{\mathfrak{p}} \neq 1$, and $\chi(\varphi_{\mathfrak{p}}) = 1$ if $\varphi_{\mathfrak{p}} = 1$. Using (12.4) this gives

$$\mathcal{L}_p(L_\chi|K, \chi, s) = \begin{cases} L_{\mathbb{C}}(s), & \text{for } p \text{ complex,} \\ L_{\mathbb{R}}(s), & \text{for } p \text{ real and } \mathfrak{P} \text{ real, i.e., } p_p = 0, \\ L_{\mathbb{R}}(s+1), & \text{for } p \text{ real and } \mathfrak{P} \text{ complex, i.e., } p_p = 1. \end{cases}$$

Hence (*) shows that indeed $\mathcal{L}_p(L|K, \chi, s) = L_p(s_p)$. $\square$

In view of the two results (12.3) and (12.5), the functional equation for Artin L -series now follows from Brauer's theorem (10.3) in a purely formal fashion, as a consequence of the functional equation for Hecke L -series, which we have already established.

(12.6) Theorem. *The Artin L -series $\Lambda(L|K, \chi, s)$ admits a meromorphic continuation to $\mathbb{C}$ and satisfies the functional equation*

$$\Lambda(L|K, \chi, s) = W(\chi) \Lambda(L|K, \bar{\chi}, 1-s)$$

with a constant $W(\chi)$ of absolute value 1.

Proof: By Brauer's theorem, the character χ is an integral linear combination

$$\chi = \sum n_i \chi_{i*},$$

where the χ_{i*} are induced from characters χ_i of degree 1 on subgroups $H_i = G(L|K_i)$. From propositions (12.3) and (12.5), it follows that

$$\begin{aligned} \Lambda(L|K, \chi, s) &= \prod_i \Lambda(L|K, \chi_{i*}, s)^{n_i} \\ &= \prod_i \Lambda(L|K_i, \chi_i, s)^{n_i} \\ &= \prod_i \Lambda(\tilde{\chi}_i, s)^{n_i}, \end{aligned}$$

where $\tilde{\chi}_i$ is the *Größencharakter* of K_i associated to χ_i . By (8.6), the Hecke L -series $\Lambda(\tilde{\chi}_i, s)$ admit meromorphic continuations to $\mathbb{C}$ and satisfy the functional equation

$$\Lambda(\tilde{\chi}_i, s) = W(\tilde{\chi}_i) \Lambda(\tilde{\chi}_i, 1 - s).$$

Therefore $\Lambda(L|K, \chi, s)$ satisfies the functional equation

$$\Lambda(L|K, \chi, s) = W(\chi) \prod_i \Lambda(\tilde{\chi}_i, 1 - s) = W(\chi) \Lambda(L|K, \bar{\chi}, 1 - s),$$

where $W(\chi) = \prod_i W(\tilde{\chi}_i)$ is of absolute value 1. $\square$

The functional equation for the Artin L -series may be given the following explicit form, which is easily deduced from (12.6) and (4.3):

$$\mathcal{L}(L|K, \chi, 1 - s) = A(\chi, s) \mathcal{L}(L|K, \bar{\chi}, s),$$

with the factor

$$\begin{aligned} A(\chi, s) &= W(\chi) \left[|d_K|^{\chi(1)} \mathfrak{N}(\mathfrak{f}(L|K, \chi)) \right]^{s - \frac{1}{2}} \\ &\quad \times (\cos \pi s / 2)^{n^+} (\sin \pi s / 2)^{n^-} \Gamma_{\mathbb{C}}(s)^{n\chi(1)} \end{aligned}$$

and the exponents

$$n^+ = \frac{n}{2} \chi(1) + \sum_{\mathfrak{p}} \frac{1}{2} \chi(\varphi_{\mathfrak{p}}), \quad n^- = \frac{n}{2} \chi(1) - \sum_{\mathfrak{p}} \frac{1}{2} \chi(\varphi_{\mathfrak{p}}).$$

Here the summations are over the real places $\mathfrak{p}$ of K . This gives immediately the zeroes of the function $\mathcal{L}(L|K, \chi, s)$ in the half-plane $\operatorname{Re}(s) \leq 0$. If χ is not the principal character, they are the following:

$$\text{at } s = 0, -2, -4, \dots \text{ zeroes of order } \frac{n}{2} \chi(1) + \sum_{\mathfrak{p} \text{ real}} \frac{1}{2} \chi(\varphi_{\mathfrak{p}}),$$

$$\text{at } s = -1, -3, -5, \dots \text{ zeroes of order } \frac{n}{2} \chi(1) - \sum_{\mathfrak{p} \text{ real}} \frac{1}{2} \chi(\varphi_{\mathfrak{p}}).$$

Remark: For the proof of the functional equation of the completed Artin L -series, we have made essential use of the fact that “Euler factors” $\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s)$ at the infinite places $\mathfrak{p}$, which are made up out of gamma functions, behave under change of fields and characters in exactly the same way as the Euler factors

$$\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s) = \det(1 - \varphi_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{-s}; V^{I_{\mathfrak{p}}})^{-1}$$

at the finite places. This uniform behaviour is in striking contrast to the great difference in the procedures that lead to the definitions of the Euler factors for $\mathfrak{p}|\infty$ and $\mathfrak{p} \nmid \infty$. It is in this context that the mathematician *CHRISTOPHER DENINGER* recently made a very interesting discovery (see [26], [27]). He shows that the Euler factors for all places $\mathfrak{p}$ can all be written in the same way:

$$\mathcal{L}_{\mathfrak{p}}(L|K, \chi, s) = \det_{\infty} \left(\frac{\log \mathfrak{N}(\mathfrak{p})}{2\pi i} (s \operatorname{id} - \Theta_{\mathfrak{p}}); H(X_{\mathfrak{p}}/\mathbb{L}_{\mathfrak{p}}) \right)^{-1}.$$

Here $H(X_{\mathfrak{p}}/\mathbb{L}_{\mathfrak{p}})$ is an *infinite dimensional* $\mathbb{C}$ -vector space which can be canonically constructed, $\Theta_{\mathfrak{p}}$ is a certain linear “Frobenius” operator on it, and $\det_{\infty}$ is a “regularized determinant” which generalizes the ordinary notion of determinant for finite dimensional vector spaces to the infinite dimensional case. The theory based on this observation is of the utmost generality, and reaches far beyond Artin L -series. It suggests a complete analogy for the theory of L -series of algebraic varieties over finite fields. The striking success which the geometric interpretation and treatment of the L -series has enjoyed in this analogous situation adds to the relevance of *DENINGER*’s theory for present-day research.

§ 13. Density Theorems

Dirichlet’s prime number theorem (5.14) says that in every *arithmetic progression*

$$a, a \pm m, a \pm 2m, a \pm 3m, \dots,$$

$a, m \in \mathbb{N}$, $(a, m) = 1$, there occur infinitely many prime numbers. Using L -series, we will now deduce a far-reaching generalization and sharpening of this theorem.

(13.1) Definition. Let M be a set of prime ideals of K . The limit

$$d(M) = \lim_{s \rightarrow 1+0} \frac{\sum_{\mathfrak{p} \in M} \mathfrak{N}(\mathfrak{p})^{-s}}{\sum_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{-s}},$$

provided it exists, is called the **Dirichlet density** of M .

From the product expansion

$$\zeta_K(s) = \prod_{\mathfrak{p}} \frac{1}{1 - \mathfrak{N}(\mathfrak{p})^{-s}}, \quad \operatorname{Re}(s) > 1,$$

we obtain as in § 8, p. 494,

$$\log \zeta_K(s) = \sum_{\mathfrak{p}, m} \frac{1}{m \mathfrak{N}(\mathfrak{p})^{ms}} = \sum_{\mathfrak{p}} \frac{1}{\mathfrak{N}(\mathfrak{p})^s} + \sum_{\mathfrak{p}, m \geq 2} \frac{1}{m \mathfrak{N}(\mathfrak{p})^{ms}}.$$

The latter sum obviously defines an analytic function at $s = 1$. We write $f(s) \sim g(s)$ if $f(s) - g(s)$ is an analytic function at $s = 1$. Then we have

$$\log \zeta_K(s) \sim \sum_{\mathfrak{p}} \frac{1}{\mathfrak{N}(\mathfrak{p})^s} \sim \sum_{\deg(\mathfrak{p})=1} \frac{1}{\mathfrak{N}(\mathfrak{p})^s},$$

because the sum $\sum_{\deg(\mathfrak{p}) \geq 2} \mathfrak{N}(\mathfrak{p})^{-s}$ taken over all $\mathfrak{p}$ of degree ≥ 2 is analytic at $s = 1$. Furthermore, by (5.11), (ii), we have $\zeta_K(s) \sim \frac{1}{s-1}$, and so

$$\sum_{\mathfrak{p}} \frac{1}{\mathfrak{N}(\mathfrak{p})^s} \sim \log \frac{1}{s-1}.$$

So we may also write the Dirichlet density as

$$d(M) = \lim_{s \rightarrow 1+0} \frac{\sum_{\mathfrak{p} \in M} \mathfrak{N}(\mathfrak{p})^{-s}}{\log \frac{1}{s-1}}.$$

Since the sum $\sum \mathfrak{N}(\mathfrak{p})^{-s}$ over all prime ideals of degree > 1 converges, the definition of Dirichlet density only depends on the prime ideals of degree 1 in M . Adding or omitting finitely many prime ideals also does not change anything as far as existence or value of the Dirichlet density is concerned. One frequently also considers the **natural density**

$$\delta(M) = \lim_{x \rightarrow \infty} \frac{\#\{\mathfrak{p} \in M \mid \mathfrak{N}(\mathfrak{p}) \leq x\}}{\#\{\mathfrak{p} \mid \mathfrak{N}(\mathfrak{p}) \leq x\}}.$$

It is not difficult to show that the existence of $\delta(M)$ implies the existence of $d(M)$, and that one has $\delta(M) = d(M)$. The converse is not always true (see [123], p. 26). In the notation of chap. VI, § 1 and § 7, we prove the generalized **Dirichlet density theorem**.

(13.2) Theorem. *Let $\mathfrak{m}$ be a module of K and $H^{\mathfrak{m}}$ an ideal group such that $J^{\mathfrak{m}} \supseteq H^{\mathfrak{m}} \supseteq P^{\mathfrak{m}}$ with index $h_{\mathfrak{m}} = (J^{\mathfrak{m}} : H^{\mathfrak{m}})$.*

For every class $\mathfrak{K} \in J^{\mathfrak{m}}/H^{\mathfrak{m}}$, the set $P(\mathfrak{K})$ of prime ideals in $\mathfrak{K}$ has density

$$d(P(\mathfrak{K})) = \frac{1}{h_{\mathfrak{m}}}.$$

For the proof we need the following

(13.3) Lemma. *Let χ be a nontrivial (irreducible) character of J^m/P^m (i.e., a character of degree 1). Then the Hecke L -series*

$$L(\chi, s) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p})\mathfrak{N}(\mathfrak{p})^{-s}}$$

($\chi(\mathfrak{p}) = 0$ for $\mathfrak{p}|\mathfrak{m}$) satisfies

$$L(\chi, 1) \neq 0.$$

Proof: By (8.5) and the remark following (5.10) (in the case $m = 1$), $L(\chi, s)$ does not have a pole at $s = 1$. Let $L|K$ be the ray class field mod $\mathfrak{m}$, so $G(L|K) \cong J^m/P^m$. Interpreting χ as a character of the Galois group $G(L|K)$, the function $L(\chi, s)$ agrees with the Artin L -series $\mathcal{L}(L|K, \chi, s)$ up to finitely many Euler factors — see (10.6). Like $L(\chi, s)$, this Artin L -series does not have a pole at $s = 1$. So all we have to show is that $\mathcal{L}(L|K, \chi, 1) \neq 0$. According to (10.5), we have

$$\zeta_L(s) = \zeta_K(s) \prod_{\chi \neq 1} \mathcal{L}(L|K, \chi, s)^{\chi(1)},$$

where χ runs through the nontrivial irreducible characters of $G(L|K)$. By (5.11), both $\zeta_K(s)$ and $\zeta_L(s)$ have simple poles at $s = 1$, i.e., the product is nonzero at $s = 1$. Since none of the factors has a pole, we find $\mathcal{L}(L|K, \chi, 1) \neq 0$. $\square$

Proof of (13.2): Exactly as for the Dedekind zeta function above, we obtain for the Dirichlet L -series

$$\log L(\chi, s) \sim \sum_{\mathfrak{p}} \frac{\chi(\mathfrak{p})}{\mathfrak{N}(\mathfrak{p})^s} = \sum_{\mathfrak{R}' \in J^m/P^m} \chi(\mathfrak{R}') \sum_{\mathfrak{p} \in \mathfrak{R}'} \frac{1}{\mathfrak{N}(\mathfrak{p})^s}.$$

Multiplying this by $\chi(\mathfrak{R}^{-1})$ and summing over all (irreducible) χ yields

$$\log \zeta_K(s) + \sum_{\chi \neq 1} \chi(\mathfrak{R}^{-1}) \log L(\chi, s) \sim \sum_{\mathfrak{R}' \in J^m/P^m} \sum_{\chi} \chi(\mathfrak{R}'\mathfrak{R}^{-1}) \sum_{\mathfrak{p} \in \mathfrak{R}'} \frac{1}{\mathfrak{N}(\mathfrak{p})^s}.$$

Since $L(\chi, 1) \neq 0$, $\log L(\chi, s)$ is analytic at $s = 1$. But

$$\sum_{\chi} \chi(\mathfrak{R}'\mathfrak{R}^{-1}) = \begin{cases} 0, & \text{if } \mathfrak{R}' \neq \mathfrak{R}, \\ h_m, & \text{if } \mathfrak{R}' = \mathfrak{R}. \end{cases}$$

Hence we get

$$\log \frac{1}{s-1} \sim \log \zeta_K(s) \sim h_m \sum_{\mathfrak{p} \in \mathfrak{R}} \frac{1}{\mathfrak{N}(\mathfrak{p})^s},$$

and the theorem is proved. $\square$

The theorem shows in particular that the density of the prime ideals in a class of $J^{\mathfrak{m}}/H^{\mathfrak{m}}$ is the same for every class, i.e., the prime ideals are **equidistributed** among the classes. In the case $K = \mathbb{Q}$, $\mathfrak{m} = (m)$, and $H^{\mathfrak{m}} = P^{\mathfrak{m}}$, we have $J^{\mathfrak{m}}/P^{\mathfrak{m}} \cong (\mathbb{Z}/m\mathbb{Z})^*$ (see chap. VI, (1.10)), and we recover the classical Dirichlet prime number theorem recalled at the beginning, in the stronger form which says that the prime numbers in an arithmetic progression, i.e., in a class $a \bmod m$, $(a, m) = 1$, have density $\frac{1}{\varphi(m)} = 1/\#(\mathbb{Z}/m\mathbb{Z})^*$.

Relating the prime ideals $\mathfrak{p}$ of a class of $J^{\mathfrak{m}}/P^{\mathfrak{m}}$, via the class field theory isomorphism $J^{\mathfrak{m}}/P^{\mathfrak{m}} \cong G(L|K)$, to the Frobenius automorphisms $\varphi_{\mathfrak{p}} = \left(\frac{L|K}{\mathfrak{p}}\right)$, gives us a Galois-theoretic interpretation of the Dirichlet density theorem. We now deduce a more general density theorem which is particularly important in that it concerns arbitrary Galois extensions (not necessarily abelian). For every $\sigma \in G(L|K)$, let us consider the set

$$P_{L|K}(\sigma)$$

of all unramified prime ideals $\mathfrak{p}$ of K such that there exists a prime ideal $\mathfrak{P}|\mathfrak{p}$ of L satisfying

$$\sigma = \left(\frac{L|K}{\mathfrak{P}}\right),$$

where $\left(\frac{L|K}{\mathfrak{P}}\right)$ is the Frobenius automorphism $\varphi_{\mathfrak{P}}$ of $\mathfrak{P}$ over K . It is clear that this set depends only on the conjugacy class

$$\langle \sigma \rangle = \{ \tau \sigma \tau^{-1} \mid \tau \in G(L|K) \}$$

of σ and that one has $P_{L|K}(\sigma) \cap P_{L|K}(\tau) = \emptyset$ if $\langle \sigma \rangle \neq \langle \tau \rangle$. What is the density of the set $P_{L|K}(\sigma)$? The answer to this question is given by the **Čebotarev density theorem**.

(13.4) Theorem. *Let $L|K$ be a Galois extension with group G . Then for every $\sigma \in G$, the set $P_{L|K}(\sigma)$ has a density, and it is given by*

$$d(P_{L|K}(\sigma)) = \frac{\#\langle \sigma \rangle}{\#G}.$$

Proof: We first assume that G is generated by σ . Let $\mathfrak{m}$ be the conductor of $L|K$. Then $L|K$ is the class field of an ideal group $H^{\mathfrak{m}}$, $J^{\mathfrak{m}} \supseteq H^{\mathfrak{m}} \supseteq P^{\mathfrak{m}}$. Let $\mathfrak{R} \in J^{\mathfrak{m}}/H^{\mathfrak{m}}$ be the class corresponding to the element σ under the isomorphism

$$J^{\mathfrak{m}}/H^{\mathfrak{m}} \xrightarrow{\sim} G, \quad \mathfrak{p} \mapsto \left(\frac{L|K}{\mathfrak{p}}\right).$$

Then $P_{L|K}(\sigma)$ consists precisely of the prime ideals $\mathfrak{p}$ which lie in the class $\mathfrak{K}$. By the Dirichlet density theorem (13.2), we conclude that $P_{L|K}(\sigma)$ has density

$$d(P_{L|K}(\sigma)) = \frac{1}{h_m} = \frac{1}{\#G} = \frac{\#\langle\sigma\rangle}{\#G}.$$

In the general case, let Σ be the fixed field of σ . If f is the order of σ , then, as we just saw, $d(P_{L|\Sigma}(\sigma)) = \frac{1}{f}$. Let $\bar{P}(\sigma)$ be the set of prime ideals $\mathfrak{P}$ of L such that $\mathfrak{P}|\mathfrak{p} \in P_{L|K}(\sigma)$ and $(\frac{L|K}{\mathfrak{P}}) = \sigma$. Then $\bar{P}(\sigma)$ corresponds bijectively to the set $P'_{L|\Sigma}(\sigma)$ of those prime ideals $\mathfrak{q}$ in $P_{L|\Sigma}(\sigma)$ such that $\Sigma_{\mathfrak{q}} = K_{\mathfrak{p}}$, $\mathfrak{q}|\mathfrak{p}$. Since the remaining prime ideals in $P_{L|\Sigma}(\sigma)$ are either ramified or have degree > 1 over $\mathbb{Q}$, we may omit them and obtain

$$d(P'_{L|\Sigma}(\sigma)) = d(P_{L|\Sigma}(\sigma)) = \frac{1}{f}.$$

Now we consider the surjective map

$$\rho : P'_{L|\Sigma}(\sigma) \rightarrow P_{L|K}(\sigma), \quad \mathfrak{q} \mapsto \mathfrak{q} \cap K.$$

As $P'_{L|\Sigma}(\sigma) \cong \bar{P}(\sigma)$, we get, for every $\mathfrak{p} \in P_{L|K}(\sigma)$,

$$\rho^{-1}(\mathfrak{p}) \cong \{\mathfrak{P} \in \bar{P}(\sigma) \mid \mathfrak{P}|\mathfrak{p}\} \cong Z(\sigma)/(\sigma),$$

where $Z(\sigma) = \{\tau \in G \mid \tau\sigma = \sigma\tau\}$ is the centralizer of σ . So we get

$$d(P_{L|K}(\sigma)) = \frac{1}{(Z(\sigma) : \langle\sigma\rangle)} d(P'_{L|\Sigma}(\sigma)) = \frac{f}{\#Z(\sigma)} \frac{1}{f} = \frac{\#\langle\sigma\rangle}{\#G}. \quad \square$$

The Čebotarev density theorem has quite a number of surprising consequences, which we will now deduce. If S and T are any two sets of primes, then let us write

$$S \subseteq T$$

to indicate that S is contained in T up to finitely many exceptional elements.

Furthermore, let us write $S = T$ if $S \subseteq T$ and $T \subseteq S$.

Let $L|K$ be a finite extension of algebraic number fields. We denote by $P(L|K)$ the set of all unramified prime ideals $\mathfrak{p}$ of K which admit in L a prime divisor $\mathfrak{P}$ of degree 1 over K . So, if $L|K$ is Galois, then $P(L|K)$ is just the set of all prime ideals of K which split completely in L .

(13.5) Lemma. *Let $N|K$ be a Galois extension containing L , and let $G = G(N|K)$, $H = G(N|L)$. Then one has*

$$P(L|K) = \bigcup_{\langle\sigma\rangle \cap H \neq \emptyset} P_{N|K}(\sigma) \quad (\text{disjoint union}).$$

Proof: A prime ideal $\mathfrak{p}$ of K which is unramified in N lies in $P(L|K)$ if and only if the conjugacy class $\langle \sigma \rangle$ of $\sigma = \left(\frac{N|K}{\mathfrak{p}} \right)$, for some prime ideal $\mathfrak{P}|\mathfrak{p}$ of N , contains an element of H , i.e., if and only if $\mathfrak{p} \in P_{N|K}(\sigma)$ for some $\sigma \in G$ such that $\langle \sigma \rangle \cap H \neq \emptyset$. $\square$

(13.6) Corollary. *If $L|K$ is an extension of degree n , then the set $P(L|K)$ has density $d(P(L|K)) \geq \frac{1}{n}$. Furthermore, one has*

$$d(P(L|K)) = \frac{1}{n} \iff L|K \text{ is Galois.}$$

Proof: Let $N|K$ be a Galois extension containing L , and let $G = G(N|K)$ and $H = G(N|L)$. By (13.5), we have

$$P(L|K) = \bigcup_{\langle \sigma \rangle \cap H \neq \emptyset} P_{N|K}(\sigma).$$

The Čebotarev density theorem (13.4) then yields

$$d(P(L|K)) = \sum_{\langle \sigma \rangle \cap H \neq \emptyset} \frac{\#\langle \sigma \rangle}{\#G} = \frac{1}{\#G} \# \left(\bigcup_{\langle \sigma \rangle \cap H \neq \emptyset} \langle \sigma \rangle \right).$$

Since $H \subseteq \bigcup_{\langle \sigma \rangle \cap H \neq \emptyset} \langle \sigma \rangle$, it follows that

$$d(P(L|K)) \geq \frac{\#H}{\#G} = \frac{1}{n}.$$

$L|K$ is Galois if and only if H is a normal subgroup of G , and this is the case if and only if $\langle \sigma \rangle \subseteq H$ whenever $\langle \sigma \rangle \cap H \neq \emptyset$, and so this holds if and only if $H = \bigcup_{\langle \sigma \rangle \cap H \neq \emptyset} \langle \sigma \rangle$. This implies the second claim. $\square$

(13.7) Corollary. *If almost all prime ideals split completely in the finite extension $L|K$, then $L = K$.*

Proof: Let $N|K$ be the normal closure of $L|K$, i.e., the smallest Galois extension containing L . A prime ideal $\mathfrak{p}$ of K splits completely in L if and only if it splits completely in $N|K$ (see chap. I, §9, exercise 4). Under the hypothesis of the corollary, we therefore have

$$1 = d(P(L|K)) = d(P(N|K)) = \frac{1}{[N : K]},$$

so that $[N : K] = 1$ and $N = L = K$. $\square$

(13.8) Corollary. *An extension $L|K$ is Galois if and only if every prime ideal in $P(L|K)$ splits completely in L .*

Proof: Let again $N|K$ be the normal closure of $L|K$. Then $P(N|K)$ consists precisely of those prime ideals which split completely in L . Hence if $P(N|K) = P(L|K)$, then by (13.6),

$$\frac{1}{[N : K]} = d(P(N|K)) = d(P(L|K)) \geq \frac{1}{[L : K]},$$

i.e., $[N : K] \leq [L : K]$, so $L = N$ is Galois. The converse is trivial. $\square$

(13.9) Proposition (M. BAUER). *If $L|K$ is Galois and $M|K$ is an arbitrary finite extension, then*

$$P(L|K) \supseteq P(M|K) \iff L \subseteq M.$$

Proof: $L \subseteq M$ trivially implies that $P(M|K) \subseteq P(L|K)$. So assume conversely that $P(L|K) \supseteq P(M|K)$. Let $N|K$ be a Galois extension containing L and M , and let $G = G(N|K)$, $H = G(N|L)$, $H' = G(N|M)$. Then we have

$$P(M|K) = \bigcup_{\langle \sigma \rangle \cap H' \neq \emptyset} P_{N|K}(\sigma) \subseteq P(L|K) = \bigcup_{\langle \sigma \rangle \cap H \neq \emptyset} P_{N|K}(\sigma).$$

Let $\sigma \in H'$. Since $P_{N|K}(\sigma)$ is infinite by (13.4), there must exist some $\mathfrak{p} \in P_{N|K}(\sigma)$ such that $\mathfrak{p} \in P_{N|K}(\tau)$ for a suitable $\tau \in G$ such that $\langle \tau \rangle \cap H \neq \emptyset$. But then σ is conjugate to τ , and since H is a normal subgroup of G , we find $\langle \sigma \rangle = \langle \tau \rangle \subseteq H$. We therefore have $H' \subseteq H$, and hence $L \subseteq M$. $\square$

(13.10) Corollary. *A Galois extension $L|K$ is uniquely determined by the set $P(L|K)$ of prime ideals which split completely in it.*

This beautiful result is the beginning of an answer to the programme formulated by LEOPOLD KRONECKER (1821–1891), of characterizing the extensions of K , with all their algebraic and arithmetic properties, solely in terms of sets of prime ideals, “*in a similar way as Cauchy’s theorem determines a function by its boundary values*”. The result raises the question of how to characterize the sets $P(L|K)$ of prime ideals solely in terms of the base field K . For abelian extensions, class field theory gives a concise

answer to this, in that it recognizes $P(L|K)$ as the set of prime ideals lying in the ideal group $H^{\mathfrak{m}}$ for any module of definition $\mathfrak{m}$ (see chap. VI, (7.3)). If for instance $L|K$ is the Hilbert class field, then $P(L|K)$ consists precisely of the prime ideals which are principal ideals. If on the other hand $K = \mathbb{Q}$ and $L = \mathbb{Q}(\mu_m)$, then $P(L|K)$ consists of all prime numbers $p \equiv 1 \pmod{m}$.

In the case of nonabelian extensions $L|K$, a characterization of the sets $P(L|K)$ is essentially not known. However, this problem is part of a much more general and far-reaching programme known as “Langlands philosophy”, which is undergoing a rapid development at the moment. For an introduction to this circle of ideas, we refer the interested reader to [106].